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# Section 1 : Simple Definition

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## PART 1 — RANDOM VARIABLE

### What It Actually Is (No Jargon First)

A **Random Variable** is just a **labeling machine**.

**Real-world example you already know:**

- You have a coin with “Heads” and “Tails” written on it
- Your computer can’t read “Heads” — it needs numbers
- So you make a rule: “Whenever I see Heads, write 1. Whenever I see Tails, write 0.”

That rule IS the random variable. Not the coin. Not the flip. Just the **rule that converts outcomes to numbers**.

**Why this matters:** Math only works with numbers. Probability theory needs numbers to add, multiply, and calculate. Random variables are the bridge.

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### The “Scary” Formal Definition (Decoded)

**What your tutorial wrote:**

| *A random variable is a measurable function  $X : \Omega \rightarrow \mathbb{R}$*

**What this actually means, word by word:**

| Symbol              | Plain English                                      | Why It Exists                                            |
|---------------------|----------------------------------------------------|----------------------------------------------------------|
| $\Omega$<br>(Omega) | “The bag of all possible things that could happen” | We need to know what outcomes exist before we label them |
| $\mathbb{R}$        | “Real numbers” (any number: 1, 3.5, -2, 0.001)     | Math needs numbers to work with                          |
| $X$                 | The labeling rule itself                           | This is what we call “the random variable”               |
| $X(\omega)$         | “Apply rule X to outcome $\omega$ ”                | Gives us the number for that specific outcome            |

### Example with dice:

- $\Omega = \{1, 2, 3, 4, 5, 6\}$  these are the faces
- Rule  $X$  says: "Whatever face shows up, use that number"
- So  $X(4) = 4, X(6) = 6$

### Another example:

- $\Omega = \{\text{rain, sun, cloud}\}$
  - Rule  $X$ : Rain 1, Sun 0, Cloud 0.5
  - $X(\text{rain}) = 1$
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## Why "Measurable Function" Sounds Scary (But Isn't)

"Measurable" just means: *"We can actually calculate probabilities for the numbers we output."*

### Example of what would NOT be measurable:

- Rule: "If outcome is weird, output a number that has no probability"
- This breaks probability theory, so we exclude it.

**For your notes:** "Measurable" = "The math works properly." You won't need to worry about this in ML.

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## Discrete vs Continuous: The Real Difference

|                     | Discrete                           | Continuous                            |
|---------------------|------------------------------------|---------------------------------------|
| What it means       | You can list all possible values   | You cannot list all values (infinite) |
| Can you count them? | Yes                                | No                                    |
| Examples            | Dice, coin flips, number of emails | Height, temperature, time             |
| Probability tool    | PMF (Probability Mass Function)    | PDF (Probability Density Function)    |
| Sum or Integral?    | $\sum$ (sum)                       | $\int$ (integral)                     |

**Key insight:** Continuous variables aren't "more complicated" — they just need calculus instead of addition. The IDEAS are identical.

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## ML Examples (Where You Actually See These)

| ML Task              | Random Variable | What It Represents   |
|----------------------|-----------------|----------------------|
| Image Classification | $Y$             | 1 = cat, 0 = not cat |

|                        |   |                             |
|------------------------|---|-----------------------------|
| Language Model         | X | Next word token (number ID) |
| Regression             | Y | Predicted house price       |
| Reinforcement Learning | R | Reward from environment     |
| GANs                   | Z | Random noise vector         |

## PART 2 — EXPECTATION

### What It Actually Means

**Expectation = “The average you get if you repeat something forever.”**

**Critical point:** It is NOT necessarily a value you can actually observe.

**Dice example:**  = 3.5

- Can you roll 3.5 on a die? **No.**
- But if you roll 1,000,000 times and average all results, you’ll get very close to 3.5.

**Another example:**

- Lottery ticket costs \$1
- 1 in 1,000,000 chance to win \$500,000
- Expectation =  $(500,000 \times \frac{1}{1,000,000}) + (0 \times \frac{999,999}{1,000,000}) = \$0.50$
- You **never** win \$0.50. You win \$0 or \$500,000.
- But the “fair price” of the ticket is \$0.50.

### Complete Dice Expectation (Every Single Step)

**Problem:** Fair die. Find  $E[X]$ .

**Step 1:** List all outcomes and probabilities

| Outcome x | Probability $P(X = x)$ |
|-----------|------------------------|
| 1         | $\frac{1}{6}$          |
| 2         | $\frac{1}{6}$          |
| 3         | $\frac{1}{6}$          |
| 4         | $\frac{1}{6}$          |
| 5         | $\frac{1}{6}$          |

|   |               |
|---|---------------|
| 6 | $\frac{1}{6}$ |
|---|---------------|

**Step 2:** Write the formula

$$\checkmark = \sum_{x} x \cdot P(X=x)$$

**Step 3:** Substitute every value (no skipping)

$$\checkmark = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6}$$

**Step 4:** Calculate each multiplication separately

$$= \frac{1}{6} + \frac{2}{6} + \frac{3}{6} + \frac{4}{6} + \frac{5}{6} + \frac{6}{6}$$

**Step 5:** Add numerators (all have same denominator)

$$= \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = \frac{21}{6}$$

**Step 6:** Divide

$$= 3.5$$

**Answer:**  $\checkmark = 3.5$

## Continuous Expectation (What Changes)

**Formula:**  $\checkmark = \int_{-\infty}^{\infty} x \cdot f(x) \, dx$

**What's different from discrete:**

- Instead of summing ( $\sum$ ), we integrate ( $\int$ )
- Instead of  $P(X = x)$ , we use  $f(x)$  (density function)
- **Why?** Because for continuous variables,  $P(X = \text{exact value}) = 0$  (infinite possibilities), so we use density instead.

**For your notes:** The integral is just “adding up infinitely many tiny pieces.” Same idea as summation.

## Where Expectation Lives in ML

| ML Concept        | Expectation Form   | What It Means                          |
|-------------------|--------------------|----------------------------------------|
| Loss Function     | $E[\text{Loss}]$   | Average error over all data            |
| Risk Minimization | $E[L(Y, \hat{Y})]$ | Expected penalty for wrong predictions |

|                               |              |                              |
|-------------------------------|--------------|------------------------------|
| Policy Gradient               | $E[R]$       | Average reward agent expects |
| VAE (Variational Autoencoder) | $E_{q(z x)}$ |                              |

## PART 3 — VARIANCE

### What Variance Actually Measures

Variance = “How much do outcomes typically deviate from the average?”

Two scenarios with same mean, different variance:

**Student A:** 80, 80, 80

- Mean = 80
- Variance = 0 (no spread at all)

**Student B:** 60, 80, 100

- Mean = 80
- Variance = ? (we'll calculate)

Same average, completely different behavior. Variance catches this.

### Why Square? (The Most Common Question)

**Problem:** If we just average deviations:  $(X - E[X])$

**Example:** Values are 2 and 4, mean is 3

- Deviations:  $2 - 3 = -1$  and  $4 - 3 = +1$
- Average deviation:  $\frac{-1+1}{2} = 0$
- **Conclusion:** “No spread!” WRONG

**Solution:** Square the deviations first

- Squared deviations:  $(-1)^2 = 1$  and  $(1)^2 = 1$
- Average:  $\frac{1+1}{2} = 1$
- **Conclusion:** “There IS spread!” CORRECT

**Why not absolute value?**  $|X - E[X]|$  would also work, but squares are easier for calculus (derivatives). That's the only reason.

### Complete Dice Variance (Every Single Step)

**Given:** Fair die,  $\mu = 3.5$  (from before)

**Formula:**  $\text{Var}(X) = E[(X - E[X])^2]$

**Step 1:** Calculate  $(X - 3.5)$  for each outcome

| X | $X - 3.5$        | $(X - 3.5)^2$     |
|---|------------------|-------------------|
| 1 | $1 - 3.5 = -2.5$ | $(-2.5)^2 = 6.25$ |
| 2 | $2 - 3.5 = -1.5$ | $(-1.5)^2 = 2.25$ |
| 3 | $3 - 3.5 = -0.5$ | $(-0.5)^2 = 0.25$ |
| 4 | $4 - 3.5 = 0.5$  | $(0.5)^2 = 0.25$  |
| 5 | $5 - 3.5 = 1.5$  | $(1.5)^2 = 2.25$  |
| 6 | $6 - 3.5 = 2.5$  | $(2.5)^2 = 6.25$  |

**Step 2:** Apply expectation (multiply each by probability  $\frac{1}{6}$  and sum)

$$\text{Var}(X) = \frac{1}{6}(6.25) + \frac{1}{6}(2.25) + \frac{1}{6}(0.25) + \frac{1}{6}(0.25) + \frac{1}{6}(2.25) + \frac{1}{6}(6.25)$$

**Step 3:** Factor out  $\frac{1}{6}$

$$= \frac{1}{6}(6.25 + 2.25 + 0.25 + 0.25 + 2.25 + 6.25)$$

**Step 4:** Add inside parentheses

$$= \frac{1}{6}(17.5)$$

**Step 5:** Divide

$$= 2.9167$$

**Answer:**  $\text{Var}(X) = 2.9167$  (or exactly  $\frac{35}{12}$ )

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## Standard Deviation (Why We Bother)

$$\sigma = \sqrt{\text{Var}(X)} = \sqrt{2.9167} \approx 1.71$$

**Why square root?**

- Variance is in “squared units” (if  $X$  is dollars, variance is “square dollars”)
- Standard deviation is in original units (dollars again)
- **Interpretation:** “Typical distance from the mean”

**For dice:**  $\sigma \approx 1.71$  means outcomes typically differ from 3.5 by about 1.71.

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## Alternative Variance Formula (Often Easier)

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

**What this means:**

1. Find average of squared values:  $E[X^2]$
2. Subtract the square of the average:  $(E[X])^2$

**Why use this?** Sometimes easier to calculate.

**Dice verification:**

- $E[X^2] = \frac{1}{6}(1 + 4 + 9 + 16 + 25 + 36) = \frac{91}{6} \approx 15.167$
- $(E[X])^2 = (3.5)^2 = 12.25$
- $\text{Var}(X) = 15.167 - 12.25 = 2.917$  (Same answer!)

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## PART 4 — CONNECTIONS & BIG PICTURE

### The Chain (Memorize This)

Random Variable    "What are we measuring?"

Expectation        "What's the center/average?"

Variance            "How much does it spread around?"

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### ML-Specific Connections

| Concept                       | Random Variable            | Expectation Role           | Variance Role                                 |
|-------------------------------|----------------------------|----------------------------|-----------------------------------------------|
| <b>Bias-Variance Tradeoff</b> | Model prediction $\hat{Y}$ | $E[\hat{Y}]$ vs true value | How much $\hat{Y}$ varies with different data |
| <b>Overfitting</b>            | Prediction error           | Low (fits training well)   | High (unstable on new data)                   |
| <b>Underfitting</b>           | Prediction error           | High (fits poorly)         | Low (stable but wrong)                        |
| <b>SGD Noise</b>              | Gradient estimate          | True gradient              | Variance of mini-batch gradients              |

|         |                   |                    |                                     |
|---------|-------------------|--------------------|-------------------------------------|
| Dropout | Neuron activation | Average activation | Increased variance (regularization) |
|---------|-------------------|--------------------|-------------------------------------|

## HANDWRITTEN NOTES CHEAT SHEET

### Random Variable

- **Definition:** Rule that maps outcomes → numbers
- **Notation:**  $X : \Omega \rightarrow \mathbb{R}$
- **Types:** Discrete (countable) vs Continuous (infinite)
- **ML use:** Convert real-world events to math

### Expectation

- **Definition:** Long-run average
- **Discrete:**  $\sum x \cdot P(X=x)$
- **Continuous:**  $\int x \cdot f(x) \cdot dx$
- **Key fact:** May not be a possible outcome
- **ML use:** Average loss, average reward

### Variance

- **Definition:** Average squared deviation from mean
- **Formula 1:**  $\text{Var}(X) = E[(X - E[X])^2]$
- **Formula 2:**  $\text{Var}(X) = E[X^2] - (E[X])^2$
- **Standard Deviation:**  $\sigma = \sqrt{\text{Var}(X)}$
- **ML use:** Model stability, uncertainty, overfitting detection

## MEMORY HOOK (Target Analogy)

Imagine shooting arrows at a target:

| Concept                | Target Analogy                                           |
|------------------------|----------------------------------------------------------|
| <b>Random Variable</b> | Where each arrow lands (the outcome itself)              |
| <b>Expectation</b>     | The bullseye center (where you aim)                      |
| <b>Variance</b>        | How scattered your arrows are around the center          |
| <b>Low Variance</b>    | Arrows clustered tight (consistent but maybe off-center) |
| <b>High Variance</b>   | Arrows spread everywhere (unpredictable)                 |



|             |                                                    |
|-------------|----------------------------------------------------|
| <b>Bias</b> | Systematically missing the bullseye (aiming wrong) |
|-------------|----------------------------------------------------|

**Bias-Variance Tradeoff:** You can have tight arrows in the wrong place (low variance, high bias = underfitting) or arrows all over the place centered on target (high variance, low bias = overfitting). Good models balance both.

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## COMMON BEGINNER MISTAKES (Avoid These)

| Mistake                                  | Why It's Wrong                                      | Correct Understanding                                              |
|------------------------------------------|-----------------------------------------------------|--------------------------------------------------------------------|
| "Expectation is the most likely outcome" | No — it's the average, not the mode                 | Dice: most likely is 1-6 equally, expectation is 3.5               |
| "Variance can be negative"               | No — it's average of squares                        | Squares are always $\geq 0$ , so variance $\geq 0$                 |
| "Standard deviation = variance"          | No — SD is square root of variance                  | $\sigma = \sqrt{\text{Var}}$ , not $\sigma = \text{Var}$           |
| "Continuous $P(X = x)$ is the density"   | No — $P(X = \text{exact value}) = 0$ for continuous | $f(x)$ is density, not probability. Probability requires intervals |

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## Section 2: WHY RANDOM VARIABLES, EXPECTATION & VARIANCE MATTER IN ML

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### THE CORE IDEA (Stated Plainly)

Machine Learning is **not** about memorizing perfect rules. It's about **finding patterns in messy, uncertain data**.

**The problem:** Real data is never clean. Every measurement has noise, errors, and randomness.

**The solution:** We use Random Variables, Expectation, and Variance to **mathematically handle uncertainty** instead of pretending it doesn't exist.

**If we ignored these concepts:** ML would fail because it would treat every noisy data point as absolute truth.

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# 1. NOISE: WHAT IT ACTUALLY IS

## Simple Definition

**Noise = Random stuff that hides the true pattern.**

**Example you can feel:**

- You study 4 hours → expect 60 marks (true pattern)
- But you got 58 instead (noise: you were tired, room was hot, question was tricky)
- The “58” is NOT the true relationship. It’s the true relationship PLUS noise.

## Why Data Looks Messy

| True Pattern (Hidden) | Noise (Random)      | What You Actually See |
|-----------------------|---------------------|-----------------------|
| Study 2h → 40 marks   | +5 (good sleep)     | 45                    |
| Study 4h → 60 marks   | -2 (bad mood)       | 58                    |
| Study 6h → 80 marks   | -3 (hard questions) | 77                    |

**What you see:** Scattered points that don’t perfectly fit a line.

**What ML must find:** The underlying line (true pattern) despite the scatter.

## Why This Forces Us to Use Random Variables

If we treated “45” as the absolute truth for 2 hours study, we’d learn the wrong pattern.

Instead, we say:

- $X$  = study hours (we control this)
- $Y$  = exam score (this is **random** because of noise)
- $Y = \text{true pattern} + \text{noise}$

**$Y$  is a random variable because its value is uncertain even when  $X$  is fixed.**

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# 2. EXPECTATION: THE TRUE TARGET OF ML

## What Optimization Actually Means

**Your tutorial says:** “Find parameters that minimize error.”

**What this actually means step-by-step:**

**Step 1:** You have a model with knobs (parameters  $\theta$ )

- Example: Linear model  $f(x) = wx + b$
- $\theta = \{w, b\}$  are the knobs

**Step 2:** You show the model one image of a cat

- Model predicts "0.7" (70% confident it's a cat)
- Truth is "1" (it IS a cat)
- Loss =  $(1 - 0.7)^2 = 0.09$

**Step 3:** But tomorrow you show the SAME model a SLIGHTLY DIFFERENT cat image

- Model predicts "0.4" (40% confident)
- Loss =  $(1 - 0.4)^2 = 0.36$

**The loss CHANGED even though you didn't touch the knobs ( $\theta$ ).**

**Why?** Because the data ( $X$ ) is random.

## Why Loss Is a Random Variable (Complete Explanation)

| Component                 | Is It Fixed or Random?        | Why?                          |
|---------------------------|-------------------------------|-------------------------------|
| Model parameters $\theta$ | <b>Fixed</b> (at this moment) | You set these                 |
| Input data $X$            | <b>Random</b>                 | Different samples each time   |
| Prediction $f(X; \theta)$ | <b>Random</b>                 | Depends on random $X$         |
| True label $Y$            | <b>Random</b>                 | Depends on random $X$         |
| Loss $L(X, Y; \theta)$    | <b>RANDOM</b>                 | Depends on random $X$ and $Y$ |

**Critical insight:** If loss is random, we CANNOT minimize "the loss" because it changes every time. We must minimize something stable.

## The Solution: Expected Loss (Risk)

**Instead of minimizing one random loss, we minimize the AVERAGE loss over ALL possible data.**

$$\text{Risk}(\theta) = E[L(X, Y; \theta)]$$

**What this means in plain English:**

- Imagine you could test your model on EVERY POSSIBLE image in the universe
- Calculate the loss for each
- Take the average
- THAT is what we want to minimize

**Why we can't do this directly:** We don't have infinite data. We have a small sample.

**What we do instead:** Use the sample average (empirical risk) and hope LLN makes it close to true risk.

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## Complete Numerical Example: Expected Loss

**Scenario:** Email spam classifier

**Possible emails in the universe:**

| Email Type                 | Loss if Wrong | Probability in Real World |
|----------------------------|---------------|---------------------------|
| Easy spam (obvious)        | 1             | 50% (0.5)                 |
| Hard spam (tricky)         | 4             | 30% (0.3)                 |
| Very hard spam (deceptive) | 8             | 20% (0.2)                 |

**Why losses differ:** Easy spam is easy to catch → small penalty if wrong. Deceptive spam is hard → big penalty if wrong.

**Step-by-step calculation:**

**Formula:**  $E[L] = \sum(\text{loss} \times \text{probability})$

**Step 1:** Multiply each loss by its probability

- Easy:  $1 \times 0.5 = 0.5$
- Hard:  $4 \times 0.3 = 1.2$
- Very hard:  $8 \times 0.2 = 1.6$

**Step 2:** Add them up

$$E[L] = 0.5 + 1.2 + 1.6 = 3.3$$

**What this number means:**

- If you use this model forever on all possible emails, your average loss per email will be 3.3
- This is what we want to minimize by adjusting model parameters

**What we DON'T care about:**

- Getting loss = 1 on one specific easy email (that's luck)
- Getting loss = 8 on one specific hard email (that's bad luck)

**What we DO care about:**

- Making the AVERAGE (3.3) as small as possible
- 

## Why "Infinite Data" Keeps Coming Up

**The gap between theory and practice:**

| What Theory Wants       | What We Actually Have                 |
|-------------------------|---------------------------------------|
| True expectation $E[L]$ | Sample average $\frac{1}{n} \sum L_i$ |
| All possible data       | 10,000 training examples              |
| Perfect knowledge       | Limited knowledge                     |

**Law of Large Numbers bridges this gap:**

- As  $n \rightarrow \infty$ , sample average  $\rightarrow$  true expectation
- More data = better estimate = better model

**This is WHY data matters in ML.** Not because “more data is better” as a vague idea, but because mathematically, more data makes our estimate of expectation more accurate.

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### 3. VARIANCE: WHY DEEP NETWORKS EXPLODE OR VANISH

#### The Problem (Explained with Actual Numbers)

**Setup:** A neural network with 100 layers. Each layer multiplies input by weights.

**Simplified example (one weight per layer):**

**Case A: Weight = 1.1**

- Layer 1 input: 1.0
- Layer 2:  $1.0 \times 1.1 = 1.1$
- Layer 3:  $1.1 \times 1.1 = 1.21$
- Layer 10:  $1.1^{10} \approx 2.59$
- Layer 100:  $1.1^{100} \approx 13,780$

**Result:** EXPLOSION. Numbers become infinity. Computer crashes.

**Case B: Weight = 0.9**

- Layer 1 input: 1.0
- Layer 2:  $1.0 \times 0.9 = 0.9$
- Layer 3:  $0.9 \times 0.9 = 0.81$
- Layer 10:  $0.9^{10} \approx 0.35$
- Layer 100:  $0.9^{100} \approx 0.000026$

**Result:** VANISHING. Numbers become zero. No signal passes through.

**The question:** What should the weight be?

**Answer:** Exactly 1.0 would be perfect, but we have MANY weights with random values. We need to control their VARIANCE so the average effect is stable.

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## Why Variance Controls This (Complete Explanation)

**In reality:** Each layer has many weights. The output is a sum of many random products.

**Key mathematical fact:** When you add random variables, their variances add (if independent).

**What happens without control:**

- Layer 1: Input variance = 1
- Layer 2: Variance grows (explosion) or shrinks (vanishing)
- Layer 100: Variance is either infinity or zero

**What Xavier/He initialization does:**

- Carefully set  $\text{Var}(W)$  so that after multiplication, variance stays approximately 1
  - This prevents both explosion and vanishing
- 

## Xavier Initialization (Complete Derivation)

**Goal:** After passing through a layer, output variance = input variance

**Setup:**

- Input  $x$  has  $n$  elements
- Each  $x_i$  has variance  $\text{Var}(x)$
- Weights  $W_{ij}$  are random, independent of  $x$

**Layer output (one neuron):**

$$y = \sum_{i=1}^n W_i x_i$$

**Calculate variance of  $y$ :**

**Step 1:** Assume  $W_i$  and  $x_i$  are independent

$$\text{Var}(y) = \text{Var}\left(\sum_{i=1}^n W_i x_i\right)$$

**Step 2:** Variance of sum = sum of variances (if independent)

$$= \sum_{i=1}^n \text{Var}(W_i x_i)$$

**Step 3:** Variance of product (independent variables)

$$\text{Var}(W_i x_i) = E[W_i^2]E[x_i^2] - (E[W_i]E[x_i])^2$$

**Step 4:** Assume zero mean weights ( $E[W_i] = 0$ ) and inputs ( $E[x_i] = 0$ )

$$\text{Var}(W_i x_i) = E[W_i^2]E[x_i^2] = \text{Var}(W_i)\text{Var}(x_i)$$

**Step 5:** Assume all weights have same variance  $\text{Var}(W)$ , all inputs have same variance  $\text{Var}(x)$

$$\text{Var}(y) = \sum_{i=1}^n \text{Var}(W)\text{Var}(x) = n \cdot \text{Var}(W) \cdot \text{Var}(x)$$

**Step 6:** We want  $\text{Var}(y) = \text{Var}(x)$  (preserve variance)

$$\text{Var}(x) = n \cdot \text{Var}(W) \cdot \text{Var}(x)$$

**Step 7:** Divide both sides by  $\text{Var}(x)$  (assuming it's not zero)

$$1 = n \cdot \text{Var}(W)$$

**Step 8:** Solve for  $\text{Var}(W)$

$$\boxed{\text{Var}(W) = \frac{1}{n}}$$

**This is Xavier initialization.** Set weight variance to  $\frac{1}{n}$  where  $n$  = number of input connections.

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## He Initialization (For ReLU)

**Problem with Xavier:** ReLU sets negative values to zero. This changes the variance.

**Fix:** Need twice the variance to compensate for “killing” half the values.

$$\boxed{\text{Var}(W) = \frac{2}{n}}$$

**Why 2 instead of 1?** Because ReLU zeros out about half the values, so we need bigger weights to maintain signal strength.

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## 4. BIAS-VARIANCE TRADEOFF (Fully Decoded)

### The Complete Picture

Your model makes predictions  $\hat{f}(x)$ . The true relationship is  $f(x)$ .

**Error decomposition:**

$$\text{Total Error} = \text{Bias}^2 + \text{Variance} + \text{Noise}$$

**What each term means:**

| Term                    | Plain English                                  | When It's High    | Visual                         |
|-------------------------|------------------------------------------------|-------------------|--------------------------------|
| <b>Bias<sup>2</sup></b> | Systematic mistake (always wrong the same way) | Model too simple  | Arrows consistently off-center |
| <b>Variance</b>         | Inconsistency (sensitive to training data)     | Model too complex | Arrows scattered everywhere    |
| <b>Noise</b>            | Unavoidable randomness in reality              | Always present    | Target itself moves            |

## Detailed Example: Predicting House Prices

**True relationship:** Price = \$100,000 + \$200 × (square feet) + random noise

### Model A: Too Simple (High Bias)

- Uses only: Price = \$150,000 (average, ignores size)
- **What happens:** Always wrong by similar amount
- **Bias:** High (systematically misses the trend)
- **Variance:** Low (same prediction every time)

### Model B: Too Complex (High Variance)

- Uses: Price = complicated function that memorizes every training house
- **What happens:** Perfect on training, terrible on new houses
- **Bias:** Low (fits training well)
- **Variance:** High (changes wildly with different training data)

### Model C: Just Right

- Uses: Price = \$100,000 + \$200 × (square feet)
- **Bias:** Low (correct trend)
- **Variance:** Low (stable predictions)

## Mathematical Derivation (Complete)

**Goal:** Show that  $E[(Y - \hat{f}(X))^2] = \text{Bias}^2 + \text{Variance} + \text{Noise}$

### Setup:

- True value:  $Y = f(X) + \mathbb{I}$  where  $\mathbb{I}$  is noise with  $E[\mathbb{I}] = 0$ ,  $\text{Var}(\mathbb{I}) = \sigma^2$
- Prediction:  $\hat{f}(X)$  (depends on training data, so it's random)

**Step 1:** Write error at point  $X = x$

$$\text{Error} = E[(Y - \hat{f}(x))^2 | X = x]$$

**Step 2:** Add and subtract  $E[\hat{f}(x)]$  (average prediction)



$$= E[(Y - E[\hat{f}(x)] + E[\hat{f}(x)] - \hat{f}(x))^2]$$

**Step 3:** Let  $a = Y - E[\hat{f}(x)]$  and  $b = E[\hat{f}(x)] - \hat{f}(x)$

$$= E[(a + b)^2] = E[a^2 + 2ab + b^2] = E[a^2] + 2E[ab] + E[b^2]$$

**Step 4:** Show  $E[ab] = 0$  (cross term vanishes)

- $E[b] = E[E[\hat{f}(x)] - \hat{f}(x)] = E[\hat{f}(x)] - E[\hat{f}(x)] = 0$
- So  $E[ab] = a \cdot E[b] = a \cdot 0 = 0$

**Step 5:** Calculate  $E[a^2]$

$$E[a^2] = E[(Y - E[\hat{f}(x)])^2] = E[(f(x) + \epsilon - E[\hat{f}(x)])^2]$$

$$= E[((f(x) - E[\hat{f}(x)]) + \epsilon)^2]$$

$$= (f(x) - E[\hat{f}(x)])^2 + 2(f(x) - E[\hat{f}(x)])E[\epsilon] + E[\epsilon^2]$$

$$= \underbrace{(f(x) - E[\hat{f}(x)])^2}_{\text{Bias}^2} + \underbrace{\sigma^2}_{\text{Noise}}$$

**Step 6:** Calculate  $E[b^2]$

$$E[b^2] = E[(E[\hat{f}(x)] - \hat{f}(x))^2] = \underbrace{\text{Var}(\hat{f}(x)))}_{\text{Variance}}$$

**Final Result:**

$$E[(Y - \hat{f}(x))^2] = \underbrace{(f(x) - E[\hat{f}(x)])^2}_{\text{Bias}^2} + \underbrace{\text{Var}(\hat{f}(x)))}_{\text{Variance}} + \underbrace{\sigma^2}_{\text{Noise}}$$

## How to Diagnose Your Model

| Symptom                       | Problem                     | Fix                                      |
|-------------------------------|-----------------------------|------------------------------------------|
| Bad on training AND test      | High bias (underfitting)    | More complex model, more features        |
| Good on training, bad on test | High variance (overfitting) | More data, regularization, simpler model |
| Good on both                  | Low bias, low variance      | Perfect!                                 |
| Bad on both but test worse    | Both high                   | Fix bias first, then variance            |

## 5. LAW OF LARGE NUMBERS (Complete Understanding)

### What It Actually Says

#### Weak Law of Large Numbers:

*As sample size increases, sample average gets closer to true expectation.*

#### Mathematically:

✓  $\text{as } n \rightarrow \infty$

**What  $\xrightarrow{P}$  means:** “Converges in probability” — the probability that  $\bar{X}_n$  differs from  $E[X]$  by more than any small amount goes to zero.

---

### Complete Numerical Example

**True expectation:** Coin flip, ✓ = 0.5

**Simulation (what you’d actually observe):**

| Number of Flips $n$ | Sample Average $\bar{X}_n$ | Distance from 0.5 |
|---------------------|----------------------------|-------------------|
| 10                  | 0.7                        | 0.2               |
| 100                 | 0.63                       | 0.13              |
| 1,000               | 0.508                      | 0.008             |
| 10,000              | 0.4972                     | 0.0028            |
| 100,000             | 0.5003                     | 0.0003            |
| 1,000,000           | 0.49998                    | 0.00002           |

**Pattern:** As  $n$  grows, the sample average gets closer to 0.5. Not perfectly, but the error shrinks.

---

### Why ML Cannot Exist Without LLN

#### The fundamental problem of ML:

- We want to minimize  $E[L(X, \theta)]$  (true risk)
- But we only have  $\frac{1}{n} \sum_{i=1}^n L(x_i, \theta)$  (empirical risk)

#### LLN guarantees:

$$\frac{1}{n} \sum_{i=1}^n L(x_i, \theta) \approx E[L(X, \theta)] \text{ when } n \text{ is large}$$

### This is WHY:

- More data → better model (usually)
- Training on sample → works on population
- Statistics/ML is possible at all

**Without LLN:** Training on 10,000 images would tell us nothing about the 10,001st image.

---

## HANDWRITTEN NOTES CHEAT SHEET

### Why This Topic Matters

- ML = learning from **uncertain** data
- Real data has **noise** (random variation)
- We need math tools to handle uncertainty

### Noise

- Definition: Random stuff hiding true pattern
- Example: Study hours → exam score (true) + mood/luck (noise)
- Result: Data looks scattered, not perfectly linear

### Expected Loss (Risk)

- Loss is **random** (depends on random data)
- Can't minimize random thing → minimize average
- Formula:  $R(\theta) = E[L(X, \theta)]$
- We use sample average as estimate (LLN makes this valid)

### Variance in Deep Networks

- Deep networks multiply signals through many layers
- Wrong variance → explosion (too big) or vanishing (too small)
- **Xavier:**  $\text{Var}(W) = \frac{1}{n}$
- **He:**  $\text{Var}(W) = \frac{2}{n}$  (for ReLU)

### Bias-Variance Tradeoff

- **Bias<sup>2</sup>:** Systematic error (model too simple)
- **Variance:** Prediction instability (model too complex)
- **Noise:** Unavoidable randomness
- Formula:  $\text{Error} = \text{Bias}^2 + \text{Variance} + \text{Noise}$

## Law of Large Numbers

- Sample average → true expectation as  $n \rightarrow \infty$
- Why more data helps
- Why training on sample generalizes to population

## MEMORY HOOK (Archery Analogy)

| Concept         | Archery Analogy                              |
|-----------------|----------------------------------------------|
| Random Variable | Where each arrow lands (uncertain)           |
| Expectation     | The center of the target (true aim)          |
| Variance        | How scattered your arrows are                |
| Bias            | Systematically aiming left (wrong direction) |
| LLN             | More arrows reveal your true aim             |
| Optimization    | Adjusting your aim toward center             |
| Noise           | Wind randomly pushing arrows                 |

**High Bias, Low Variance:** All arrows hit the same wrong spot.

**Low Bias, High Variance:** Arrows scattered around center.

**Goal:** Arrows clustered tightly at center.

## CONNECTIONS TO NEXT TOPICS

| This Topic                 | Leads To                                            |
|----------------------------|-----------------------------------------------------|
| Expected loss minimization | Gradient descent, SGD                               |
| Variance control           | Batch normalization, initialization schemes         |
| Bias-variance              | Regularization, ensemble methods, cross-validation  |
| LLN                        | Central Limit Theorem, confidence intervals         |
| Random variables           | Probability distributions (Normal, Bernoulli, etc.) |

## Section 3 : REAL-WORLD CASES IN AI

## THE UNIFYING PATTERN (Memorize This First)

Every AI system in the world follows this same pipeline:

1. REALITY IS UNCERTAIN
2. MODEL IT AS A RANDOM VARIABLE
3. USE EXPECTATION FOR "WHAT TYPICALLY HAPPENS"
4. USE VARIANCE FOR "HOW RISKY/UNSTABLE IS THIS?"
5. MAKE DECISION BASED ON BOTH

**This pattern repeats in LLMs, trading, self-driving cars, medicine, weather, and robotics.**  
Once you see it, you'll recognize it everywhere.

---

## CASE 1 — LARGE LANGUAGE MODELS (Complete Mechanism)

### What Actually Happens Inside an LLM (Step-by-Step)

**Step 1:** You type: "The sky is"

**Step 2:** The model converts your text to numbers (embeddings)

- "The" vector [0.2, -0.5, 1.1, ...]
- "sky" vector [0.8, 0.1, -0.3, ...]
- "is" vector [-0.1, 0.9, 0.4, ...]

**Step 3:** Neural network processes these vectors through many layers

- Each layer: matrix multiplications + non-linearities
- Final layer outputs a vector of size = vocabulary size (e.g., 50,000)

**Step 4:** Convert output vector to probabilities

- Apply "softmax" function: turns any numbers into probabilities that sum to 1
- Result: probability distribution over all possible next words

**Step 5:** The model does NOT "choose" the word yet

- It only produced probabilities

- The actual word selection happens in a separate step (sampling/decoding)
- 

### The Random Variable (Precisely Defined)

**Random Variable  $X$  = The next token generated**

**Possible outcomes:** All words in vocabulary  $\{w_1, w_2, \dots, w_V\}$  where  $V \approx 50,000$

**Probability distribution:**  $P(X = w_i) = p_i$  (output by softmax)

**Example with your input “The sky is”:**

| Token $w_i$ | Probability $P(X = w_i)$ | Why This Probability?               |
|-------------|--------------------------|-------------------------------------|
| blue        | 0.70                     | Very common completion              |
| dark        | 0.15                     | Common at night                     |
| clear       | 0.10                     | Less common but valid               |
| green       | 0.03                     | Rare (pollution?)                   |
| pizza       | 0.02                     | Nonsense but grammatically possible |

**Important:** The model itself is deterministic (same input → same probabilities). The randomness comes from HOW we sample from these probabilities.

---

### Two Ways to Generate (Critical Distinction)

| Method                 | What Happens                                   | Random Variable Role                 |
|------------------------|------------------------------------------------|--------------------------------------|
| <b>Greedy decoding</b> | Always pick highest probability word           | $X$ is deterministic (no randomness) |
| <b>Sampling</b>        | Roll a weighted die according to probabilities | $X$ is truly random                  |

**Example:**

- Greedy: Always pick “blue” (0.70)
- Sampling: 70% chance “blue”, 15% chance “dark”, etc.

**Why sampling?** Greedy produces boring, repetitive text. Sampling introduces creativity and variety.

---

### Expectation in LLMs (Training Objective)

**What the model actually minimizes:**

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \log P(X = x_i^{\text{true}} | \text{context}_i)$$

**What this means in plain English:**

**Step 1:** Take a training sentence: “The sky is blue”

**Step 2:** Show model “The sky is”    model predicts probabilities

**Step 3:** Check probability assigned to TRUE next word “blue”

**Step 4:** If  $P(\text{blue}) = 0.70$ , then  $\log(0.70) \approx -0.357$

**Step 5:** Loss =  $-\log(0.70) = 0.357$  (lower is better)

**Step 6:** Adjust model weights to make this probability HIGHER next time

**This is cross-entropy loss = negative log-likelihood**

**Why expectation?** Over millions of sentences, we’re minimizing the AVERAGE loss. That’s expectation.

---

## Complete Numerical Example: Cross-Entropy Calculation

**Training data:** Three sentences

1. “The sky is blue”    true next word: “blue”
2. “The sky is dark”    true next word: “dark”
3. “The sky is clear”    true next word: “clear”

**Model predictions for “The sky is”:**

| True Word | Model Probability | $-\log(P)$            |
|-----------|-------------------|-----------------------|
| blue      | 0.70              | $-\log(0.70) = 0.357$ |
| dark      | 0.20              | $-\log(0.20) = 1.609$ |
| clear     | 0.10              | $-\log(0.10) = 2.303$ |

**Average loss (expectation over training data):**

$$\mathcal{L} = \frac{0.357 + 1.609 + 2.303}{3} = \frac{4.269}{3} = 1.423$$

**What this tells us:** The model is “uncertain” on average. Perfect model would have probabilities [1.0, 0, 0] for each case, giving loss = 0.

---

## Variance and Uncertainty in LLMs

**Your tutorial mentions “high variance means unsure.” Let me clarify precisely:**

| Concept                   | What It Measures                         | Formula                             |
|---------------------------|------------------------------------------|-------------------------------------|
| Entropy                   | Overall uncertainty in distribution      | $H = - \sum p_i \log p_i$           |
| Variance of probabilities | How spread out probabilities are         | $\text{Var}(p) = E[p^2] - (E[p])^2$ |
| Predictive variance       | Model's confidence in its own prediction | More complex (Bayesian)             |

**For your notes:** When people say “high variance” in LLMs, they usually mean **high entropy** (flat distribution) or **the model gives different answers to the same question** (sampling variance).

**Complete entropy calculation for “The sky is”:**

**Low uncertainty case:**

| Word  | Probability |
|-------|-------------|
| blue  | 0.99        |
| dark  | 0.005       |
| clear | 0.005       |

$$\begin{aligned}
 H &= -(0.99 \log 0.99 + 0.005 \log 0.005 + 0.005 \log 0.005) \\
 &= -(0.99 \times (-0.010) + 0.005 \times (-5.298) + 0.005 \times (-5.298)) \\
 &= -(-0.0099 - 0.0265 - 0.0265) \\
 &= 0.0629 \text{ bits}
 \end{aligned}$$

**High uncertainty case:**

| Word  | Probability |
|-------|-------------|
| go    | 0.25        |
| stay  | 0.20        |
| work  | 0.18        |
| sleep | 0.17        |
| leave | 0.20        |

$$\begin{aligned}
 H &= -(0.25 \log 0.25 + 0.20 \log 0.20 + 0.18 \log 0.18 + 0.17 \log 0.17 + 0.20 \log 0.20) \\
 &= -(0.25 \times (-2) + 0.20 \times (-2.322) + 0.18 \times (-2.474) + 0.17 \times (-2.555) + 0.20 \times (-2.
 \end{aligned}$$



$$= -(-0.5 - 0.464 - 0.445 - 0.434 - 0.464)$$

$$= 2.307 \text{ bits}$$

**Interpretation:** Higher entropy = more uncertainty. The model is “more confused” in the second case.

## Hallucinations Explained (No Ambiguity)

**Definition:** Hallucination = Model generates text that is confident-sounding but factually wrong.

**Why this happens (complete explanation):**

| Cause                        | Mechanism                                                             | Example                                                                    |
|------------------------------|-----------------------------------------------------------------------|----------------------------------------------------------------------------|
| <b>Training data gap</b>     | Model never saw correct fact                                          | Claims “Einstein won Nobel for relativity” (actually photoelectric effect) |
| <b>Optimization mismatch</b> | Model trained to predict likely text, not true text                   | “The capital of France is London” (grammatically plausible)                |
| <b>Exposure bias</b>         | Model generates based on its own previous outputs, compounding errors | Small mistake early   bigger mistake later                                 |
| <b>Overconfidence</b>        | Softmax can produce high probabilities for wrong answers              | 99% confident in wrong answer                                              |

**Important:** Hallucination is NOT the same as high variance/uncertainty. A model can hallucinate with 99% confidence (low uncertainty, wrong answer).

**Detection strategy:** High uncertainty (flat distribution) often indicates “I don’t know.” But low uncertainty doesn’t guarantee correctness.

## CASE 2 — ALGORITHMIC TRADING (Complete Math)

### Why Stock Prices Are Random Variables (Fundamental Reason)

**The efficient market hypothesis (simplified):**

- All known information is already reflected in current price
- Future price changes depend on NEW information
- New information is unpredictable by definition (if predictable, it would already be known)
- Therefore: Future price changes are random

**This does NOT mean “completely random.”** It means “unpredictable using current information.”

---

## Complete Expected Return Calculation

**Scenario:** You buy a stock at \$100. Tomorrow's price is uncertain.

**Possible outcomes:**

| Price $x$ | Profit $x - 100$ | Probability $P(X = x)$ | Source of Probability |
|-----------|------------------|------------------------|-----------------------|
| \$90      | -\$10            | 0.20                   | Bad earnings report   |
| \$100     | \$0              | 0.30                   | No change             |
| \$110     | \$10             | 0.35                   | Good news             |
| \$120     | \$20             | 0.15                   | Excellent news        |

**Step 1:** Calculate expected price

✓  $= \sum x \cdot P(X=x)$

$$= 90 \times 0.20 + 100 \times 0.30 + 110 \times 0.35 + 120 \times 0.15$$

$$= 18 + 30 + 38.5 + 18$$

$$= 104.5$$

**Step 2:** Calculate expected profit

✓  $- 100 = 104.5 - 100 = 4.5$

**Interpretation:** On average, you expect to make \$4.50 per share. But on any single day, you might lose \$10 or gain \$20.

---

## Complete Variance and Risk Calculation

**Step 1:** Calculate deviations from expected price ( ✓ = 104.5)

| Price $x$ | $x - E[X]$ | $(x - E[X])^2$ | Probability |
|-----------|------------|----------------|-------------|
| 90        | -14.5      | 210.25         | 0.20        |
| 100       | -4.5       | 20.25          | 0.30        |

|     |      |        |      |
|-----|------|--------|------|
| 110 | 5.5  | 30.25  | 0.35 |
| 120 | 15.5 | 240.25 | 0.15 |

**Step 2:** Calculate variance

$$\text{Var}(X) = \sum (x - E[X])^2 \cdot P(X = x)$$

$$= 210.25 \times 0.20 + 20.25 \times 0.30 + 30.25 \times 0.35 + 240.25 \times 0.15$$

$$= 42.05 + 6.075 + 10.5875 + 36.0375$$

$$= 94.75$$

**Step 3:** Calculate standard deviation

$$\sigma = \sqrt{94.75} \approx 9.73$$

**Interpretation:** Typical price fluctuation is about \$9.73 around the expected \$104.50.

---

## Comparing Two Investments (Why Variance = Risk)

**Investment A: Government Bond**

| Return | Probability |
|--------|-------------|
| 5%     | 1.0         |

$$E[R_A] = 5\% \times 1.0 = 5\%$$

$$\text{Var}(R_A) = (5\% - 5\%)^2 \times 1.0 = 0$$

**Investment B: Tech Stock**

| Return | Probability |
|--------|-------------|
| -20%   | 0.3         |
| 5%     | 0.4         |
| 30%    | 0.3         |

$$E[R_B] = (-20\% \times 0.3) + (5\% \times 0.4) + (30\% \times 0.3)$$

$$= -6\% + 2\% + 9\% = 5\%$$

$$\text{Var}(R_B) = (-20\% - 5\%)^2 \times 0.3 + (5\% - 5\%)^2 \times 0.4 + (30\% - 5\%)^2 \times 0.3$$

$$= (-25\%)^2 \times 0.3 + 0 + (25\%)^2 \times 0.3$$

$$= 625\% \times 0.3 + 625\% \times 0.3$$

$$= 187.5\% + 187.5\% = 375\%^2$$

$$\sigma_B = \sqrt{375\%} \approx 19.36\%$$

#### Comparison:

| Metric             | Bond (A) | Tech Stock (B)    |
|--------------------|----------|-------------------|
| Expected return    | 5%       | 5%                |
| Variance           | 0        | 375% <sup>2</sup> |
| Standard deviation | 0%       | 19.36%            |

**Same expected return, vastly different risk.** This is why variance matters.

---

## Mean-Variance Optimization (Complete Formula)

**Objective:** Maximize return while minimizing risk

$$\max_{\text{portfolio}} (E[R_{\text{portfolio}}] - \lambda \cdot \text{Var}(R_{\text{portfolio}}))$$

**What each term means:**

| Symbol                             | Meaning                      | Typical Value                   |
|------------------------------------|------------------------------|---------------------------------|
| $E[R_{\text{portfolio}}]$          | Expected return of portfolio | 8-12% annually                  |
| $\text{Var}(R_{\text{portfolio}})$ | Risk (variance) of portfolio | Varies widely                   |
| $\lambda$                          | Risk aversion coefficient    | 1-5 (higher = more risk-averse) |

#### Example calculation:

- Portfolio A:  $E[R] = 10\%$ ,  $\text{Var} = 100\%^2$ ,  $\lambda = 2$
- Score:  $10\% - 2 \times 100\%^2 = 10 - 200 = -190$
- Portfolio B:  $E[R] = 8\%$ ,  $\text{Var} = 25\%^2$ ,  $\lambda = 2$
- Score:  $8\% - 2 \times 25\%^2 = 8 - 50 = -42$

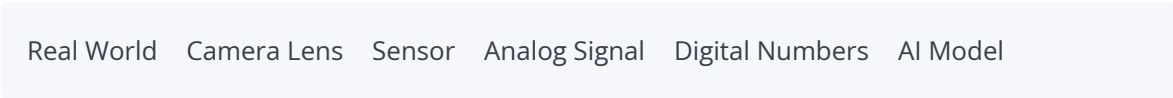
**Portfolio B wins** because its risk-adjusted return is better, even though raw return is lower.

---

## CASE 3 — SELF-DRIVING CARS (Safety-Critical Probability)

### Why Vision Is Fundamentally Uncertain

The sensor pipeline:



At EVERY step, uncertainty enters:

| Stage          | Source of Uncertainty   | Effect                  |
|----------------|-------------------------|-------------------------|
| Real world     | Object partially hidden | Missing information     |
| Camera lens    | Dirt, scratches         | Blurred image           |
| Sensor         | Electronic noise        | Random pixel variations |
| Analog Digital | Quantization            | Loss of fine detail     |
| AI model       | Limited training data   | Unfamiliar scenarios    |

**Result:** The AI never sees “reality.” It sees a noisy, incomplete estimate.

---

### Object Detection as Random Variable (Complete)

Random Variable  $Y$  = Object category detected

Possible outcomes: {human, car, cyclist, tree, road, unknown}

Example prediction (clear day):

| Object  | Probability | Confidence Level |
|---------|-------------|------------------|
| human   | 0.98        | Very high        |
| cyclist | 0.01        | Very low         |
| tree    | 0.005       | Negligible       |
| unknown | 0.005       | Negligible       |

**Decision:** Brake immediately (high confidence human)

Example prediction (foggy night):

| Object  | Probability | Confidence Level |
|---------|-------------|------------------|
| human   | 0.40        | Uncertain        |
| tree    | 0.35        | Uncertain        |
| unknown | 0.25        | High uncertainty |

**Decision:** Slow down, alert driver, use other sensors (radar/LiDAR)

---

## Complete Expected Loss in Autonomous Driving

**Scenario:** The car must decide whether to brake

**Possible actions:**

- Action 1: Brake hard
- Action 2: Brake gently
- Action 3: Continue

**Loss matrix (safety-critical):**

| True State \ Action | Brake Hard      | Brake Gentle | Continue       |
|---------------------|-----------------|--------------|----------------|
| Human (prob 0.4)    | 1 (safe)        | 3 (risky)    | 100 (disaster) |
| Tree (prob 0.35)    | 2 (unnecessary) | 1 (smooth)   | 0 (fine)       |
| Unknown (prob 0.25) | 2 (safe)        | 2 (cautious) | 50 (dangerous) |

**Expected loss for each action:**

**Brake Hard:**

$$E[L] = 1 \times 0.4 + 2 \times 0.35 + 2 \times 0.25$$

$$= 0.4 + 0.7 + 0.5 = 1.6$$

**Brake Gentle:**

$$E[L] = 3 \times 0.4 + 1 \times 0.35 + 2 \times 0.25$$

$$= 1.2 + 0.35 + 0.5 = 2.05$$

**Continue:**

$$E[L] = 100 \times 0.4 + 0 \times 0.35 + 50 \times 0.25$$

$$= 40 + 0 + 12.5 = 52.5$$

**Optimal decision:** Brake hard (lowest expected loss: 1.6)

**Why this matters:** Even though “human” is only 40% likely, the cost of being wrong (100) makes braking the safest choice.

---

## Sensor Fusion (Bayesian Combination)

**Multiple sensors observe the same object:**

| Sensor | Detects       | Confidence |
|--------|---------------|------------|
| Camera | human         | 0.40       |
| Radar  | moving object | 0.80       |
| LiDAR  | human-shaped  | 0.70       |

**Bayesian update (simplified):**

- Prior (camera only):  $P(\text{human}) = 0.40$
- Likelihood (radar supports): multiply by 0.80
- Likelihood (LiDAR supports): multiply by 0.70
- Posterior: proportional to  $0.40 \times 0.80 \times 0.70 = 0.224$

**Normalized probability:**  $\frac{0.224}{0.224 + \text{other hypotheses}} \approx 0.85$

**Result:** Combined confidence 85%   much safer to brake

---

## ADDITIONAL CASES (Expanded)

### CASE 4 — MEDICAL AI (Complete Example)

**Problem:** Detect cancer from X-ray

**Random Variable  $\mathcal{D}$  = Diagnosis**

**Model output:**

| Diagnosis    | Probability | Action Required |
|--------------|-------------|-----------------|
| Healthy      | 0.15        | None            |
| Benign tumor | 0.20        | Monitor         |

|                  |      |                      |
|------------------|------|----------------------|
| Malignant cancer | 0.65 | Immediate treatment  |
| Unclear          | 0.00 | (not an option here) |

**Expected loss calculation:**

| True State \ Action | Treat                      | Monitor                | Ignore                |
|---------------------|----------------------------|------------------------|-----------------------|
| Healthy (0.15)      | 10 (unnecessary treatment) | 2 (anxiety)            | 0 (correct)           |
| Benign (0.20)       | 8 (overtreatment)          | 1 (correct)            | 5 (missed monitoring) |
| Malignant (0.65)    | 1 (correct)                | 50 (delayed treatment) | 100 (fatal miss)      |

**Expected loss:**

**Treat:**

$$E[L] = 10 \times 0.15 + 8 \times 0.20 + 1 \times 0.65 = 1.5 + 1.6 + 0.65 = 3.75$$

**Monitor:**

$$E[L] = 2 \times 0.15 + 1 \times 0.20 + 50 \times 0.65 = 0.3 + 0.2 + 32.5 = 33.0$$

**Ignore:**

$$E[L] = 0 \times 0.15 + 5 \times 0.20 + 100 \times 0.65 = 0 + 1 + 65 = 66.0$$

**Decision:** Treat (lowest expected loss, despite 15% chance of healthy patient)

**Medical principle:** "Better to overtreat than undertreat" is encoded in the loss matrix.

---

## CASE 5 — WEATHER FORECASTING

**Random Variable R = Rainfall tomorrow (in mm)**

**Forecast model output:**

| Rainfall (mm) | Probability |
|---------------|-------------|
| 0             | 0.30        |
| 5             | 0.25        |
| 15            | 0.25        |



|    |      |
|----|------|
| 30 | 0.15 |
| 50 | 0.05 |

**Expected rainfall:**

$$E[R] = 0 \times 0.30 + 5 \times 0.25 + 15 \times 0.25 + 30 \times 0.15 + 50 \times 0.05$$

$$= 0 + 1.25 + 3.75 + 4.5 + 2.5 = 12.0 \text{ mm}$$

**Variance:**

$$\text{Var}(R) = (0 - 12)^2 \times 0.30 + (5 - 12)^2 \times 0.25 + (15 - 12)^2 \times 0.25 + (30 - 12)^2 \times 0.15 +$$

$$= 144 \times 0.30 + 49 \times 0.25 + 9 \times 0.25 + 324 \times 0.15 + 1444 \times 0.05$$

$$= 43.2 + 12.25 + 2.25 + 48.6 + 72.2 = 178.5 \text{ mm}^2$$

**Standard deviation:**  $\sigma = \sqrt{178.5} \approx 13.36 \text{ mm}$

**Interpretation:** Expected rain is 12mm, but typical deviation is 13.36mm. Forecast is quite uncertain!

---

## CASE 6 — REINFORCEMENT LEARNING (Robot Learning)

**Scenario:** Robot learns to walk

**Random Variable R = Reward per step**

**Possible rewards:**

| Event           | Reward | Probability |
|-----------------|--------|-------------|
| Falls over      | -100   | 0.1         |
| Stumbles        | -10    | 0.3         |
| Walks awkwardly | 1      | 0.4         |
| Walks smoothly  | 10     | 0.2         |

**Expected reward:**

$$E[R] = (-100) \times 0.1 + (-10) \times 0.3 + 1 \times 0.4 + 10 \times 0.2$$

$$= -10 - 3 + 0.4 + 2 = -10.6$$

**Negative!** Robot is losing on average. Must improve policy.

**Variance:**

$$\text{Var}(R) = (-100 - (-10.6))^2 \times 0.1 + (-10 - (-10.6))^2 \times 0.3 + (1 - (-10.6))^2 \times 0.4 +$$

$$= (-89.4)^2 \times 0.1 + (0.6)^2 \times 0.3 + (11.6)^2 \times 0.4 + (20.6)^2 \times 0.2$$

$$= 7992.36 \times 0.1 + 0.36 \times 0.3 + 134.56 \times 0.4 + 424.36 \times 0.2$$

$$= 799.236 + 0.108 + 53.824 + 84.872 = 938.04$$

**High variance** makes learning difficult (unstable gradients). Techniques like reward shaping or baseline subtraction reduce variance.

---

## HANDWRITTEN NOTES CHEAT SHEET

### LLMs

- **Random Variable:** Next token  $X$  with vocabulary probabilities
- **Expectation:** Minimize average cross-entropy loss over all text
- **Variance/Uncertainty:** Entropy of output distribution (high = unsure)
- **Hallucination:** Confident but wrong (low entropy, incorrect fact)

### Finance

- **Random Variable:** Future price/return  $X$
- **Expectation:**  $\mathbb{E}[X] = \sum x P(x)$  = long-run average profit
- **Variance:**  $\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]$  = risk/volatility
- **Mean-Variance:** Maximize  $\mathbb{E}[R] - \lambda \text{Var}(R)$

### Self-Driving Cars

- **Random Variable:** Object category  $Y$  (human, car, tree)
- **Expectation:** Minimize expected safety loss
- **Variance:** Sensor uncertainty (fog, blur increase variance)
- **Decision:** Conservative action when uncertainty high

## Medical AI

- **Random Variable:** Diagnosis  $D$
- **Expectation:** Minimize expected harm to patient
- **Variance:** Diagnostic uncertainty    request more tests
- **Loss matrix:** Asymmetric (missing cancer >> false alarm)

## Weather

- **Random Variable:** Rainfall  $R$
- **Expectation:** Average predicted amount
- **Variance:** Forecast confidence (high variance = unreliable)

## Robotics (RL)

- **Random Variable:** Reward  $R$
  - **Expectation:**  $E[R]$  = average performance
  - **Variance:** Learning stability (high = hard to learn)
- 

# UNIFIED MEMORY HOOK

Every AI system asks three questions:

| Question                | Math Tool       | Example                                         |
|-------------------------|-----------------|-------------------------------------------------|
| What is uncertain?      | Random Variable | Next word, stock price, object type             |
| What typically happens? | Expectation     | Average loss, average profit, average rain      |
| How risky is this?      | Variance        | Model confidence, investment risk, sensor noise |

The decision pipeline:

Uncertainty    Model with RV    Calculate Expectation    Assess Variance    Act Safely

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## Section 4: THEORY (MID-LONG DEPTH)

## PART 1 — INTUITIVE THEORY (Building Mental Models)

### The Hat Analogy (Fully Expanded)

**The setup:** Imagine a magical hat containing numbers, but NOT equally likely.

**Example hat:**

| Number in Hat | Probability | Why This Probability?              |
|---------------|-------------|------------------------------------|
| 1             | 0.50        | Most common (like “easy” outcomes) |
| 5             | 0.30        | Moderately common                  |
| 10            | 0.20        | Rare but possible                  |

**You reach in blindly and pull one number.** Before pulling, you don’t know which you’ll get. **This uncertainty is the entire point.**

**Random Variable  $X$  = The number you pull**

**Possible outcomes:** {1, 5, 10}

**Why this matters:** We can’t just list “1, 5, 10” and call it done. We need to ANSWER QUESTIONS about this hat.

---

### The Two Fundamental Questions

| Question              | What It Means                               | Math Tool   |
|-----------------------|---------------------------------------------|-------------|
| Where is the center?  | “If I pull many times, what’s the average?” | Expectation |
| How spread out is it? | “Are results clustered or scattered?”       | Variance    |

**These are the ONLY two questions we need to summarize uncertainty.** Everything else (skewness, kurtosis) is extra detail.

---

### Expectation = Center of Gravity (Complete Physics Analogy)

**Physical setup:** Imagine a seesaw with weights.

| Position on Seesaw | Weight | Influence on Balance |
|--------------------|--------|----------------------|
|--------------------|--------|----------------------|

|           |        |                     |
|-----------|--------|---------------------|
| 1 meter   | 0.8 kg | Strong pull (heavy) |
| 10 meters | 0.2 kg | Weak pull (light)   |

**Where does the seesaw balance?**

**Physics formula:**

$$x_{\text{center}} = \frac{\sum (\text{weight} \times \text{position})}{\sum \text{weight}}$$

**Probability version:**

$$\checkmark = \sum x \cdot P(X=x)$$

**They are IDENTICAL formulas.** Probability IS mass. The “balance point” IS the expectation.

---

## Complete Calculation: Center of Gravity

**Hat contents:**

| Value x | Probability P (X = x) |
|---------|-----------------------|
| 1       | 0.8                   |
| 10      | 0.2                   |

**Step 1:** Multiply each value by its probability (weight)

- $1 \times 0.8 = 0.8$
- $10 \times 0.2 = 2.0$

**Step 2:** Add them up

$$\checkmark = 0.8 + 2.0 = 2.8$$

**Result:**  $\checkmark = 2.8$

**Why closer to 1?** Because 1 has “heavier probability” (0.8 vs 0.2). Just like a heavy weight pulls the seesaw closer to itself.

**Visual:**

0----1----2----3----4----5----6----7----8----9----10

$E[X]=2.8$  (pulled toward 1 by heavy probability)

---

**Variance = Width of the Hat (Two Hats Comparison)**

### Hat A (Clustered):

| Value | Probability |
|-------|-------------|
| 49    | 0.33        |
| 50    | 0.34        |
| 51    | 0.33        |

### Hat B (Spread out):

| Value | Probability |
|-------|-------------|
| 1     | 0.33        |
| 50    | 0.34        |
| 100   | 0.33        |

### Both have same expectation:

- Hat A:  $\checkmark = 49(0.33) + 50(0.34) + 51(0.33) = 16.17 + 17 + 16.83 = 50$
- Hat B:  $\checkmark = 1(0.33) + 50(0.34) + 100(0.33) = 0.33 + 17 + 33 = 50.33 \approx 50$

### But behavior is COMPLETELY different:

- Hat A: You always get ~50. Very predictable.
- Hat B: You might get 1 or 100. Very unpredictable.

### Variance captures this difference.

---

## Complete Variance Calculation: Hat A vs Hat B

### Hat A (Clustered):

**Step 1:** We know  $\checkmark = 50$

**Step 2:** Calculate  $(X - E[X])^2$  for each outcome

| X  | $X - 50$ | $(X - 50)^2$ | P(X) | Weighted |
|----|----------|--------------|------|----------|
| 49 | -1       | 1            | 0.33 | 0.33     |
| 50 | 0        | 0            | 0.34 | 0        |
| 51 | 1        | 1            | 0.33 | 0.33     |

**Step 3:** Sum the weighted squared deviations

$$\text{Var}(X) = 0.33 + 0 + 0.33 = 0.66$$

**Hat B (Spread out):**

| X   | $X - 50$ | $(X - 50)^2$ | $P(X)$ | Weighted |
|-----|----------|--------------|--------|----------|
| 1   | -49      | 2401         | 0.33   | 792.33   |
| 50  | 0        | 0            | 0.34   | 0        |
| 100 | 50       | 2500         | 0.33   | 825.00   |

$$\text{Var}(X) = 792.33 + 0 + 825.00 = 1617.33$$

**Comparison:**

| Hat | Expectation | Variance | Standard Deviation |
|-----|-------------|----------|--------------------|
| A   | 50          | 0.66     | 0.81               |
| B   | 50          | 1617.33  | 40.22              |

**Hat B has 2450× more variance!** Yet same average. This is why variance is essential.

---

## Moment of Inertia Analogy (Complete Physics Connection)

**Physics:** Moment of inertia measures how hard it is to spin an object.

**Setup:** Two rods with weights.

**Rod A:** Weights at center (easy to spin)

**Rod B:** Weights at ends (hard to spin)

| Rod | Weight Placement | Moment of Inertia | Variance Analogy            |
|-----|------------------|-------------------|-----------------------------|
| A   | Center           | Small             | Small variance (clustered)  |
| B   | Ends             | Large             | Large variance (spread out) |

**Physics formula:**

$$I = \sum m_i r_i^2$$

**Probability formula:**

$$\text{Var}(X) = E[(X - \mu)^2] = \sum P(x_i)(x_i - \mu)^2$$

**They are the SAME formula with different names:**

- $m_i$  (mass)    $P(x_i)$  (probability)

- $r_i$  (distance from axis)     $(x_i - \mu)$  (distance from mean)

**Key insight:** Variance = “How hard would it be to spin this probability distribution?” More spread = harder to spin = larger variance.

---

## PART 2 — FORMAL THEORY (Complete Mathematical Foundations)

### Random Variable Is NOT an Algebra Variable (Critical Distinction)

| Algebra Variable                   | Random Variable                                     |
|------------------------------------|-----------------------------------------------------|
| Unknown constant                   | Function that assigns numbers to outcomes           |
| Example: $x + 2 = 5$ solve for $x$ | Example: $X(\text{Heads}) = 1, X(\text{Tails}) = 0$ |
| Has ONE true value                 | Has MANY possible values with probabilities         |
| “What is the value?”               | “What are the possible values and their chances?”   |
| Deterministic                      | Probabilistic                                       |

**Why beginners confuse them:** Both use letters like  $X$ . But they are completely different concepts.

**Algebra variable:** “I don’t know the number yet, but it has one fixed value.”

**Random variable:** “The value IS uncertain. Even after observing, different trials give different values.”

---

### The Function View (Complete Decoding)

**Formal definition:**

$$X : \Omega \rightarrow \mathbb{R}$$

**What each symbol means:**

| Symbol              | Name            | What It Actually Is                         | Example                         |
|---------------------|-----------------|---------------------------------------------|---------------------------------|
| $X$                 | Random Variable | The labeling rule itself                    | “Assign 1 to Heads, 0 to Tails” |
| $\Omega$<br>(Omega) | Sample Space    | “The bag of all possible physical outcomes” | {Heads, Tails}                  |
| $\mathbb{R}$        | Real Numbers    | Any number on the number line               | 1, 0, 3.5, $-2$ , $\pi$         |



|                     |                      |                                     |                       |
|---------------------|----------------------|-------------------------------------|-----------------------|
| $\omega$<br>(omega) | Single Outcome       | One specific thing from the bag     | "Heads"               |
| $X(\omega)$         | Function Application | "Apply rule X to outcome $\omega$ " | $X(\text{Heads}) = 1$ |

### Complete example:

**Physical experiment:** Flip a coin

**Sample space:**  $\Omega = \{\text{Heads}, \text{Tails}\}$

**Define random variable X:**

- $X(\text{Heads}) = 1$
- $X(\text{Tails}) = 0$

**Define random variable Y (different rule!):**

- $Y(\text{Heads}) = 100$
- $Y(\text{Tails}) = -50$

**Same experiment, different random variables.** This is why  $X$  is a FUNCTION, not the outcome itself.

## Why Mapping Is Absolutely Necessary

**Question:** Why can't we just work with "Heads" and "Tails" directly?

**Answer:** Math needs numbers. Specifically, we need to:

| Operation         | Needs Numbers? | Example                                      |
|-------------------|----------------|----------------------------------------------|
| Add outcomes      | Yes            | "Heads + Tails" = meaningless                |
| Calculate average | Yes            | Average of "Heads, Heads, Tails" = undefined |
| Calculate spread  | Yes            | Variance of words = undefined                |
| Optimize          | Yes            | "Minimize Heads" = undefined                 |

**Random variables are the BRIDGE:**

Physical World (words, objects, events)   Random Variable   Math World (numbers)

**Without this bridge:** Probability theory cannot exist. We couldn't calculate anything.

## Probability Distribution (The Second Half)

**Random variable alone is INCOMPLETE.** It tells us WHAT values are possible, but not HOW LIKELY.

**Example:** Two dice. Same random variable  $X$  = “sum of two dice,” but different distributions:

**Fair dice:**

| Sum $x$    | 2    | 3    | 4    | 5    | 6    | 7    | 8    | 9    | 10   | 11   | 12   |
|------------|------|------|------|------|------|------|------|------|------|------|------|
| $P(X = x)$ | 1/36 | 2/36 | 3/36 | 4/36 | 5/36 | 6/36 | 5/36 | 4/36 | 3/36 | 2/36 | 1/36 |

**Loaded dice (always sum to 7):**

| Sum $x$    | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|------------|---|---|---|---|---|---|---|---|----|----|----|
| $P(X = x)$ | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0  | 0  | 0  |

**Same random variable, completely different behavior.** You NEED the distribution.

---

## Complete Dice Distribution Example

**Experiment:** Roll two fair dice.  $X$  = sum.

**Step 1:** List all 36 outcomes (each equally likely: 1/36)

| Die 1 | Die 2 | Sum $X$ |
|-------|-------|---------|
| 1     | 1     | 2       |
| 1     | 2     | 3       |
| ...   | ...   | ...     |
| 6     | 6     | 12      |

**Step 2:** Count how many ways to get each sum

| Sum $x$ | Number of Ways | Probability $P(X = x)$ |
|---------|----------------|------------------------|
| 2       | 1              | 1/36                   |
| 3       | 2              | 2/36                   |
| 4       | 3              | 3/36                   |
| 5       | 4              | 4/36                   |

|    |   |      |
|----|---|------|
| 6  | 5 | 5/36 |
| 7  | 6 | 6/36 |
| 8  | 5 | 5/36 |
| 9  | 4 | 4/36 |
| 10 | 3 | 3/36 |
| 11 | 2 | 2/36 |
| 12 | 1 | 1/36 |

**Check:** Sum of probabilities =  $\frac{1+2+3+4+5+6+5+4+3+2+1}{36} = \frac{36}{36} = 1$

---

## Expectation = First Moment (Complete Explanation)

### What is a “moment”?

In physics: A moment describes how mass is distributed in space.

In probability: A moment describes how probability is distributed over numbers.

| Moment                | Formula                                                   | What It Measures          |
|-----------------------|-----------------------------------------------------------|---------------------------|
| First moment          | <input checked="" type="checkbox"/> $= \sum x \cdot P(x)$ | Center (location)         |
| Second moment         | $E[X^2] = \sum x^2 P(x)$                                  | Spread + center combined  |
| Second central moment | $E[(X - \mu)^2]$                                          | Pure spread (variance)    |
| Third moment          | $E[(X - \mu)^3]$                                          | Skewness (asymmetry)      |
| Fourth moment         | $E[(X - \mu)^4]$                                          | Kurtosis (tail heaviness) |

**Why “first”?** Because it uses  $x^1$  (first power).

**Why “moment”?** Borrowed from physics where  $E[X^n]$  describes rotational moments.

---

## Complete Dice Expectation (Two Methods)

### Method 1: Direct formula

$$\begin{aligned}
 E[X] &= \sum_{x=2}^{12} x \cdot P(X = x) \\
 &= 2\left(\frac{1}{36}\right) + 3\left(\frac{2}{36}\right) + 4\left(\frac{3}{36}\right) + 5\left(\frac{4}{36}\right) + 6\left(\frac{5}{36}\right) + 7\left(\frac{6}{36}\right) + 8\left(\frac{5}{36}\right) + 9\left(\frac{4}{36}\right) + 10\left(\frac{3}{36}\right) + 11\left(\frac{2}{36}\right) + 12\left(\frac{1}{36}\right)
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{2 + 6 + 12 + 20 + 30 + 42 + 40 + 36 + 30 + 22 + 12}{36} \\
 &= \frac{252}{36} = 7
 \end{aligned}$$

### Method 2: Using linearity (easier)

Let  $X = D_1 + D_2$  where  $D_1, D_2$  are individual dice.

$$E[X] = E[D_1 + D_2] = E[D_1] + E[D_2]$$

For one die:

$$E[D_1] = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = \frac{21}{6} = 3.5$$

Therefore:

$$\checkmark = 3.5 + 3.5 = 7$$

**Both methods give 7.** Method 2 is easier because it uses a powerful property.

---

## Variance = Second Central Moment (Complete Derivation)

**The shortcut formula derivation (every single step):**

**Goal:** Show that  $\text{Var}(X) = E[X^2] - (E[X])^2$

**Step 1:** Start with definition

$$\text{Var}(X) = E[(X - \mu)^2]$$

where  $\mu = E[X]$

**Step 2:** Expand the square inside the expectation

$$(X - \mu)^2 = X^2 - 2X\mu + \mu^2$$

**Step 3:** Apply expectation to each term (linearity of expectation)

$$E[(X - \mu)^2] = E[X^2 - 2X\mu + \mu^2]$$

$$= E[X^2] - E[2X\mu] + E[\mu^2]$$

**Step 4:** Simplify each term

- $E[X^2]$  stays as is
- $E[2X\mu] = 2\mu E[X]$  because  $\mu$  is a constant (it's  $E[X]$ , a fixed number)
- $E[\mu^2] = \mu^2$  because  $\mu^2$  is a constant

**Step 5:** Substitute back

✓  $+ \mu^2$

**Step 6:** Replace  $E[X]$  with  $\mu$

$$= E[X^2] - 2\mu \cdot \mu + \mu^2$$

$$= E[X^2] - 2\mu^2 + \mu^2$$

**Step 7:** Combine like terms

$$= E[X^2] - \mu^2$$

**Step 8:** Replace  $\mu$  with  $E[X]$

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

**This is the shortcut formula.** It says: "Average of squares minus square of average."

---

## Complete Dice Variance (Using Shortcut)

**From before:** ✓ = 7

**Step 1:** Calculate  $E[X^2]$

$$\begin{aligned} E[X^2] &= \sum_{x=2}^{12} x^2 \cdot P(X = x) \\ &= 4\left(\frac{1}{36}\right) + 9\left(\frac{2}{36}\right) + 16\left(\frac{3}{36}\right) + 25\left(\frac{4}{36}\right) + 36\left(\frac{5}{36}\right) + 49\left(\frac{6}{36}\right) + 64\left(\frac{5}{36}\right) + 81\left(\frac{4}{36}\right) + 100\left(\frac{3}{36}\right) \\ &= \frac{4 + 18 + 48 + 100 + 180 + 294 + 320 + 324 + 300 + 242 + 144}{36} \\ &= \frac{1974}{36} = 54.833 \end{aligned}$$

**Step 2:** Apply shortcut formula

$$\text{Var}(X) = E[X^2] - (E[X])^2 = 54.833 - 49 = 5.833$$

**Step 3:** Standard deviation

$$\sigma = \sqrt{5.833} \approx 2.415$$

**Interpretation:** Typical sum deviates from 7 by about 2.4.

---

## PART 3 — WHAT THIS THEORY IS GOOD AT

## When Probability Theory Excels

| Condition              | Why It Works                                  | Example                             |
|------------------------|-----------------------------------------------|-------------------------------------|
| Large data             | Law of Large Numbers makes estimates reliable | Image recognition with 1M images    |
| Stable patterns        | Probabilities don't change over time          | Dice, fair coins                    |
| Repeatable             | Can observe many trials                       | A/B testing, medical trials         |
| Measurable frequencies | Can count occurrences                         | Click-through rates, spam detection |

**The common thread:** When you can observe many examples and the underlying system doesn't change, expectation and variance give you reliable summaries.

---

## Why ML Works (The Hidden Assumption)

**Every ML algorithm assumes:**

1. Training data comes from some distribution  $P(X, Y)$
2. Future data comes from the SAME distribution  $P(X, Y)$
3. We have enough samples to estimate  $P(X, Y)$  accurately

**If these hold:** Expectation-based training → good real-world performance.

**If these fail:** Model breaks (see "Where Theory Fails" below).

---

## PART 4 — WHERE THEORY FAILS (Critical for Real-World ML)

### What Is Stationarity? (Complete Definition)

**Stationarity means:** The probability rules DON'T CHANGE over time.

**Mathematically:**

$P(X \leq x)$  is the same at time  $t_1$  and time  $t_2$

**Simple examples:**

| Stationary                  | Non-Stationary                     |
|-----------------------------|------------------------------------|
| Fair coin (always 50/50)    | Coin that wears out (bias changes) |
| Dice (always 1/6 each face) | Dice that gets loaded over time    |

|                                      |                                         |
|--------------------------------------|-----------------------------------------|
| Customer preferences (stable market) | Customer preferences (trending product) |
| Medical symptoms (stable disease)    | New disease variant (symptoms change)   |

## Why Expectation Fails Without Stationarity

### Example: Stock Market

#### Before 2008 crisis:

- Historical data: Average annual return = 8%
- Expectation based on this:  $E[R] = 8\%$

#### After 2008 crisis:

- Market structure changed completely
- Old expectation NO LONGER VALID
- Actual returns: -20%, +5%, -10% (very different)

**The expectation “8%” became MEANINGLESS because the underlying distribution changed.**

**Key insight:** Expectation is only valid for the distribution it was calculated from. Change the distribution, and the old expectation is garbage.

## Data Drift: The Silent Killer of ML Systems

**Definition:** Data drift = The statistical properties of data change over time, making trained models obsolete.

### Three types (complete breakdown):

#### Type 1: Covariate Drift (Input Changes)

**What changes:**  $P(X)$  — the distribution of inputs

#### Example:

- **Training:** Photos from professional cameras (high resolution, good lighting)
- **Deployment:** Photos from cheap phones (low resolution, blurry)
- **Result:** Model trained on sharp images fails on blurry ones

#### Mathematically:

$$P_{\text{train}}(X) \neq P_{\text{deploy}}(X)$$

#### Type 2: Label Drift (Output Changes)

**What changes:**  $P(Y)$  — the distribution of labels

**Example:**

- **Training:** Medical dataset with 50% healthy, 50% sick
- **Deployment:** Screening program where 95% healthy, 5% sick
- **Result:** Model expects 50/50 but sees 95/5, makes many false alarms

**Mathematically:**

$$P_{\text{train}}(Y) \neq P_{\text{deploy}}(Y)$$

---

### Type 3: Concept Drift (Relationship Changes) — MOST DANGEROUS

**What changes:**  $P(Y|X)$  — the relationship between input and output

**Example:**

- **Training:** “Buy product X” predicts customer satisfaction
- **Deployment:** New competitor launches better product X
- **Result:** Same customer behavior now means DISSATISFACTION

**Mathematically:**

$$P_{\text{train}}(Y|X) \neq P_{\text{deploy}}(Y|X)$$

**This is the worst kind** because the model’s fundamental understanding becomes wrong.

---

## Complete Numerical Example: Concept Drift

**Problem:** Predict if customer will buy (1) or not (0) based on age.

**Training data (2020):**

| Age | Buy? | Pattern            |
|-----|------|--------------------|
| 20  | 0    | Young = no money   |
| 30  | 1    | Middle = has money |
| 40  | 1    | Middle = has money |
| 50  | 0    | Old = retired      |

**Model learns:** Middle age (30-40) Buy

**Deployment data (2024):** Economy crashed, young people now have crypto wealth, middle-aged lost savings.



| Age | Buy? | Pattern               |
|-----|------|-----------------------|
| 20  | 1    | Young = crypto wealth |
| 30  | 0    | Middle = lost savings |
| 40  | 0    | Middle = lost savings |
| 50  | 1    | Old = pension safe    |

**Model prediction for age 30:** Buy (based on 2020 data) **WRONG!** Actual: Don't buy.

**Why it failed:**  $P(\text{Buy} | \text{Age} = 30)$  changed from 1 to 0. Concept drift.

---

## Why Drift Is So Dangerous (The Mechanism)

**What the model learned:**

$$\theta^* = \arg \min_{\theta} E_{\text{train}}[L(X, Y; \theta)]$$

**What it actually needs:**

$$\theta^* = \arg \min_{\theta} E_{\text{deploy}}[L(X, Y; \theta)]$$

**When drift occurs:**

$$E_{\text{train}}[L] \neq E_{\text{deploy}}[L]$$

**Result:** Model optimized for wrong expectation. Performance crashes.

**Real-world failures caused by drift:**

- COVID-19 broke demand forecasting models (behavior changed overnight)
  - Spam filters fail when attackers change tactics
  - Credit scoring fails during financial crises
  - Recommendation systems fail when trends shift
- 

## HANDWRITTEN NOTES CHEAT SHEET

### Random Variable

- **Definition:** Function  $X : \Omega \rightarrow \mathbb{R}$  that maps outcomes to numbers
- **NOT an algebra variable** — it's a function with multiple possible values
- **Why needed:** Math requires numbers, reality gives words/objects/events

### Probability Distribution

- **Definition:**  $P(X = x)$  for all possible  $x$
- **Gives likelihood** of each outcome
- **Required alongside RV** — RV alone is incomplete

## Expectation (First Moment)

- **Definition:**  $E[X] = \sum x P(x)$
- **Physics analogy:** Center of gravity / balance point
- **What it tells us:** Long-run average, center of distribution
- **Linear property:**  $E[aX + b] = aE[X] + b$

## Variance (Second Central Moment)

- **Definition:**  $\text{Var}(X) = E[(X - \mu)^2]$
- **Shortcut:**  $\text{Var}(X) = E[X^2] - (E[X])^2$
- **Physics analogy:** Moment of inertia
- **What it tells us:** Spread, uncertainty, risk
- **Properties:**
  - $\text{Var}(aX + b) = a^2 \text{Var}(X)$  (scale matters, shift doesn't)
  - $\text{Var}(X) \geq 0$  always

## Standard Deviation

- **Definition:**  $\sigma = \sqrt{\text{Var}(X)}$
- **Units:** Same as original data (unlike variance)
- **Interpretation:** Typical distance from mean

## Stationarity

- **Definition:** Probability structure stable over time
- **Required for:** Reliable estimation of expectation/variance
- **If violated:** Historical statistics become misleading

## Data Drift

- **Covariate drift:**  $P(X)$  changes
- **Label drift:**  $P(Y)$  changes
- **Concept drift:**  $P(Y | X)$  changes (most dangerous)
- **Result:** Trained models fail in deployment

---

## MEMORY HOOK (Galaxy Analogy)

| Concept                  | Galaxy Analogy                            |
|--------------------------|-------------------------------------------|
| Random Variable          | Individual stars (possible outcomes)      |
| Sample Space $\Omega$    | The entire universe of stars              |
| Probability Distribution | How much mass each star has               |
| Expectation              | Center of the galaxy (balance point)      |
| Variance                 | How spread out the galaxy is              |
| Standard Deviation       | Typical distance of stars from center     |
| Stationarity             | Galaxy shape stays the same               |
| Data Drift               | Galaxy is changing while you use old maps |

**Imagine you're an astronomer:**

- You map the galaxy (train your model)
- You assume the galaxy stays the same (stationarity)
- You navigate using your map (deploy model)
- But the galaxy is actually moving/changing (data drift)
- Your map becomes wrong (model fails)

## CONNECTIONS TO ADVANCED TOPICS

| This Topic                      | Leads To                                          |
|---------------------------------|---------------------------------------------------|
| Moments (expectation, variance) | Skewness, kurtosis, moment-generating functions   |
| Linearity of expectation        | Covariance, correlation, independence             |
| Variance properties             | Chebyshev's inequality, concentration bounds      |
| Stationarity                    | Time series analysis, ARIMA models                |
| Data drift                      | Domain adaptation, continual learning, monitoring |
| $E[X^2] - (E[X])^2$             | Bias-variance decomposition                       |

## Section 5: EXPLANATION OF EVERY TERM

## THE BIG PICTURE FIRST

These 11 terms are the **alphabet of probability**. You cannot read any ML paper without them. Here's how they connect:

SAMPLE SPACE ( $\Omega$ )   What CAN happen?

RANDOM VARIABLE ( $X$ )   Turn it into numbers

SUPPORT   Where does probability actually live?

PMF (discrete) or PDF (continuous)   How likely is each value?

EXPECTATION ( $E[X]$ )   What's the center?

VARIANCE ( $\text{Var}(X)$ )   How spread out is it?

STANDARD DEVIATION ( $\sigma$ )   Spread in original units

i.i.d.   Can we learn from multiple samples?

---

## 1. SAMPLE SPACE ( $\Omega$ )

### What It Actually Is

**Sample space** = The complete list of **EVERYTHING** that could possibly happen before you do the experiment.

**Critical point:** This is **BEFORE** any probability is assigned. Just the list of possibilities.

### Why It Must Come First

**You cannot talk about probability without knowing what you're talking about.**

**Example:** Someone says "Probability of rain is 30%."

**Your first question should be:** "What are the possible outcomes?"

- Just {rain, no rain}?
- Or {sunny, cloudy, rainy, snowy}?

- The answer changes everything.

## Complete Examples

| Experiment       | Sample Space $\Omega$  | Why These Outcomes?                              |
|------------------|------------------------|--------------------------------------------------|
| Coin flip        | {Heads, Tails}         | Only two sides                                   |
| Two coins        | {HH, HT, TH, TT}       | Each coin has 2 outcomes, $2 \times 2 = 4$ total |
| Die roll         | {1, 2, 3, 4, 5, 6}     | Six faces                                        |
| Weather tomorrow | {Sunny, Cloudy, Rainy} | Depends on what you care about                   |
| Stock price      | $[0, \infty)$          | Any non-negative number                          |

**Important:** The sample space depends on HOW YOU DEFINE the problem. There's no single "correct" sample space — it depends on what question you're asking.

## Two Coins Example (Complete)

**Experiment:** Flip two coins simultaneously.

**Wrong sample space:** {HH, HT, TT} Missing TH!

**Correct sample space:** {HH, HT, TH, TT}

**Why order matters:** HT (first Heads, second Tails) is different from TH (first Tails, second Heads). They are physically different outcomes.

**Check:** Total outcomes =  $2 \times 2 = 4$ . Each coin has 2 possibilities.

## Key Properties of Sample Space

1. **Mutually exclusive:** Only one outcome can happen at a time
2. **Collectively exhaustive:** At least one outcome MUST happen
3. **No probabilities yet:** Just the list of what's possible

## 2. SUPPORT

### What It Actually Is

**Support = The subset of the sample space where probability is actually GREATER THAN ZERO.**

**In plain English:** "These are the values that can ACTUALLY happen, not just theoretically."

## Why Support Matters

**Example:** You define a random variable for “outcome of a die roll” but you’re only interested in even numbers.

| All Possible Values | Support        | Why?                            |
|---------------------|----------------|---------------------------------|
| {1, 2, 3, 4, 5, 6}  | {2, 4, 6}      | We only care about even numbers |
| $P(X=1) = 0$        | $P(X=2) = 1/3$ | 1 is not in support             |
| $P(X=2) = 1/3$      | $P(X=4) = 1/3$ | 2 is in support                 |
| $P(X=3) = 0$        | $P(X=6) = 1/3$ | 3 is not in support             |

**Mathematical definition:**

$$\text{Support}(X) = \{x : P(X = x) > 0\}$$

## Continuous Support Example

**Human height:**

- Theoretically: Could be any real number from 0 to  $\infty$
- Actually: Support is approximately [120, 220] cm
- $P(X = 500 \text{ cm}) = 0$  Not in support
- $P(X = 170 \text{ cm}) > 0$  In support

**For continuous variables:** Support is where the PDF is strictly positive.

---

## 3. RANDOM VARIABLE (X)

### The Most Misunderstood Concept

**Beginners think:** “Random variable = a random number”

**Correct understanding:** “Random variable = a FUNCTION that assigns numbers to outcomes”

### Complete Definition

$$X : \Omega \rightarrow \mathbb{R}$$

**What this means step-by-step:**

| Step | What Happens | Example |
|------|--------------|---------|
|------|--------------|---------|

|   |                               |                              |
|---|-------------------------------|------------------------------|
| 1 | Physical outcome occurs       | You flip a coin, get “Heads” |
| 2 | Function X takes that outcome | $X(\text{“Heads”})$          |
| 3 | Function outputs a number     | $X(\text{“Heads”}) = 1$      |

**The randomness comes from the OUTCOME, not from the function.** The function  $X$  is completely deterministic — same input always gives same output.

## Multiple Random Variables on Same Experiment

**Experiment:** Roll a die

**Random Variable A:** “The number showing”

- $A(1) = 1, A(2) = 2, \dots, A(6) = 6$

**Random Variable B:** “Is it even?”

- $B(1) = 0, B(2) = 1, B(3) = 0, B(4) = 1, B(5) = 0, B(6) = 1$

**Random Variable C:** “Square of the number”

- $C(1) = 1, C(2) = 4, C(3) = 9, \dots, C(6) = 36$

**Same experiment, three different random variables.** This proves  $X$  is a function, not the outcome itself.

## ML Example: Email Classifier

**Experiment:** An email arrives

**Sample space:** {spam, not spam, phishing, newsletter, ...}

**Random Variable X:** “Is this spam?”

- $X(\text{spam}) = 1$
- $X(\text{not spam}) = 0$
- $X(\text{phishing}) = 1$  (treat as spam)
- $X(\text{newsletter}) = 0$  (not spam)

**This is exactly what ML models do** — they create numerical mappings from real-world events.

---

## 4. DISCRETE RANDOM VARIABLE

### What “Discrete” Actually Means

**Discrete = You can LIST all possible values, even if the list is infinite.**

**Key test:** Can you count the values? 1st, 2nd, 3rd, ...?

**Complete Examples**

| Discrete                     | Why Discrete?          | Values              |
|------------------------------|------------------------|---------------------|
| Die roll                     | Finite, listable       | {1, 2, 3, 4, 5, 6}  |
| Coin flips until first Heads | Infinite but countable | {1, 2, 3, ...}      |
| Number of emails today       | Countable              | {0, 1, 2, 3, ...}   |
| Students in class            | Finite                 | {0, 1, 2, ..., 500} |

**What Discrete Looks Like on a Number Line**

Discrete: • • • • • •  
          1  2  3  4  5  6

Gaps exist between values  
No value at 2.5, 4.7, etc.

**5. PROBABILITY MASS FUNCTION (PMF)**

**What “Mass” Means**

**Think of probability as physical mass:**

- Each outcome has a “lump” of probability
- The PMF tells you how big each lump is
- Total mass = 1 (certainty)

**Complete PMF Example: Fair Die**

| Value x | $P(X = x)$          | Visual Mass |
|---------|---------------------|-------------|
| 1       | $1/6 \approx 0.167$ |             |
| 2       | $1/6 \approx 0.167$ |             |
| 3       | $1/6 \approx 0.167$ |             |



|   |                     |  |
|---|---------------------|--|
| 4 | $1/6 \approx 0.167$ |  |
| 5 | $1/6 \approx 0.167$ |  |
| 6 | $1/6 \approx 0.167$ |  |

**Total mass:**  $6 \times \frac{1}{6} = 1$

## PMF Properties (Must Satisfy)

### Property 1: Non-negative

$$P(X = x) \geq 0 \text{ for all } x$$

**Why:** Negative probability makes no sense.

### Property 2: Sum to 1

$$\sum_x P(X = x) = 1$$

**Why:** Something must happen (total certainty).

## Complete Biased Coin PMF

**Coin with  $P(\text{Heads}) = 0.7$ ,  $P(\text{Tails}) = 0.3$**

| Outcome      | $P(X = x)$ | Check      |
|--------------|------------|------------|
| Heads (1)    | 0.7        | $\geq 0$   |
| Tails (0)    | 0.3        | $\geq 0$   |
| <b>Total</b> | <b>1.0</b> | <b>= 1</b> |

## ML Example: Spam Classifier PMF

| Class        | Probability | Interpretation          |
|--------------|-------------|-------------------------|
| Spam         | 0.7         | 70% confident it's spam |
| Not Spam     | 0.3         | 30% confident it's not  |
| <b>Total</b> | <b>1.0</b>  | Must sum to 1           |

---

## 6. CONTINUOUS RANDOM VARIABLE

### What “Continuous” Actually Means

**Continuous** = You **CANNOT** list all values. There are infinitely many, with no gaps.

**Key test:** Between any two values, there’s always another value.

### Complete Examples

| Continuous  | Why Continuous?                   | Range            |
|-------------|-----------------------------------|------------------|
| Height      | Infinite precision possible       | $[0, 300]$ cm    |
| Time        | Can always measure more precisely | $[0, \infty)$    |
| Temperature | Smooth transition                 | $[-273, \infty)$ |
| Weight      | No smallest increment             | $[0, \infty)$    |

### What Continuous Looks Like on a Number Line

Continuous:

0   50   100   150   200

No gaps, no jumps

Every point has a value

Cannot list them all

## 7. PROBABILITY DENSITY FUNCTION (PDF)

### The Most Confusing Concept for Beginners

**Critical fact:** For continuous variables,  $P(X = \text{exact value}) = 0$

**This seems impossible but is mathematically true.**

### Why Exact Probability Is Zero

**Example:** What’s the probability a person is EXACTLY 170.000000000... cm tall?

- Could be 170.0000000001
- Could be 169.9999999999

- There are infinitely many numbers near 170

**Probability is “spread” over infinitely many points each point gets zero probability.**

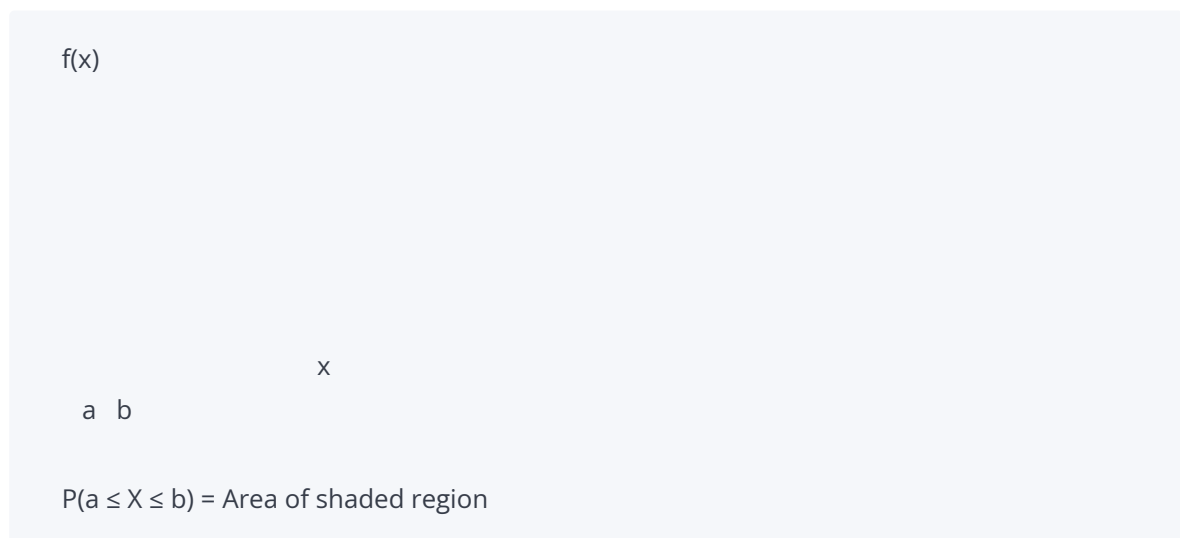
## What PDF Actually Represents

**PDF  $f(x)$  does NOT give probability directly. It gives DENSITY.**

**The rule:** Probability = Area under the PDF curve

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

## Visual Explanation



## Complete PDF Example: Uniform [0, 1]

**Definition:**

$$f(x) = \begin{cases} 1 & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

**Check total area:**

$$\int_0^1 1 dx = [x]_0^1 = 1 - 0 = 1$$

**Calculate  $P(0.2 \leq X \leq 0.5)$ :**

$$\int_{0.2}^{0.5} 1 dx = [x]_{0.2}^{0.5} = 0.5 - 0.2 = 0.3$$

**Answer:** Probability is 0.3 (30%)

## PMF vs PDF Comparison

| Feature                     | PMF (Discrete)      | PDF (Continuous)           |
|-----------------------------|---------------------|----------------------------|
| Gives probability directly? | Yes: $P(X = x)$     | No: $f(x)$ is density      |
| Probability of exact value  | Can be $> 0$        | Always $= 0$               |
| How to get probability      | Read directly       | Calculate area under curve |
| Total                       | $\sum P(X = x) = 1$ | $\int f(x)dx = 1$          |
| Visual                      | Bars (histogram)    | Smooth curve               |
| Example                     | Die roll, coin flip | Height, temperature, time  |

✓ or  $\mu$ )

## What It Actually Computes

Expectation = The “balance point” if probability were physical mass

## Complete Discrete Calculation

Hat example from tutorial:

| Value $x$ | Probability $P(X = x)$ | Weighted Value $x \cdot P(X = x)$ |
|-----------|------------------------|-----------------------------------|
| 1         | 0.5                    | 0.5                               |
| 5         | 0.3                    | 1.5                               |
| 10        | 0.2                    | 2.0                               |

**Step 1:** Calculate each weighted value

- $1 \times 0.5 = 0.5$
- $5 \times 0.3 = 1.5$
- $10 \times 0.2 = 2.0$

**Step 2:** Sum them up

✓  $= 0.5 + 1.5 + 2.0 = 4.0$

**Interpretation:** If you pull from this hat millions of times, your average will be 4.0.

## Complete Continuous Calculation

**Uniform [0, 1] random variable:**

$$\begin{aligned} E[X] &= \int_0^1 x \cdot f(x) dx = \int_0^1 x \cdot 1 dx \\ &= \left[ \frac{x^2}{2} \right]_0^1 = \frac{1}{2} - 0 = 0.5 \end{aligned}$$

**Interpretation:** The “center” of [0, 1] is 0.5. Makes sense.

---

## 9. VARIANCE (Var(X) or $\sigma^2$ )

### What It Actually Measures

**Variance** = “If I repeated this experiment forever, how far would typical outcomes be from the average, ON AVERAGE?”

### Complete Calculation: Hat Example

From before: ☒ = 4.0

| Value x | $x - E[X]$   | $(x - E[X])^2$ | $P(X = x)$ | Weighted |
|---------|--------------|----------------|------------|----------|
| 1       | $1 - 4 = -3$ | 9              | 0.5        | 4.5      |
| 5       | $5 - 4 = 1$  | 1              | 0.3        | 0.3      |
| 10      | $10 - 4 = 6$ | 36             | 0.2        | 7.2      |

**Step 1:** Calculate deviation from mean for each value

**Step 2:** Square each deviation (makes all positive)

**Step 3:** Multiply by probability

**Step 4:** Sum them up

$$\text{Var}(X) = 4.5 + 0.3 + 7.2 = 12.0$$

**Interpretation:** Typical squared deviation from mean is 12.

---

## 10. STANDARD DEVIATION ( $\sigma$ )

### Why We Need It

**Problem:** Variance is in “squared units”

- If  $X$  is in dollars, variance is in “square dollars”

- If  $\bar{X}$  is in meters, variance is in “square meters”

**Solution:** Take square root to get back to original units

$$\sigma = \sqrt{\text{Var}(\bar{X})} = \sqrt{12.0} \approx 3.46$$

**Interpretation:** Typical distance from mean is about 3.46 units.

## Complete Comparison

| Metric                | Value | Units             | Interpretation               |
|-----------------------|-------|-------------------|------------------------------|
| $E[\bar{X}]$          | 4.0   | Same as $\bar{X}$ | Center/average               |
| $\text{Var}(\bar{X})$ | 12.0  | Squared units     | Spread (hard to interpret)   |
| $\sigma$              | 3.46  | Same as $\bar{X}$ | Typical distance from center |

## 11. i.i.d. (Independent and Identically Distributed)

### The Most Important Assumption in ML

i.i.d. = Two separate conditions that **MUST** both hold

### 11.1 Independent (First Condition)

**Definition:** Knowing one sample tells you NOTHING about another.

**Mathematical test:**

$$P(X_2 | X_1) = P(X_2)$$

**Example where independence HOLDS:**

- Coin flips: First flip Heads doesn't change second flip
- Random survey responses from different people

**Example where independence FAILS:**

- Stock prices today and tomorrow (today affects tomorrow)
- Weather today and tomorrow (today's rain affects tomorrow's)
- Words in a sentence (“The” makes “cat” more likely than “quantum”)

### 11.2 Identically Distributed (Second Condition)

**Definition:** Every sample comes from the SAME probability distribution.

**Mathematical test:**

$$P(X_1 = x) = P(X_2 = x) = \dots = P(X_n = x) \text{ for all } x$$

#### Example where identical distribution HOLDS:

- Flipping the same fair coin 100 times
- Measuring height of 1000 people from same population

#### Example where identical distribution FAILS:

- Training on summer photos, testing on winter photos
- Training on young patients, testing on elderly patients
- Training before COVID, testing during COVID

## 11.3 Why ML Needs Both

#### If independent but NOT identically distributed:

- Samples don't influence each other
- But they come from different distributions
- Model learns wrong pattern

#### If identically distributed but NOT independent:

- Same distribution
- But samples influence each other
- Standard error calculations wrong

#### Only when BOTH hold:

- Mathematics of generalization works
- Law of Large Numbers applies
- Confidence intervals valid
- Model can generalize

## 11.4 Complete Example: i.i.d. vs Non-i.i.d.

**Scenario:** Predicting house prices

| Condition            | Training Data                                | Test Data                                   | i.i.d.? | Problem                       |
|----------------------|----------------------------------------------|---------------------------------------------|---------|-------------------------------|
| <b>i.i.d.</b>        | 1000 houses from same city, random selection | 200 houses from same city, random selection | Yes     | None                          |
| <b>Not identical</b> | Houses from New York                         | Houses from rural India                     | No      | Different price distributions |

|                        |                                             |                        |    |                            |
|------------------------|---------------------------------------------|------------------------|----|----------------------------|
| <b>Not independent</b> | 1000 houses, but 500 are pairs of neighbors | Same                   | No | Neighbor prices correlate  |
| <b>Neither</b>         | Time series: house prices 2010-2020         | House prices 2021-2023 | No | Both correlation and drift |

## 11.5 What Happens When i.i.d. Fails

| Violation              | Effect on ML                         | Solution                           |
|------------------------|--------------------------------------|------------------------------------|
| <b>Not independent</b> | Standard errors wrong, overconfident | Time series models, spatial models |
| <b>Not identical</b>   | Model fails on new data              | Domain adaptation, retraining      |
| <b>Both violated</b>   | Complete breakdown                   | Specialized architectures          |

## HANDWRITTEN NOTES CHEAT SHEET

### Sample Space ( $\Omega$ )

- **Definition:** Complete list of all possible outcomes
- **Comes FIRST:** Before any probability is assigned
- **Properties:** Mutually exclusive, collectively exhaustive
- **Example:** Coin = {H, T}, Two coins = {HH, HT, TH, TT}

### Support

- **Definition:** Values where  $P(X = x) > 0$  (actually possible)
- **Subset of sample space:** Not all theoretical values are possible
- **Example:** Die support = {1,2,3,4,5,6},  $P(X=7)=0$  so 7 not in support

### Random Variable ( $X$ )

- **Definition:** Function  $X : \Omega \rightarrow \mathbb{R}$
- **NOT a number:** It's a rule that assigns numbers to outcomes
- **Same experiment, different  $X$ 's:** Possible and common
- **Example:**  $X(\text{Heads}) = 1, X(\text{Tails}) = 0$

### Discrete Random Variable

- **Definition:** Countable values (can list them: 1st, 2nd, 3rd...)
- **Visual:** Dots on number line with gaps



- **Examples:** Die, coin, number of emails, students in class

## PMF (Probability Mass Function)

- **Definition:**  $P(X = x)$  for discrete variables
- **Gives exact probability:** Yes, directly
- **Properties:**  $P(X = x) \geq 0, \sum P(X = x) = 1$
- **Visual:** Bars of different heights

## Continuous Random Variable

- **Definition:** Uncountable values, no gaps
- **Visual:** Solid line on number line
- **Examples:** Height, temperature, time, weight

## PDF (Probability Density Function)

- **Definition:**  $f(x)$  for continuous variables
- **Gives exact probability:** NO! Gives density only
- **Probability = Area under curve:**  $P(a \leq X \leq b) = \int_a^b f(x)dx$
- **Properties:**  $f(x) \geq 0, \int f(x)dx = 1$
- **Critical:**  $P(X = \text{exact value}) = 0$  always

## ✓ or $\mu$ )

- **Definition:** Probability-weighted average
- **Discrete:**  $\checkmark = \sum x P(X=x)$
- **Continuous:**  $\checkmark = \int x f(x)dx$
- **Interpretation:** Long-run average, center of mass

## Variance ( $\text{Var}(X)$ or $\sigma^2$ )

- **Definition:** Average squared deviation from mean
- **Formula:**  $\text{Var}(X) = E[(X - \mu)^2] = E[X^2] - (E[X])^2$
- **Interpretation:** Spread, uncertainty, risk

## Standard Deviation ( $\sigma$ )

- **Definition:**  $\sigma = \sqrt{\text{Var}(X)}$
- **Units:** Same as original data (unlike variance)
- **Interpretation:** Typical distance from mean

## i.i.d. (Independent and Identically Distributed)

- **Independent:**  $P(X_2|X_1) = P(X_2)$  — samples don't influence each other
- **Identically Distributed:** Same probability distribution for all samples
- **Both needed:** For LLN, CLT, and most ML theory
- **Reality:** Often violated (time series, drift, correlation)

## MEMORY HOOK (City Analogy)

| Term                                      | City Analogy                                                          |
|-------------------------------------------|-----------------------------------------------------------------------|
| <b>Sample Space (<math>\Omega</math>)</b> | The entire city map (all possible locations)                          |
| <b>Support</b>                            | Actually inhabited neighborhoods (where people live)                  |
| <b>Random Variable</b>                    | Street address numbering system (turns locations into numbers)        |
| <b>Discrete</b>                           | Houses with specific numbers (1, 2, 3... with gaps between)           |
| <b>PMF</b>                                | Population count at each house address                                |
| <b>Continuous</b>                         | Every point on a road (infinite points between any two)               |
| <b>PDF</b>                                | Population density (people per square mile, not count at exact point) |
| <b>Expectation</b>                        | City center (balance point of population)                             |
| <b>Variance</b>                           | How spread out the city is                                            |
| <b>Standard Deviation</b>                 | Typical distance from city center                                     |
| <b>i.i.d.</b>                             | Assumption that neighborhoods are independent and have same rules     |

## CONNECTIONS TO ADVANCED TOPICS

| Term               | Leads To                                          |
|--------------------|---------------------------------------------------|
| Sample Space + RV  | Measure theory, sigma-algebras                    |
| PMF/PDF            | Maximum likelihood estimation, Bayesian inference |
| Expectation        | Conditional expectation, martingales              |
| Variance           | Covariance, correlation, PCA                      |
| Standard Deviation | Z-scores, confidence intervals                    |
| i.i.d.             | Law of Large Numbers, Central Limit Theorem       |

# Section 6: MAIN EQUATIONS & THEIR MEANING

*(Zero Hidden Meanings Edition — Every Step Shown)*

## THE EQUATION MAP (Memorize This First)

| Question                            | Equation                                            | When to Use            |
|-------------------------------------|-----------------------------------------------------|------------------------|
| What’s the average?                 | $\checkmark = \sum x_i P(x_i)$                      | Discrete variables     |
| What’s the average?<br>(continuous) | $\checkmark = \int x f(x) dx$                       | Continuous variables   |
| What if I scale/shift?              | $\checkmark + b$                                    | Transforming variables |
| What’s the spread?                  | $\text{Var}(X) = E[(X - \mu)^2]$                    | Definition of variance |
| What’s the spread? (shortcut)       | $\text{Var}(X) = E[X^2] - (E[X])^2$                 | Faster calculation     |
| What if I scale/shift?              | $\text{Var}(aX + b) = a^2 \text{Var}(X)$            | Transforming variables |
| Sum of independent variables?       | $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$ | Independent only!      |

## PART 1 — EXPECTATION (DISCRETE CASE)

### The Equation

$$E[X] = \sum_i x_i P(X = x_i)$$

### What Problem It Solves

You have uncertain outcomes. You want ONE number that represents the “center” or “typical” value.

**Not the simple average** (that assumes all outcomes equally likely).

**The probability-weighted average** (some outcomes more likely than others).

## Complete Symbol Breakdown

| Symbol       | What It Is                           | Example                                    |
|--------------|--------------------------------------|--------------------------------------------|
| $E[X]$       | The expected value we're calculating | $\checkmark = 3.5$ for fair die            |
| $\sum$       | "Add up all these terms"             | Sum over all possible outcomes             |
| $x_i$        | The $i$ -th possible value           | $x_1 = 1, x_2 = 2, \dots, x_6 = 6$         |
| $P(X = x_i)$ | Probability of that specific value   | $P(X = 4) = \frac{1}{6}$                   |
| $i$          | Just a counter (index)               | Goes from 1 to however many outcomes exist |

The subscript  $i$  is NOT an exponent. It's just a label: "the  $i$ -th outcome."

## Complete Dice Example (Every Single Step)

**Problem:** Fair die. Find  $E[X]$ .

**Step 0:** Identify what we know

- Outcomes:  $x_1 = 1, x_2 = 2, x_3 = 3, x_4 = 4, x_5 = 5, x_6 = 6$
- Probabilities:  $P(X = x_i) = \frac{1}{6}$  for all  $i$

**Step 1:** Write the formula

$$\checkmark = \sum_{i=1}^6 x_i \cdot P(X=x_i)$$

**Step 2:** Expand the sum (write out every term)

$$\checkmark = x_1 \cdot P(X=x_1) + x_2 \cdot P(X=x_2) + x_3 \cdot P(X=x_3) + x_4 \cdot P(X=x_4) + x_5 \cdot P(X=x_5) + x_6 \cdot P(X=x_6)$$

**Step 3:** Substitute values

$$\checkmark = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6}$$

**Step 4:** Perform each multiplication

$$\checkmark = \frac{1}{6} + \frac{2}{6} + \frac{3}{6} + \frac{4}{6} + \frac{5}{6} + \frac{6}{6}$$

**Step 5:** Add numerators (all have same denominator)

$$\checkmark = \frac{1 + 2 + 3 + 4 + 5 + 6}{6}$$

**Step 6:** Calculate numerator

$$1 + 2 + 3 + 4 + 5 + 6 = 21$$

**Step 7:** Divide

$$\checkmark = \frac{21}{6} = 3.5$$

**Answer:** ☒ = 3.5

**Critical insight:** A die can NEVER show 3.5. Expectation is NOT a possible outcome. It's the long-run average.

---

## PART 2 — EXPECTATION (CONTINUOUS CASE)

### Why We Need a Different Formula

**Discrete:** Values are separate, countable. We can list them: 1, 2, 3, 4, 5, 6.

**Continuous:** Values are infinite, uncountable. Between any two values, there's another value. We CANNOT list them all.

**Example:** Height can be 170.0, 170.01, 170.001, 170.0001, ... infinitely many values.

**Solution:** Replace summation ( $\sum$ ) with integration ( $\int$ ) — “continuous summation.”

### The Equation

$$E[X] = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

### Complete Symbol Breakdown

| Symbol                | What It Is                          | Analogy                                                 |
|-----------------------|-------------------------------------|---------------------------------------------------------|
| $\int$                | Integration symbol                  | “Continuous $\sum$ ” — adds infinitely many tiny pieces |
| $-\infty$ to $\infty$ | Limits of integration               | “From smallest possible to largest possible”            |
| $x$                   | The variable we're integrating over | Current value being considered                          |
| $f(x)$                | Probability Density Function (PDF)  | “How dense is probability at point $x$ ?”               |
| $dx$                  | Infinitesimal interval              | “Tiny slice of width $dx$ ”                             |

### What $f(x)$ Actually Is (Critical)

$f(x)$  is **NOT probability**. It is DENSITY.

**Probability = Area under  $f(x)$  curve**

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

**Why can't  $f(x)$  be probability?** Because for continuous variables,  $P(X = \text{exact value}) = 0$ .

### Complete Example: Uniform [0, 2]

**PDF:**  $f(x) = \begin{cases} \frac{1}{2} & \text{if } 0 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$

**Check it's valid:**  $\int_0^2 \frac{1}{2} dx = \frac{1}{2} \times 2 = 1$

**Calculate  $E[X]$ :**

**Step 1:** Write formula

☒  $= \int_0^2 x \cdot \frac{1}{2} dx$

**Step 2:** Pull out constant

☒  $= \frac{1}{2} \int_0^2 x dx$

**Step 3:** Integrate  $x$  (power rule:  $\int x^n dx = \frac{x^{n+1}}{n+1}$ )

$$\int x dx = \frac{x^2}{2}$$

**Step 4:** Apply limits

☒  $= \frac{1}{2} \left[ \frac{x^2}{2} \right]_0^2$

**Step 5:** Substitute upper limit

$$= \frac{1}{2} \cdot \frac{2^2}{2} = \frac{1}{2} \cdot \frac{4}{2} = \frac{1}{2} \cdot 2 = 1$$

**Step 6:** Substitute lower limit

$$= \frac{1}{2} \cdot \frac{0^2}{2} = 0$$

**Step 7:** Subtract

☒  $= 1 - 0 = 1$

**Answer:** ☒  $= 1$

**Check:** Center of  $[0, 2]$  is 1. Makes sense!

---

## PART 3 — LINEARITY OF EXPECTATION

### Rule 1: Scaling and Shifting

$$E[aX + b] = aE[X] + b$$

## What This Means in Plain English

If you:

1. Multiply a random variable by constant  $a$
2. Add constant  $b$

Then the expectation:

1. Gets multiplied by  $a$
2. Gets  $b$  added to it

The expectation transforms the **SAME WAY** as the variable.

## Complete Proof (Every Step)

**Given:**  $Y = aX + b$  where  $a, b$  are constants,  $X$  is random variable.

**Goal:** Show  $\mathbb{E}[Y] = a\mathbb{E}[X] + b$

**Step 1:** Start with definition

$$\mathbb{E}[Y] = \mathbb{E}[aX + b]$$

**Step 2:** Use definition of expectation

$$= \sum_i (ax_i + b)P(X = x_i)$$

**Step 3:** Distribute the sum

$$= \sum_i ax_iP(X = x_i) + \sum_i bP(X = x_i)$$

**Step 4:** Pull constants out of sums

$$= a \sum_i x_iP(X = x_i) + b \sum_i P(X = x_i)$$

**Step 5:** Recognize the pieces

- $\sum_i x_iP(X = x_i) = \mathbb{E}[X]$  This is the definition of expectation!
- $\sum_i P(X = x_i) = 1$  Probabilities must sum to 1!

**Step 6:** Substitute

$$= a\mathbb{E}[X] + b \cdot 1$$

**Step 7:** Simplify

$$= a\mathbb{E}[X] + b$$

**Proved.**

## Complete Numerical Example

Given:  $\checkmark = 10$

Define:  $Y = 3X + 7$

Find:  $E[Y]$

Step 1: Apply formula

$$E[Y] = E[3X + 7]$$

Step 2: Use linearity

$$\checkmark + 7$$

Step 3: Substitute

$$= 3(10) + 7$$

Step 4: Calculate

$$= 30 + 7 = 37$$

Answer:  $E[Y] = 37$

**No PMF needed!** This is why linearity is powerful — you don't need to know the full distribution.

---

## Rule 2: Sum of Random Variables

$$E[X + Y] = E[X] + E[Y]$$

### What This Means

**The average of a sum equals the sum of the averages.**

**Critical point:** This works EVEN IF  $X$  and  $Y$  are DEPENDENT!

### Complete Proof

Step 1: Definition

$$E[X + Y] = \sum_i \sum_j (x_i + y_j) P(X = x_i, Y = y_j)$$

Step 2: Split the sum

$$= \sum_i \sum_j x_i P(X = x_i, Y = y_j) + \sum_i \sum_j y_j P(X = x_i, Y = y_j)$$

Step 3: Factor out  $x_i$  from inner sum (doesn't depend on  $j$ )



$$= \sum_i x_i \sum_j P(X = x_i, Y = y_j) + \sum_j y_j \sum_i P(X = x_i, Y = y_j)$$

**Step 4:** Use marginal probability

- $\sum_j P(X = x_i, Y = y_j) = P(X = x_i)$
- $\sum_i P(X = x_i, Y = y_j) = P(Y = y_j)$

**Step 5:** Substitute

$$= \sum_i x_i P(X = x_i) + \sum_j y_j P(Y = y_j)$$

**Step 6:** Recognize expectations

$$\checkmark + E[Y]$$

**Proved.** Notice: No independence assumption used!

## Complete Numerical Example

**Given:**  $\checkmark = 5$ ,  $E[Y] = 8$

**Find:**  $E[X + Y]$

**Step 1:** Apply formula

$$\checkmark + E[Y]$$

**Step 2:** Substitute

$$= 5 + 8 = 13$$

**Answer:**  $E[X + Y] = 13$

**Even if X and Y are correlated!** This always works.

---

## PART 4 — VARIANCE (DEFINITION)

### The Equation

$$\text{Var}(X) = E[(X - \mu)^2]$$

where  $\mu = E[X]$

### What Problem It Solves

Expectation tells us the center. But HOW SPREAD OUT are the values around that center?

**Example:**

- Dataset A: 49, 50, 51    Mean = 50, very clustered
- Dataset B: 0, 50, 100    Mean = 50, very spread out

Same mean, completely different behavior. Variance captures this.

## Complete Symbol Breakdown

| Symbol          | What It Is                     | Why It's There                                        |
|-----------------|--------------------------------|-------------------------------------------------------|
| $\text{Var}(X)$ | The variance we're calculating | Spread measure                                        |
| $X$             | The random variable            | The uncertain quantity                                |
| $\mu$           | $E[X]$ , the mean              | Center point to measure spread from                   |
| $X - \mu$       | Deviation from mean            | "How far is this outcome from center?"                |
| $(X - \mu)^2$   | Squared deviation              | Makes all values positive; penalizes large deviations |
| $E[\cdot]$      | Average                        | "Average squared deviation"                           |

## Why Square? (Two Complete Reasons)

### Reason 1: Prevent Cancellation

#### Without squaring:

- Deviations: -5, +5
- Average:  $\frac{-5+5}{2} = 0$
- **Wrong!** There IS spread, but it cancels out.

#### With squaring:

- Squared deviations: 25, 25
- Average:  $\frac{25+25}{2} = 25$
- **Correct!** Captures the spread.

### Reason 2: Penalize Large Deviations

| Deviation | Squared Deviation | Ratio |
|-----------|-------------------|-------|
| 2         | 4                 | 2×    |
| 10        | 100               | 10×   |

**Deviation of 10 is penalized 25× more than deviation of 2** (because  $10^2 = 100$  vs  $2^2 = 4$ ).

**This is useful in ML:** Large errors are much worse than small errors.

## Complete Calculation Example

**Distribution:**

| X | P(X) |
|---|------|
| 1 | 0.5  |
| 5 | 0.5  |

**Step 1:** Find mean

☑  $= 1(0.5) + 5(0.5) = 0.5 + 2.5 = 3$

**Step 2:** Calculate deviations and squared deviations

| X | $X - \mu$    | $(X - \mu)^2$ | P(X) | Weighted           |
|---|--------------|---------------|------|--------------------|
| 1 | $1 - 3 = -2$ | $(-2)^2 = 4$  | 0.5  | $4 \times 0.5 = 2$ |
| 5 | $5 - 3 = 2$  | $2^2 = 4$     | 0.5  | $4 \times 0.5 = 2$ |

**Step 3:** Sum weighted squared deviations

$$\text{Var}(X) = 2 + 2 = 4$$

**Answer:**  $\text{Var}(X) = 4$

**Step 4:** Standard deviation

$$\sigma = \sqrt{4} = 2$$

**Interpretation:** Typical distance from mean is 2 units.

---

## PART 5 — COMPUTATIONAL VARIANCE FORMULA

### The Shortcut

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

### Why This Is Faster

**Definition formula:** Requires calculating  $(X - \mu)^2$  for every outcome. You need  $\mu$  first.

**Shortcut formula:** Calculate  $E[X^2]$  and  $(E[X])^2$  separately. Often easier.

## Complete Proof (Every Single Step)

**Goal:** Show  $\text{Var}(X) = E[X^2] - (E[X])^2$

**Step 1:** Start with definition

$$\text{Var}(X) = E[(X - \mu)^2]$$

**Step 2:** Expand the square using  $(a - b)^2 = a^2 - 2ab + b^2$

$$(X - \mu)^2 = X^2 - 2X\mu + \mu^2$$

**Step 3:** Substitute into expectation

$$\text{Var}(X) = E[X^2 - 2X\mu + \mu^2]$$

**Step 4:** Use linearity of expectation (split into three expectations)

$$= E[X^2] - E[2X\mu] + E[\mu^2]$$

**Step 5:** Pull out constants ( $\mu$  is a constant — it's  $E[X]$ , a fixed number)

- $E[2X\mu] = 2\mu E[X]$   $\mu$  is constant, so it comes out
- $E[\mu^2] = \mu^2$   $\mu^2$  is just a number, expectation of a constant is itself

**Step 6:** Substitute back

✓ +  $\mu^2$

**Step 7:** Replace  $E[X]$  with  $\mu$

$$= E[X^2] - 2\mu \cdot \mu + \mu^2$$

$$= E[X^2] - 2\mu^2 + \mu^2$$

**Step 8:** Combine like terms

$$= E[X^2] - \mu^2$$

**Step 9:** Replace  $\mu$  with  $E[X]$

$$\boxed{\text{Var}(X) = E[X^2] - (E[X])^2}$$

**Proved.**

## Complete Numerical Example Using Shortcut

**Same distribution:**

|   |      |
|---|------|
| X | P(X) |
|---|------|

|   |     |
|---|-----|
| 1 | 0.5 |
| 5 | 0.5 |

**Step 1:** Calculate  $E[X]$  (we know this:  $\mu = 3$ )

**Step 2:** Calculate  $E[X^2]$

$$E[X^2] = \sum x_i^2 P(X = x_i)$$

$$= 1^2(0.5) + 5^2(0.5)$$

$$= 1(0.5) + 25(0.5)$$

$$= 0.5 + 12.5 = 13$$

**Step 3:** Calculate  $(E[X])^2$

$$(E[X])^2 = 3^2 = 9$$

**Step 4:** Apply shortcut formula

$$\text{Var}(X) = E[X^2] - (E[X])^2 = 13 - 9 = 4$$

**Answer:**  $\text{Var}(X) = 4$  (Same as before!)

---

## PART 6 — VARIANCE PROPERTIES

### Property 1: Scaling and Shifting

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

#### What This Means

| Operation     | Effect on Variance           | Why?                                             |
|---------------|------------------------------|--------------------------------------------------|
| Multiply by a | Variance multiplied by $a^2$ | Spread scales by factor a, variance uses squares |
| Add b         | Variance UNCHANGED           | Shifting everything doesn't change spread        |

**Critical insight:** Adding a constant does NOTHING to variance. Only multiplication matters.

## Complete Proof (Every Step)

**Given:**  $Y = aX + b$

**Goal:** Show  $\text{Var}(Y) = a^2 \text{Var}(X)$

**Step 1:** Use definition of variance

$$\text{Var}(Y) = E[(Y - E[Y])^2]$$

**Step 2:** Substitute  $Y = aX + b$

$$= E[(aX + b - E[aX + b])^2]$$

**Step 3:** Use linearity of expectation on  $E[aX + b]$

✓  $+ b = a\mu + b$

**Step 4:** Substitute back

$$= E[(aX + b - (a\mu + b))^2]$$

**Step 5:** Simplify inside parentheses

$$= E[(aX + b - a\mu - b)^2]$$

$$= E[(aX - a\mu)^2]$$

$b$  and  $-b$  cancel!

**Step 6:** Factor out  $a$

$$= E[(a(X - \mu))^2]$$

$$= E[a^2(X - \mu)^2]$$

**Step 7:** Pull constant  $a^2$  out of expectation

$$= a^2 E[(X - \mu)^2]$$

**Step 8:** Recognize variance definition

$$\boxed{\text{Var}(aX + b) = a^2 \text{Var}(X)}$$

**Proved.** Notice:  $b$  completely disappeared! Shifting doesn't affect spread.

## Complete Numerical Example

**Given:**  $\text{Var}(X) = 4$

**Define:**  $Y = 2X + 5$

**Find:**  $\text{Var}(Y)$

**Step 1:** Identify  $a = 2$ ,  $b = 5$

**Step 2:** Apply formula

$$\text{Var}(Y) = a^2 \text{Var}(X) = 2^2 \times 4 = 4 \times 4 = 16$$

**Answer:**  $\text{Var}(Y) = 16$

**Check with definition:**

- If  $X$  takes values  $\{1, 5\}$ , then  $Y = 2X + 5$  takes values  $\{7, 15\}$
  - $E[Y] = 7(0.5) + 15(0.5) = 11$
  - $\text{Var}(Y) = (7 - 11)^2(0.5) + (15 - 11)^2(0.5) = 16(0.5) + 16(0.5) = 8 + 8 = 16$
- 

## Property 2: Sum of Independent Variables

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$$

**ONLY when  $X$  and  $Y$  are INDEPENDENT!**

## Why Independence Matters

**Expectation of sum:** Always equals sum of expectations. No conditions.

**Variance of sum:** Only equals sum of variances if independent. If dependent, there's an extra term.

**The extra term (when NOT independent):**

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$$

Where  $\text{Cov}(X, Y)$  is covariance. If independent,  $\text{Cov}(X, Y) = 0$ .

## Complete Proof for Independent Case

**Step 1:** Definition

$$\text{Var}(X + Y) = E[(X + Y) - E[X + Y]]^2$$

**Step 2:** Use linearity of expectation

✓  $E[Y] = \mu_X + \mu_Y$

**Step 3:** Substitute

$$= E[(X + Y - \mu_X - \mu_Y)^2]$$

$$= E[(X - \mu_X) + (Y - \mu_Y)]^2$$

**Step 4:** Expand square  $(a + b)^2 = a^2 + 2ab + b^2$

$$= E[(X - \mu_X)^2 + 2(X - \mu_X)(Y - \mu_Y) + (Y - \mu_Y)^2]$$

**Step 5:** Split expectation

$$= E[(X - \mu_X)^2] + 2E[(X - \mu_X)(Y - \mu_Y)] + E[(Y - \mu_Y)^2]$$

**Step 6:** Recognize pieces

- $E[(X - \mu_X)^2] = \text{Var}(X)$
- $E[(Y - \mu_Y)^2] = \text{Var}(Y)$
- $E[(X - \mu_X)(Y - \mu_Y)] = \text{Cov}(X, Y)$

**Step 7:** If independent,  $\text{Cov}(X, Y) = 0$

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$$

**Proved.**

## Complete Numerical Example

**Independent coin flips:**

- $X$  = first flip (1=Heads, 0=Tails),  $\text{Var}(X) = 0.25$
- $Y$  = second flip (1=Heads, 0=Tails),  $\text{Var}(Y) = 0.25$

**Find  $\text{Var}(X + Y)$ :**

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) = 0.25 + 0.25 = 0.5$$

**Verify directly:**

- $X + Y$  can be 0, 1, or 2
- $P(X + Y = 0) = 0.25$ ,  $P(X + Y = 1) = 0.5$ ,  $P(X + Y = 2) = 0.25$
- $E[X + Y] = 0(0.25) + 1(0.5) + 2(0.25) = 1$
- $\text{Var}(X + Y) = (0 - 1)^2(0.25) + (1 - 1)^2(0.5) + (2 - 1)^2(0.25) = 0.25 + 0 + 0.25 = 0.5$

## HANDWRITTEN NOTES CHEAT SHEET

**Expectation (Discrete)**

$$E[X] = \sum_i x_i P(X = x_i)$$



- Multiply each outcome by its probability
- Add everything up
- Result: long-run average

## Expectation (Continuous)

$$E[X] = \int_{-\infty}^{\infty} xf(x)dx$$

- Replace sum with integral
- $f(x)$  is density, NOT probability
- Probability = area under  $f(x)$

## Linearity of Expectation

$$E[aX + b] = aE[X] + b$$

$$E[X + Y] = E[X] + E[Y] \text{ (always, even if dependent!)}$$

## Variance (Definition)

$$\text{Var}(X) = E[(X - \mu)^2]$$

- Average squared deviation from mean
- $\mu = E[X]$

## Variance (Shortcut)

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

- Often faster to compute
- $E[X^2] \neq (E[X])^2$  in general!

## Variance Properties

$$\text{Var}(aX + b) = a^2\text{Var}(X)$$

- Scaling: multiply variance by  $a^2$
- Shifting ( $+b$ ): NO EFFECT on variance

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$$

- **ONLY if independent!**
  - If dependent: add  $2\text{Cov}(X, Y)$
-

# MEMORY HOOK

| Concept              | Physical Analogy                                           |
|----------------------|------------------------------------------------------------|
| Expectation          | Balance point of a seesaw                                  |
| Variance             | Moment of inertia (how hard to spin)                       |
| Linearity            | Seesaw scales naturally with weights                       |
| $\text{Var}(aX + b)$ | Stretching the seesaw makes it harder to balance ( $a^2$ ) |
| $\text{Var}(X + Y)$  | Two independent seesaws = sum of instabilities             |

# CONNECTIONS TO ADVANCED TOPICS

| Equation                                            | Leads To                                      |
|-----------------------------------------------------|-----------------------------------------------|
| $\checkmark = \sum x P(x)$                          | Conditional expectation, martingales          |
| $\checkmark + b$                                    | Affine transformations, feature scaling       |
| $\text{Var}(X) = E[X^2] - (E[X])^2$                 | Bias-variance decomposition                   |
| $\text{Var}(aX + b) = a^2\text{Var}(X)$             | Xavier/He initialization, batch normalization |
| $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$ | Ensemble methods, random forests              |

# Section 7 : STEP-BY-STEP NUMERICAL EXAMPLES

*(Zero Hidden Meanings Edition — Every Calculation Shown)*

# THE CORE LESSON (Memorize This)

Two distributions can have the SAME expectation but COMPLETELY DIFFERENT variance.  
This is why variance matters in ML: same average accuracy, completely different risk.

# PART 1 — SETUP

**Random Variable:**  $X$  = outcome of a 4-sided die

**Possible values:**  $\{1, 2, 3, 4\}$

**Two scenarios to compare:**

| Scenario | Name       | Description                              |
|----------|------------|------------------------------------------|
| A        | Fair die   | Every outcome equally likely             |
| B        | Biased die | Middle values more likely, extremes rare |

## SCENARIO A — FAIR DIE (Complete Calculation)

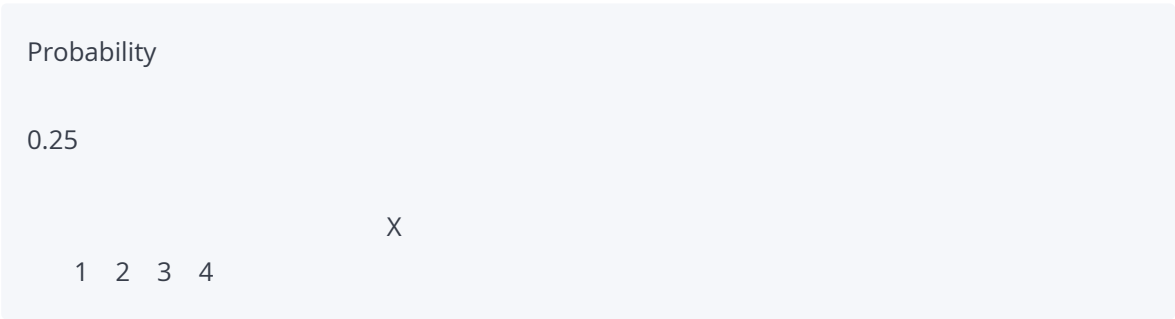
### Step 1: Define the Distribution

**What “fair” means:** Every face has equal probability.

| X | $P(X = x)$ | Why?             |
|---|------------|------------------|
| 1 | 0.25       | 1 out of 4 faces |
| 2 | 0.25       | 1 out of 4 faces |
| 3 | 0.25       | 1 out of 4 faces |
| 4 | 0.25       | 1 out of 4 faces |

**Check:**  $0.25 + 0.25 + 0.25 + 0.25 = 1.0$

**Visual (probability mass):**



### Step 2: Calculate Expectation $E[X]$

**Formula:**  $E[X] = \sum x \cdot P(X=x)$

**Step 2a:** Write out every term

$$\checkmark = 1 \cdot P(X=1) + 2 \cdot P(X=2) + 3 \cdot P(X=3) + 4 \cdot P(X=4)$$

**Step 2b:** Substitute probabilities

$$\checkmark = 1(0.25) + 2(0.25) + 3(0.25) + 4(0.25)$$

**Step 2c:** Calculate each multiplication separately

- $1 \times 0.25 = 0.25$
- $2 \times 0.25 = 0.50$
- $3 \times 0.25 = 0.75$
- $4 \times 0.25 = 1.00$

**Step 2d:** Add them up

$$\checkmark = 0.25 + 0.50 + 0.75 + 1.00 = 2.50$$

**Answer:**  $\checkmark = 2.5$

**Important:** 2.5 is NOT a possible outcome (die shows 1, 2, 3, or 4). But it's the balance point.

**Visual (balance point):**

1   2   3   4

$E[X]=2.5$  (balance point)

### Step 3: Calculate $E[X^2]$ (Second Moment)

**Why we need this:** For the variance shortcut formula.

**Formula:**  $E[X^2] = \sum x^2 \cdot P(X = x)$

**Step 3a:** Write out every term

$$E[X^2] = 1^2 \cdot P(X = 1) + 2^2 \cdot P(X = 2) + 3^2 \cdot P(X = 3) + 4^2 \cdot P(X = 4)$$

**Step 3b:** Calculate squares first

- $1^2 = 1$
- $2^2 = 4$
- $3^2 = 9$
- $4^2 = 16$

**Step 3c:** Multiply by probabilities

- $1 \times 0.25 = 0.25$
- $4 \times 0.25 = 1.00$

- $9 \times 0.25 = 2.25$
- $16 \times 0.25 = 4.00$

**Step 3d:** Add them up

$$E[X^2] = 0.25 + 1.00 + 2.25 + 4.00 = 7.50$$

**Answer:**  $E[X^2] = 7.5$

---

### Step 4: Calculate Variance $\text{Var}(X)$

**Formula:**  $\text{Var}(X) = E[X^2] - (E[X])^2$

**Step 4a:** We know  $E[X^2] = 7.5$  and  $\checkmark = 2.5$

**Step 4b:** Calculate  $(E[X])^2$

$$(2.5)^2 = 6.25$$

**Step 4c:** Subtract

$$\text{Var}(X) = 7.5 - 6.25 = 1.25$$

**Answer:**  $\text{Var}(X) = 1.25$

**Step 4d:** Standard deviation

$$\sigma = \sqrt{1.25} \approx 1.118$$


---

### Scenario A Summary Table

| Quantity        | Formula Used           | Value | Interpretation               |
|-----------------|------------------------|-------|------------------------------|
| $E[X]$          | $\sum xP(x)$           | 2.5   | Center of distribution       |
| $E[X^2]$        | $\sum x^2P(x)$         | 7.5   | Average of squared values    |
| $\text{Var}(X)$ | $E[X^2] - (E[X])^2$    | 1.25  | Spread measure               |
| $\sigma$        | $\sqrt{\text{Var}(X)}$ | 1.118 | Typical distance from center |

---

## SCENARIO B — BIASED DIE (Complete Calculation)

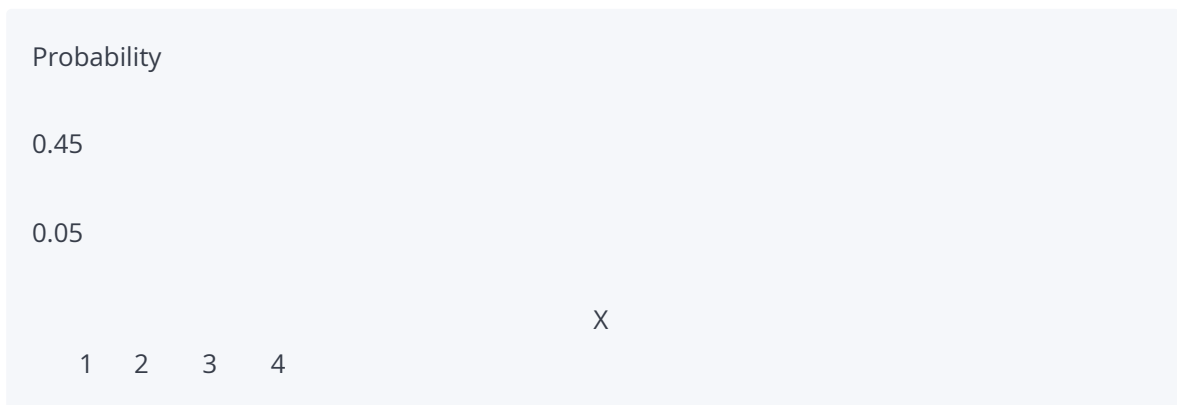
### Step 1: Define the Distribution

**What “biased” means:** Middle values (2 and 3) are much more likely. Extremes (1 and 4) are rare.

| X | P(X = x) | Why?        |
|---|----------|-------------|
| 1 | 0.05     | Very rare   |
| 2 | 0.45     | Very common |
| 3 | 0.45     | Very common |
| 4 | 0.05     | Very rare   |

**Check:**  $0.05 + 0.45 + 0.45 + 0.05 = 1.0$

**Visual (probability mass):**



**Notice:** Probability is concentrated in the middle.

## Step 2: Calculate Expectation $E[X]$

**Formula:**  $E[X] = \sum x \cdot P(X=x)$

**Step 2a:** Write out every term

$$E[X] = 1 \cdot P(X=1) + 2 \cdot P(X=2) + 3 \cdot P(X=3) + 4 \cdot P(X=4)$$

**Step 2b:** Substitute probabilities

$$E[X] = 1(0.05) + 2(0.45) + 3(0.45) + 4(0.05)$$

**Step 2c:** Calculate each multiplication separately

- $1 \times 0.05 = 0.05$
- $2 \times 0.45 = 0.90$
- $3 \times 0.45 = 1.35$
- $4 \times 0.05 = 0.20$

**Step 2d:** Add them up

$$E[X] = 0.05 + 0.90 + 1.35 + 0.20 = 2.50$$

**Answer:**  $E[X] = 2.5$

## CRITICAL OBSERVATION

| Scenario   | $E[X]$ |
|------------|--------|
| A (Fair)   | 2.5    |
| B (Biased) | 2.5    |

### THEY ARE IDENTICAL!

**Why?** Both distributions are symmetric around 2.5. The “balance point” doesn’t change even when we move probability mass around symmetrically.

### Visual comparison:

Fair: (spread out)

Biased: (concentrated)

Both balance at 2.5

### Step 3: Calculate $E[X^2]$

**Formula:**  $E[X^2] = \sum x^2 \cdot P(X = x)$

**Step 3a:** Calculate squares and multiply by probabilities

- $1^2 \times 0.05 = 1 \times 0.05 = 0.05$
- $2^2 \times 0.45 = 4 \times 0.45 = 1.80$
- $3^2 \times 0.45 = 9 \times 0.45 = 4.05$
- $4^2 \times 0.05 = 16 \times 0.05 = 0.80$

**Step 3b:** Add them up

$$E[X^2] = 0.05 + 1.80 + 4.05 + 0.80 = 6.70$$

**Answer:**  $E[X^2] = 6.70$

### Step 4: Calculate Variance $\text{Var}(X)$

**Formula:**  $\text{Var}(X) = E[X^2] - (E[X])^2$

**Step 4a:** We know  $E[X^2] = 6.70$  and  $\checkmark = 2.5$

**Step 4b:** Calculate  $(E[X])^2$

$$(2.5)^2 = 6.25$$

Step 4c: Subtract

$$\text{Var}(X) = 6.70 - 6.25 = 0.45$$

Answer:  $\text{Var}(X) = 0.45$

Step 4d: Standard deviation

$$\sigma = \sqrt{0.45} \approx 0.671$$

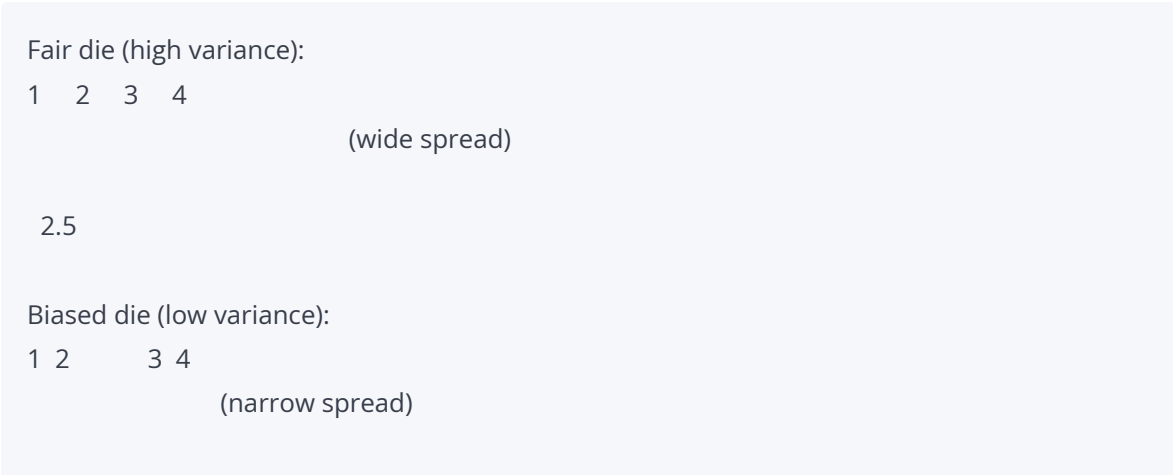
Scenario B Summary Table

| Quantity        | Formula Used           | Value | Interpretation               |
|-----------------|------------------------|-------|------------------------------|
| $E[X]$          | $\sum xP(x)$           | 2.5   | Center of distribution       |
| $E[X^2]$        | $\sum x^2P(x)$         | 6.7   | Average of squared values    |
| $\text{Var}(X)$ | $E[X^2] - (E[X])^2$    | 0.45  | Spread measure               |
| $\sigma$        | $\sqrt{\text{Var}(X)}$ | 0.671 | Typical distance from center |

SIDE-BY-SIDE COMPARISON

| Property        | Fair Die (A) | Biased Die (B) | Difference                      |
|-----------------|--------------|----------------|---------------------------------|
| $E[X]$          | 2.5          | 2.5            | <b>SAME</b>                     |
| $E[X^2]$        | 7.5          | 6.7            | Different                       |
| $\text{Var}(X)$ | 1.25         | 0.45           | <b>B is 2.78× less variable</b> |
| $\sigma$        | 1.118        | 0.671          | <b>B is 1.67× more stable</b>   |

Visual comparison of spread:





## WHY VARIANCE DIFFERS (Complete Explanation)

### The Mathematical Reason

**Fair die:** Probability spread evenly → outcomes far from center (1 and 4) are common → large squared deviations → high variance.

**Biased die:** Probability concentrated near center → outcomes far from center (1 and 4) are rare → small squared deviations → low variance.

### Complete Deviation Table

**Fair Die:**

| X     | $X - 2.5$ | $(X - 2.5)^2$ | P (X) | Contribution |
|-------|-----------|---------------|-------|--------------|
| 1     | -1.5      | 2.25          | 0.25  | 0.5625       |
| 2     | -0.5      | 0.25          | 0.25  | 0.0625       |
| 3     | 0.5       | 0.25          | 0.25  | 0.0625       |
| 4     | 1.5       | 2.25          | 0.25  | 0.5625       |
| Total |           |               |       | 1.25         |

**Biased Die:**

| X     | $X - 2.5$ | $(X - 2.5)^2$ | P (X) | Contribution |
|-------|-----------|---------------|-------|--------------|
| 1     | -1.5      | 2.25          | 0.05  | 0.1125       |
| 2     | -0.5      | 0.25          | 0.45  | 0.1125       |
| 3     | 0.5       | 0.25          | 0.45  | 0.1125       |
| 4     | 1.5       | 2.25          | 0.05  | 0.1125       |
| Total |           |               |       | 0.45         |

**Key difference:** In the biased die, the large deviations ( $\pm 1.5$ ) have tiny probability (0.05), so they contribute little to variance. In the fair die, they have large probability (0.25), so they contribute heavily.

# ML CONNECTIONS (Why This Matters)

## Connection 1: Model Stability

| ML Concept          | Fair Die Analogy          | Biased Die Analogy    |
|---------------------|---------------------------|-----------------------|
| High variance model | Predictions spread widely | Unstable, overfitting |
| Low variance model  | Predictions concentrated  | Stable, generalizes   |
| Same accuracy       | Same $E[X]$               | Same $E[X]$           |

**In ML:** Two models can have the same average accuracy (same expectation) but one is reliable (low variance) and one is unpredictable (high variance).

## Connection 2: Risk vs Return (Finance)

| Investment | Expected Return | Risk (Variance)       |
|------------|-----------------|-----------------------|
| A          | 10%             | High (like fair die)  |
| B          | 10%             | Low (like biased die) |

**Rational choice:** Pick B. Same return, less risk.

## Connection 3: Regularization

| Technique         | Effect           | Die Analogy                              |
|-------------------|------------------|------------------------------------------|
| No regularization | High variance    | Fair die                                 |
| L2 regularization | Reduces variance | Biased die (pulls weights toward center) |
| Dropout           | Reduces variance | Biased die (randomly disables neurons)   |

**What regularization does:** It concentrates probability mass near the center (like making the die biased toward middle values), reducing variance without changing the mean much.

## Connection 4: Ensemble Methods

| Method       | How It Works          | Variance Effect                    |
|--------------|-----------------------|------------------------------------|
| Single model | One prediction        | High variance                      |
| Bagging      | Average many models   | Reduces variance (like biased die) |
| Boosting     | Sequential correction | Reduces bias                       |

**Why bagging works:** Averaging many predictions is like creating a “biased die” — extreme predictions cancel out, leaving concentrated predictions near the center.

---

## ADDITIONAL EXAMPLE: Same Variance, Different Expectation

To complete your understanding, let's see the opposite case:

### Scenario C: Shifted Fair Die

Distribution:

| X | P(X = x) |
|---|----------|
| 3 | 0.25     |
| 4 | 0.25     |
| 5 | 0.25     |
| 6 | 0.25     |

This is the fair die shifted by +2.

Calculate  $E[X]$ :

✓  $= 3(0.25) + 4(0.25) + 5(0.25) + 6(0.25) = 0.75 + 1.0 + 1.25 + 1.5 = 4.5$

Calculate  $\text{Var}(X)$  using property:

$$\text{Var}(X + 2) = \text{Var}(X) = 1.25$$

(shifting doesn't change variance!)

Comparison:

| Property        | Fair Die (A) | Shifted Fair Die © |
|-----------------|--------------|--------------------|
| $E[X]$          | 2.5          | 4.5                |
| $\text{Var}(X)$ | 1.25         | 1.25               |

**Lesson:** Shifting changes expectation but NOT variance. Scaling changes both.

---

## HANDWRITTEN NOTES CHEAT SHEET

## Fair Die (Scenario A)

- $P(X = x) = 0.25$  for all  $x \in \{1, 2, 3, 4\}$
- $\checkmark = 2.5$
- $E[X^2] = 7.5$
- $\text{Var}(X) = 1.25$
- $\sigma = 1.118$

## Biased Die (Scenario B)

- $P(X = 1) = 0.05, P(X = 2) = 0.45, P(X = 3) = 0.45, P(X = 4) = 0.05$
- $\checkmark = 2.5$  (SAME as fair!)
- $E[X^2] = 6.7$
- $\text{Var}(X) = 0.45$  (MUCH smaller!)
- $\sigma = 0.671$

## Key Insight

*Same expectation  $\neq$  same behavior*

*Variance reveals the hidden difference*

## ML Takeaway

- Two models with same accuracy can have very different stability
- Low variance = reliable predictions
- High variance = unpredictable, risky

## MEMORY HOOK

### The Seesaw Analogy:

| Scenario            | Visual                          | Meaning             |
|---------------------|---------------------------------|---------------------|
| Fair die            | Weights spread evenly on seesaw | Balanced but wobbly |
| Biased die          | Weights concentrated at center  | Balanced and stable |
| Same center         | Both balance at 2.5             | Same expectation    |
| Different stability | Fair wobbles more               | Different variance  |

### In ML terms:

- **Fair die** = Overfit model (high variance, unstable)
- **Biased die** = Regularized model (low variance, stable)

- **Same center** = Same average accuracy
- **Different stability** = Different generalization

---

## CONNECTIONS TO ADVANCED TOPICS

| Concept                       | Leads To                                              |
|-------------------------------|-------------------------------------------------------|
| Same mean, different variance | Bias-variance tradeoff, model selection               |
| Variance reduction            | Regularization, ensemble methods, dropout             |
| Variance properties           | Chebyshev's inequality, concentration bounds          |
| $E[X^2]$ calculation          | Moment generating functions, characteristic functions |

---

## Section 8: VISUAL & INTUITIVE EXPLANATION

*(Zero Hidden Meanings Edition — Visual Everything)*

---

### THE INFORMATION PIPELINE (Master This First)

Every probability problem in ML follows the same 5-step pipeline. Memorize this:

STEP 1: RAW WORLD EVENT

"Something uncertain happens in reality"

Example: Flip 2 coins

Outcomes: {HH, HT, TH, TT}

STEP 2: RANDOM VARIABLE (X)

"Create a rule that turns reality into numbers"

Example:  $X = \text{"Count number of Heads"}$

Mapping: HH 2, HT 1, TH 1, TT 0

STEP 3: PROBABILITY DISTRIBUTION

"Assign likelihood to each numerical outcome"

Example:  $P(X=0)=0.25$ ,  $P(X=1)=0.50$ ,  $P(X=2)=0.25$

STEP 4: EXPECTATION ( $E[X]$ )

"Find the center/balance point"

Example:  $E[X] = 1.0$

STEP 5: VARIANCE ( $\text{Var}(X)$ )

"Measure how spread out things are"

Example:  $\text{Var}(X) = 0.5$

---

## COMPLETE TWO-COIN EXAMPLE (Every Step Visualized)

### Step 1: Raw World Event

**Physical action:** You flip two coins into the air.

**What could happen:**

- Coin 1: Heads, Coin 2: Heads    HH
- Coin 1: Heads, Coin 2: Tails    HT
- Coin 1: Tails, Coin 2: Heads    TH
- Coin 1: Tails, Coin 2: Tails    TT

**Visual:**

Coin 1: H   H   T   T

Coin 2: H   T   H   T

Outcome: HH   HT   TH   TT

**Total outcomes:** 4 (and all equally likely if coins are fair)

---

### Step 2: Random Variable (The Rule)

**We define:**  $X$  = "Number of Heads showing"

### Complete mapping table:

| Physical Outcome | X (Number of Heads) | How Many Ways? |
|------------------|---------------------|----------------|
| HH               | 2                   | 1 way          |
| HT               | 1                   | 1 way          |
| TH               | 1                   | 1 way          |
| TT               | 0                   | 1 way          |

### Visual mapping:

| Reality      | Numbers |
|--------------|---------|
| HH (2 Heads) | $X = 2$ |
| HT (1 Head)  | $X = 1$ |
| TH (1 Head)  | $X = 1$ |
| TT (0 Heads) | $X = 0$ |

**Important:** The random variable is the RULE “count Heads,” not the outcomes themselves.

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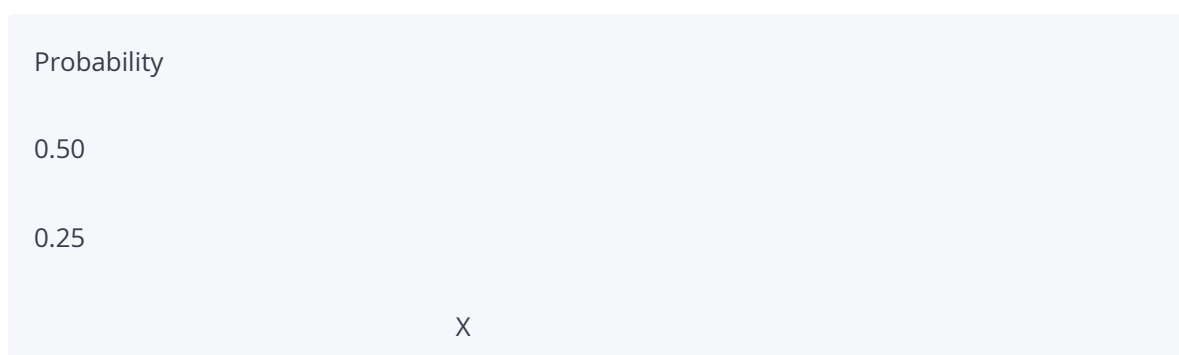
## Step 3: Probability Distribution (PMF)

Count how many ways each X value occurs:

| X | Ways to Get It | Probability  |
|---|----------------|--------------|
| 0 | TT             | $1/4 = 0.25$ |
| 1 | HT, TH         | $2/4 = 0.50$ |
| 2 | HH             | $1/4 = 0.25$ |

**Check:**  $0.25 + 0.50 + 0.25 = 1.00$

**Visual PMF (Probability Mass Function):**



0    1    2

$P(X=0)=0.25$   $P(X=1)=0.50$   $P(X=2)=0.25$

**Shape observation:** Symmetric around 1. Highest bar at 1 (most likely outcome).

---

## Step 4: Expectation (Center of Gravity)

**Formula:**  $\checkmark = \sum x \cdot P(X=x)$

**Step-by-step calculation:**

| X | $P(X = x)$ | $x \cdot P(X = x)$     | Visual Weight     |
|---|------------|------------------------|-------------------|
| 0 | 0.25       | $0 \times 0.25 = 0$    | Small weight at 0 |
| 1 | 0.50       | $1 \times 0.50 = 0.50$ | Heavy weight at 1 |
| 2 | 0.25       | $2 \times 0.25 = 0.50$ | Small weight at 2 |

**Sum:**  $\checkmark = 0 + 0.50 + 0.50 = 1.0$

**Visual balance point:**

0    1    2

0.25   0.50   0.25    Weights (probabilities)

$E[X]=1.0$

(Balance point)

The heavy weight at 1 pulls the center to 1.  
The smaller weights at 0 and 2 balance each other.

**Physical analogy:** Imagine a seesaw with:

- Weight 0.25 at position 0
- Weight 0.50 at position 1
- Weight 0.25 at position 2

Where does it balance? At position 1.

---

## Step 5: Variance (Spread Around Center)



**Formula:**  $\text{Var}(X) = E[(X - \mu)^2]$  where  $\mu = 1$

**Step-by-step calculation:**

| X | $X - \mu$    | $(X - \mu)^2$ | $P(X = x)$ | Weighted               |
|---|--------------|---------------|------------|------------------------|
| 0 | $0 - 1 = -1$ | $(-1)^2 = 1$  | 0.25       | $1 \times 0.25 = 0.25$ |
| 1 | $1 - 1 = 0$  | $0^2 = 0$     | 0.50       | $0 \times 0.50 = 0$    |
| 2 | $2 - 1 = 1$  | $1^2 = 1$     | 0.25       | $1 \times 0.25 = 0.25$ |

**Sum:**  $\text{Var}(X) = 0.25 + 0 + 0.25 = 0.50$

**Standard deviation:**  $\sigma = \sqrt{0.5} \approx 0.707$

**Visual spread:**

0   1   2

-1   0   +1

Deviations from mean

Distance from center:

- Position 0 is 1 unit away

- Position 1 is 0 units away (at center)

- Position 2 is 1 unit away

Variance = average of (distance)<sup>2</sup> = 0.5

## VISUAL COMPARISON: DIFFERENT VARIANCES

### Low Variance (Concentrated)

**Distribution:**

| X | $P(X = x)$ |
|---|------------|
| 0 | 0.05       |
| 1 | 0.90       |
| 2 | 0.05       |

**Visual:**

Probability

0.90

0.05

0

1

2

X

VERY concentrated at center

Most outcomes are exactly 1

**Expectation:**  $\checkmark = 0(0.05) + 1(0.90) + 2(0.05) = 0 + 0.90 + 0.10 = 1.0$

**Variance:**

$$\text{Var}(X) = (0 - 1)^2(0.05) + (1 - 1)^2(0.90) + (2 - 1)^2(0.05) = 0.05 + 0 + 0.05 = 0.10$$

**Standard deviation:**  $\sigma = \sqrt{0.1} \approx 0.316$

---

## High Variance (Spread Out)

**Distribution:**

| X | P(X = x) |
|---|----------|
| 0 | 0.40     |
| 1 | 0.20     |
| 2 | 0.40     |

**Visual:**

Probability

0.40

0.20

0

1

2

X

VERY spread out

Extremes are common

**Expectation:**  $\checkmark = 0(0.40) + 1(0.20) + 2(0.40) = 0 + 0.20 + 0.80 = 1.0$

**Variance:**

$$\text{Var}(X) = (0 - 1)^2(0.40) + (1 - 1)^2(0.20) + (2 - 1)^2(0.40) = 0.40 + 0 + 0.40 = 0.80$$

**Standard deviation:**  $\sigma = \sqrt{0.8} \approx 0.894$

---

## Side-by-Side Comparison

| Property        | Low Variance | High Variance | Same? |
|-----------------|--------------|---------------|-------|
| $E[X]$          | 1.0          | 1.0           | YES   |
| $\text{Var}(X)$ | 0.10         | 0.80          | NO    |
| $\sigma$        | 0.316        | 0.894         | NO    |
| Shape           | Narrow peak  | Flat/wide     | NO    |

**Visual:**

Low Variance:

High Variance:

0 1 2

0 1 2

Same center (1.0)

Same center (1.0)

Very narrow

Very wide

---

## CONTINUOUS DISTRIBUTIONS (Bell Curve Visual)

### The Normal Distribution

The most important continuous distribution in ML:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

**Don't memorize this formula.** Just understand what it looks like.

**Visual: Three Normal Curves**

Same Mean ( $\mu=0$ ), Different Variances:

Low Variance ( $\sigma=0.5$ ):    Medium ( $\sigma=1$ ):    High Variance ( $\sigma=2$ ):

-1 0 1

-2 -1 0 1 2

-4 -2 0 2 4

Tall and narrow

Medium height/width

Short and wide

$\sigma = 0.5$

$\sigma = 1.0$

$\sigma = 2.0$

Var = 0.25

Var = 1.0

Var = 4.0

**Key observation:** As variance increases, the curve becomes:

- **Shorter** (probability spreads out)
- **Wider** (more values become likely)
- **Flatter** (peak decreases)

**The area under ALL curves = 1** (total probability)

---

## The 68-95-99.7 Rule (Visual)

For normal distributions:

$\mu-3\sigma$   $\mu-2\sigma$   $\mu-\sigma$   $\mu$   $\mu+\sigma$   $\mu+2\sigma$   $\mu+3\sigma$

= 68% of data (within  $\mu \pm \sigma$ )

= 95% of data (within  $\mu \pm 2\sigma$ )

= 99.7% of data (within  $\mu \pm 3\sigma$ )

**What this means:**

- About 2/3 of outcomes fall within 1 standard deviation of mean

- About 19/20 outcomes fall within 2 standard deviations
  - Almost ALL outcomes fall within 3 standard deviations
- 

## EXTREME CASE: ZERO VARIANCE (Perfect Certainty)

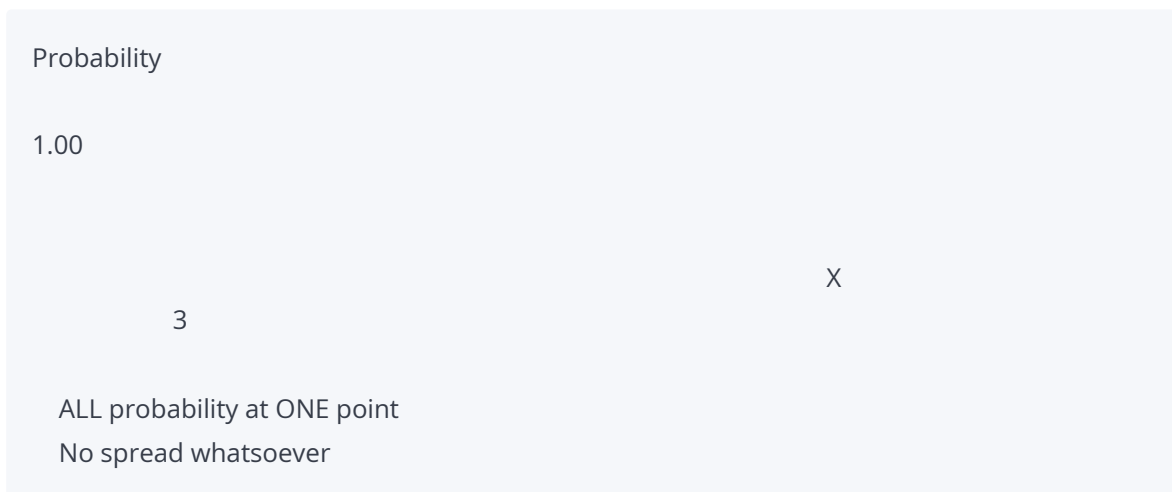
### What It Means

**Zero variance = NO randomness whatsoever**

**Distribution:**

| X             | P (X = x) |
|---------------|-----------|
| 3             | 1.00      |
| anything else | 0.00      |

**Visual:**



**Properties:**

- ☒ = 3
- $\text{Var}(X) = 0$
- $\sigma = 0$

**Physical analogy:** A point mass. All mass concentrated at one point. No “wobble” at all.

**In ML:** A model with zero variance would give the EXACT same prediction every time. Usually bad (underfitting), unless the problem is truly deterministic.

---

## THE LASER BEAM ANALOGY (Master Memory Hook)

| Probability Concept     | Laser Beam Analogy          | ML Application                 |
|-------------------------|-----------------------------|--------------------------------|
| Expectation ( $\mu$ )   | Direction the laser points  | Model's average prediction     |
| Variance ( $\sigma^2$ ) | Thickness of the laser beam | Model's prediction uncertainty |
| Low variance            | Thin, focused beam          | Confident predictions          |
| High variance           | Wide, diffuse beam          | Uncertain predictions          |
| Zero variance           | Perfectly thin line         | Deterministic (no uncertainty) |

Visual:

Low Variance Model:

High Variance Model:

) ) ) )

) ) ) )

) ) ) )

Thin, focused beam

Wide, scattered beam

Confident predictions

Uncertain predictions

ML PIPELINE VISUAL (Complete System)

How This All Fits in a Neural Network

INPUT LAYER

Image of a cat

(Raw world event)

FEATURE EXTRACTION

Convert pixels to features (edges, textures)

(Random variable: numerical representation)

PROBABILITY OUTPUT

Softmax layer: P(cat)=0.85, P(dog)=0.10, P(bird)=0.05

(Probability distribution)

EXPECTATION (Training)

Minimize  $E[\text{Loss}]$  over training data

(Find center of optimal behavior)

VARIANCE (Generalization)

Measure prediction stability across different inputs

(Control spread to prevent overfitting)

---

## HANDWRITTEN NOTES CHEAT SHEET

### The 5-Step Pipeline

1. **Raw event** Something uncertain happens
2. **Random variable** Turn it into numbers
3. **Probability distribution** Assign likelihoods
4. **Expectation** Find the center
5. **Variance** Measure the spread

### Two-Coin Example

- Outcomes: {HH, HT, TH, TT}
- $X$  = count Heads {0, 1, 2}
- PMF:  $P(0)=0.25$ ,  $P(1)=0.50$ ,  $P(2)=0.25$
- ☒ = 1.0
- $\text{Var}(X) = 0.5$

### Visual Shapes

- **Low variance:** Tall, narrow peak
- **High variance:** Short, wide curve
- **Zero variance:** Single spike (no spread)

### The 68-95-99.7 Rule

- 68% within  $\mu \pm \sigma$
- 95% within  $\mu \pm 2\sigma$
- 99.7% within  $\mu \pm 3\sigma$

## ML Connection

- **Expectation**    What model predicts on average
  - **Variance**    How much predictions vary
  - **Low variance**    Stable, reliable
  - **High variance**    Unstable, overfitting
- 

## MEMORY HOOKS

### The Seesaw

- Weights = probabilities
- Positions = outcome values
- Balance point = expectation
- How spread the weights are = variance

### The Galaxy

- Stars = possible outcomes
- Star brightness = probability
- Galaxy center = expectation
- Galaxy size = variance

### The Laser

- Beam direction = expectation
  - Beam thickness = variance
  - Focused beam = low variance (confident)
  - Diffuse beam = high variance (uncertain)
- 

## CONNECTIONS TO ADVANCED TOPICS

| Visual Concept | Leads To                                         |
|----------------|--------------------------------------------------|
| PMF shapes     | Maximum likelihood estimation                    |
| Bell curves    | Gaussian processes, variational inference        |
| Variance width | Confidence intervals, uncertainty quantification |
| Zero variance  | Deterministic models, degenerate distributions   |
| Pipeline view  | Probabilistic graphical models                   |



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## Section 9: COMMON MISTAKES

*(Zero Hidden Meanings Edition — Every Mistake Explained with Numbers)*

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### THE ROOT CAUSE OF ALL MISTAKES

*You are treating probability words like everyday English words.*

*"Expected" ≠ "what I expect to happen"*

*"Average" ≠ "typical single outcome"*

*"Random" ≠ "completely unpredictable"*

**Fix:** Every time you see a probability term, translate it to its mathematical definition FIRST, then interpret.

---

### MISTAKE 1 — "Expectation Is a Guaranteed Outcome"

#### What Beginners Think

*"If  $\frac{1}{6}$  = 2.5, then when I roll the die, I should get 2.5"*

#### What Is Actually True

*$\frac{1}{6}$  = 2.5 means: "If you roll the die a million times and average all results, you'll get very close to 2.5"*

**A single roll will NEVER give 2.5.** The die only shows 1, 2, 3, 4, 5, or 6.

#### Complete Numerical Demonstration

**Experiment:** Roll a fair die 10 times

| Roll | Result |
|------|--------|
| 1    | 4      |
| 2    | 1      |

|    |   |
|----|---|
| 3  | 6 |
| 4  | 2 |
| 5  | 5 |
| 6  | 3 |
| 7  | 1 |
| 8  | 6 |
| 9  | 4 |
| 10 | 2 |

**Running average after each roll:**

| After Roll | Sum | Average |
|------------|-----|---------|
| 1          | 4   | 4.0     |
| 2          | 5   | 2.5     |
| 3          | 11  | 3.67    |
| 4          | 13  | 3.25    |
| 5          | 18  | 3.6     |
| 6          | 21  | 3.5     |
| 7          | 22  | 3.14    |
| 8          | 28  | 3.5     |
| 9          | 32  | 3.56    |
| 10         | 34  | 3.4     |

**Notice:**

- No single roll was 2.5
- The average bounces around
- With only 10 rolls, average is 3.4 (not close to 2.5)

**Now imagine 1,000,000 rolls:**

- The average will be VERY close to 3.5
- But you still never rolled 3.5 on any single throw

## The Correct Mental Model

| Wrong Model                               | Correct Model                                          |
|-------------------------------------------|--------------------------------------------------------|
| "Expectation = prediction for next trial" | "Expectation = limit of averages over infinite trials" |
| ☑ tells me what I'll get"                 | ☑ tells me where the center of mass is"                |
| ☑ in one try"                             | "Individual outcomes can be far from $E[X]$ "          |

## ML Connection

**Wrong thinking:** "My model's expected accuracy is 85%, so this prediction should be 85% correct."

**Correct thinking:** "My model's expected accuracy is 85%, meaning over 1000 predictions, about 850 will be correct. This single prediction might be right or wrong."

---

## MISTAKE 2 — Assuming $E[X^2] = (E[X])^2$

### What Beginners Think

"If I square the values and take the average, it's the same as squaring the average."

### What Is Actually True

$$E[X^2] \geq (E[X])^2$$

Equality holds **ONLY** when  $\text{Var}(X) = 0$  (no randomness).

## Complete Numerical Proof That They're Different

Distribution:

| X | $P(X = x)$ |
|---|------------|
| 1 | 0.5        |
| 5 | 0.5        |

**Step 1: Calculate  $E[X]$**

☑  $= 1(0.5) + 5(0.5) = 0.5 + 2.5 = 3$

**Step 2: Calculate  $(E[X])^2$**

$$(E[X])^2 = 3^2 = 9$$

### Step 3: Calculate $E[X^2]$

$$E[X^2] = 1^2(0.5) + 5^2(0.5) = 1(0.5) + 25(0.5) = 0.5 + 12.5 = 13$$

### Step 4: Compare

$$E[X^2] = 13 \quad \text{vs} \quad (E[X])^2 = 9$$

$$13 \neq 9$$

**They are NOT equal!**

## Why This Happens (Complete Explanation)

**The difference is exactly the variance:**

$$\text{Var}(X) = E[X^2] - (E[X])^2 = 13 - 9 = 4$$

**Rearranging:**

$$E[X^2] = \text{Var}(X) + (E[X])^2 = 4 + 9 = 13$$

**Intuition:** Squaring BEFORE averaging gives more weight to large values. The number 5 becomes 25, which pulls the average up. Squaring AFTER averaging doesn't get this "amplification effect."

## Visual Explanation

Values: 1      5

$$E[X]=3$$

Squares: 1      25

$$E[X^2]=13 \text{ (pulled toward 25)}$$

Average then square:  $(3)^2 = 9$

Square then average:  $E[X^2] = 13$

The "5" becomes "25" and pulls the average upward!

## When ARE They Equal?

**Only when there's no spread:**

| X | P(X = x) |
|---|----------|
| 3 | 1.0      |

$$E[X] = 3$$

$$(E[X])^2 = 9$$

$$E[X^2] = 3^2(1.0) = 9$$

**Now they ARE equal!** Because  $\text{Var}(X) = 0$ .

## ML Connection

**Wrong calculation:** You compute  $E[\text{Loss}^2]$  and think it's  $(E[\text{Loss}])^2$ . This makes your variance calculation negative (impossible!), breaking your model.

**Correct approach:** Always use  $\text{Var}(X) = E[X^2] - (E[X])^2$  and compute BOTH terms separately.

---

## MISTAKE 3 — Forgetting to Subtract $(E[X])^2$ in Variance

### What Beginners Do

“Variance is just  $E[X^2]$ . I'll calculate that and stop.”

**Wrong answer:**  $\text{Var}(X) = E[X^2] = 13$  (from previous example)

### What Is Actually True

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

**You MUST subtract the squared mean!**

## Complete Correct Calculation

**Using same distribution:**

| X | P(X = x) |
|---|----------|
| 1 | 0.5      |
| 5 | 0.5      |

**Step 1:**  $\mu = 3$  (calculated above)

**Step 2:**  $E[X^2] = 13$  (calculated above)

**Step 3:**  $(E[X])^2 = 3^2 = 9$

**Step 4:**  $\text{Var}(X) = 13 - 9 = 4$

**Step 5:**  $\sigma = \sqrt{4} = 2$

## What Happens If You Forget the Subtraction

| What You Calculate | Value | What It Actually Is                       |
|--------------------|-------|-------------------------------------------|
| $E[X^2]$ only      | 13    | "Total energy" — includes center + spread |
| Correct variance   | 4     | "Pure spread" — deviation from center     |
| Forgot subtraction | 13    | WRONG — inflated by 9                     |

**The number 13 includes:**

- The "spread" (4)
- PLUS the "center effect" (9)

**Variance should measure ONLY spread.** That's why we subtract  $(E[X])^2$ .

## Visual Explanation

$$E[X^2] = 13$$

$$\begin{array}{l} \text{Center effect} \\ \text{(squared mean)} \end{array} = (E[X])^2 = 9$$

$$\begin{array}{l} \text{Spread} \\ \text{(variance)} \end{array} = \text{Var}(X) = 4$$

$$0 \quad 9 \quad 13$$

$$\text{Variance} = E[X^2] - (E[X])^2 = 13 - 9 = 4$$

## ML Connection

**Wrong:** You report model uncertainty as 13 when it's actually 4. Your model seems much more unstable than it is.

**Correct:** Always compute both  $E[X^2]$  and  $(E[X])^2$ , then subtract.

---

## MISTAKE 4 — Adding Variances Incorrectly

### What Beginners Think

*"Var(X + Y) = Var(X) + Var(Y) always. It's just like expectation."*

### What Is Actually True

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$$

The simple addition **ONLY** works when  $X$  and  $Y$  are **INDEPENDENT** (then  $\text{Cov}(X, Y) = 0$ ).

### Complete Numerical Counter-Example

**Scenario:**  $Y = X$  (perfectly dependent — they're the SAME variable)

**Distribution of X:**

| X | P(X = x) |
|---|----------|
| 1 | 0.5      |
| 3 | 0.5      |

**Calculate:**

- $\checkmark = 1(0.5) + 3(0.5) = 2$
- $E[X^2] = 1(0.5) + 9(0.5) = 5$
- $\text{Var}(X) = 5 - 4 = 1$

**Now compute**  $\text{Var}(X + Y) = \text{Var}(X + X) = \text{Var}(2X)$ :

**Method 1: Direct calculation**

- $X + Y = X + X = 2X$
- When  $X = 1$ ,  $2X = 2$
- When  $X = 3$ ,  $2X = 6$

| 2X | P(2X = z) |
|----|-----------|
| 2  | 0.5       |

|   |     |
|---|-----|
| 6 | 0.5 |
|---|-----|

- $E[2X] = 2(0.5) + 6(0.5) = 4$
- $E[(2X)^2] = 4(0.5) + 36(0.5) = 20$
- $\text{Var}(2X) = 20 - 16 = 4$

#### Method 2: Using the formula

$$\text{Var}(2X) = 2^2 \cdot \text{Var}(X) = 4 \cdot 1 = 4$$

#### Method 3: WRONG beginner approach

$$\text{Var}(X) + \text{Var}(Y) = 1 + 1 = 2$$

**WRONG!**

The correct answer is 4, not 2!

#### Why the Wrong Approach Fails

| Approach                          | Result             | Correct? |
|-----------------------------------|--------------------|----------|
| $\text{Var}(X) + \text{Var}(Y)$   | $1 + 1 = 2$        | WRONG    |
| $\text{Var}(2X) = 4\text{Var}(X)$ | $4 \times 1 = 4$   | CORRECT  |
| Full formula with covariance      | $1 + 1 + 2(1) = 4$ | CORRECT  |

**Covariance when  $Y = X$ :**

$$\text{Cov}(X, Y) = \text{Cov}(X, X) = \text{Var}(X) = 1$$

$$\text{Var}(X + Y) = 1 + 1 + 2(1) = 4$$

#### Another Example: Negative Covariance

**Scenario:**  $Y = -X$  (perfectly negatively dependent)

| X | $Y = -X$ | $X + Y$ |
|---|----------|---------|
| 1 | -1       | 0       |
| 3 | -3       | 0       |

**Result:**  $X + Y = 0$  ALWAYS. So  $\text{Var}(X + Y) = 0$ .

**Check with formula:**



$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$$

- $\text{Var}(X) = 1$
- $\text{Var}(Y) = \text{Var}(-X) = (-1)^2 \text{Var}(X) = 1$
- $\text{Cov}(X, -X) = -\text{Var}(X) = -1$

$$\text{Var}(X + Y) = 1 + 1 + 2(-1) = 0$$

**Wrong approach:**  $\text{Var}(X) + \text{Var}(Y) = 1 + 1 = 2$  **WRONG AGAIN!**

## Visual: How Covariance Changes Variance of Sum

Positive Covariance (X and Y move together):

|   |   |     |
|---|---|-----|
| X | Y | X+Y |
| X | Y | X+Y |

Sum has MORE variance than individual

Negative Covariance (X and Y move opposite):

|   |   |     |
|---|---|-----|
| X | Y | X+Y |
| X | Y | X+Y |

Sum has LESS variance than individual

Zero Covariance (Independent):

|   |    |              |
|---|----|--------------|
| X | Y? | X+Y moderate |
|---|----|--------------|

Sum has variance = sum of individual variances

## ML Connection

**Neural network example:** Two neurons in the same layer often have correlated activations. If you assume their variances add simply, you'll underestimate the total variance, leading to unstable training.

**Ensemble methods:** When combining models, if they're correlated, the variance reduction is less than expected. You need to account for covariance.

## BONUS MISTAKE 5 — Confusing $E[X]$ with “Most Likely Value”

### What Beginners Think

“Expectation is the value with highest probability.”

## What Is Actually True

**Expectation = weighted average**

**Mode = most likely value**

These are DIFFERENT concepts!

## Complete Numerical Example

**Distribution:**

| X   | P (X = x) |
|-----|-----------|
| 1   | 0.1       |
| 2   | 0.1       |
| 3   | 0.1       |
| 4   | 0.1       |
| 5   | 0.1       |
| 6   | 0.1       |
| 7   | 0.1       |
| 8   | 0.1       |
| 9   | 0.1       |
| 100 | 0.1       |

**Mode (most likely):** ALL values have equal probability (0.1). No single mode.

**Expectation:**

$$\checkmark = (1+2+3+4+5+6+7+8+9)(0.1) + 100(0.1)$$

$$= 45(0.1) + 10 = 4.5 + 10 = 14.5$$

**The value 100 is just as likely as 1, but it pulls the expectation WAY up.**

## Another Example

**Distribution:**

| X | P (X = x) |
|---|-----------|
| 1 | 0.6       |

|     |     |
|-----|-----|
| 100 | 0.4 |
|-----|-----|

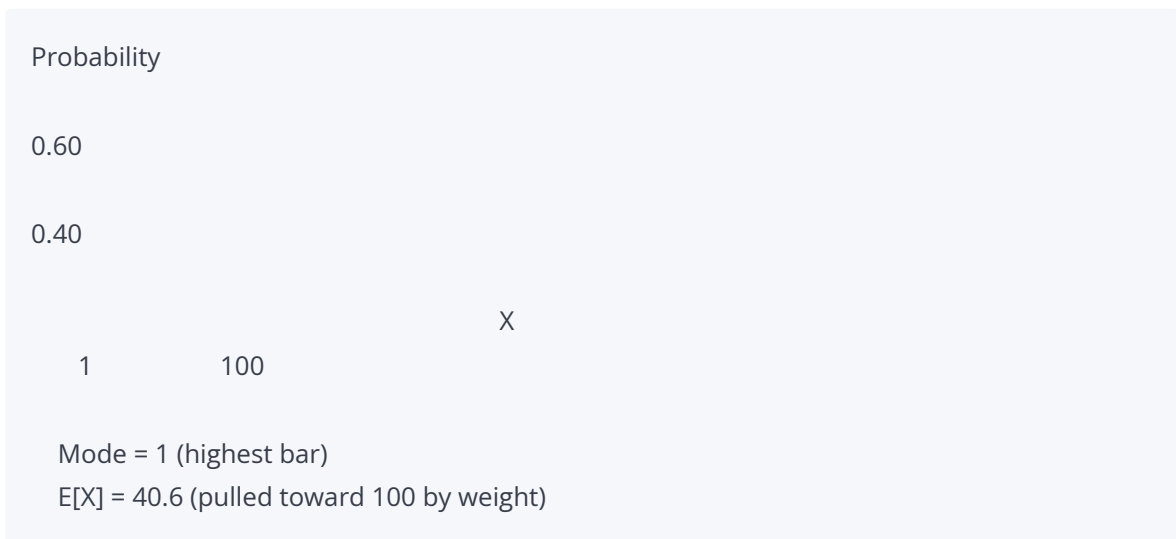
**Mode:** 1 (most likely, 60% chance)

**Expectation:**

$$\checkmark = 1(0.6) + 100(0.4) = 0.6 + 40 = 40.6$$

**The most likely value is 1, but the expectation is 40.6!**

## Visual



## BONUS MISTAKE 6 — Thinking $P(X = x) > 0$ for Continuous Variables

### What Beginners Think

*"The PDF  $f(x)$  gives the probability of  $X = x$ ."*

### What Is Actually True

**For continuous variables:**

$$P(X = \text{exact value}) = 0$$

**The PDF gives DENSITY, not probability. Probability requires an INTERVAL:**

$$P(a \leq X \leq b) = \int_a^b f(x)dx$$

### Complete Example

**PDF:**  $f(x) = 1$  for  $0 \leq x \leq 1$  (uniform)

**Wrong question:** "What is  $P(X = 0.5)$ ?"

**Wrong answer:** " $f(0.5) = 1$ , so probability is 1."

**Correct answer:** " $P(X = 0.5) = 0$ . The probability of EXACTLY 0.5 is zero."

**Correct question:** "What is  $P(0.4 \leq X \leq 0.6)$ ?"

**Correct answer:**  $\int_{0.4}^{0.6} 1 \, dx = 0.6 - 0.4 = 0.2$

## Visual

PDF  $f(x)$

1

x

0      0.5      1

$f(0.5) = 1$  (density at point)

But  $P(X=0.5) = 0$  (area of a line = 0)

$P(0.4 \leq X \leq 0.6) = \text{area of rectangle}$   
 $= 0.2 \times 1 = 0.2$

---

## HANDWRITTEN NOTES CHEAT SHEET

### Mistake 1: Expectation = Guaranteed

- **WRONG:**  $\checkmark = 2.5$  means "I'll get 2.5"
- **CORRECT:**  $\checkmark = 2.5$  means average over infinite trials is 2.5"
- **Single outcome:** Can be far from expectation

### Mistake 2: $E[X^2] = (E[X])^2$

- **WRONG:** Always equal
- **CORRECT:**  $E[X^2] \geq (E[X])^2$
- **Equal only when:**  $\text{Var}(X) = 0$  (no randomness)
- **Formula:**  $E[X^2] = \text{Var}(X) + (E[X])^2$

### Mistake 3: Forgetting $(E[X])^2$ in Variance

- **WRONG:**  $\text{Var}(X) = E[X^2]$
- **CORRECT:**  $\text{Var}(X) = E[X^2] - (E[X])^2$
- **Why subtract:** Remove “center effect,” keep only “spread”

### Mistake 4: Adding Variances Blindly

- **WRONG:**  $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$  always
- **CORRECT:**  $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$
- **Only when independent:**  $\text{Cov}(X, Y) = 0$ , so simple addition works

### Mistake 5: Expectation = Mode

- **WRONG:** “Most likely value = expectation”
- **CORRECT:** Mode = most likely, Expectation = weighted average
- **Can be very different:** See  $X \in \{1, 100\}$  example

### Mistake 6: PDF = Probability

- **WRONG:** “ $f(x)$  gives probability of  $X = x$ ”
- **CORRECT:** “ $f(x)$  gives density.  $P(X = x) = 0$  for continuous”
- **Probability:** Only defined for intervals (area under curve)

---

## MEMORY HOOK (Mistake Prevention)

### The “Always Check” List

Before submitting any probability calculation, ask:

| Check | Question                                              | If Wrong                     |
|-------|-------------------------------------------------------|------------------------------|
|       | Did I confuse expectation with a single outcome?      | Re-read definition           |
|       | Did I compute $E[X^2]$ and $(E[X])^2$ separately?     | Redo variance                |
|       | Did I subtract $(E[X])^2$ from $E[X^2]$ ?             | Fix variance formula         |
|       | Are my variables independent before adding variances? | Add covariance term          |
|       | Am I using PDF correctly (area, not height)?          | Use integral for probability |

---

## CONNECTIONS TO ADVANCED TOPICS

| Mistake                    | Why It Matters in Advanced ML                        |
|----------------------------|------------------------------------------------------|
| Expectation $\neq$ outcome | Understanding prediction uncertainty, calibration    |
| $E[X^2] \neq (E[X])^2$     | Correct loss computation, second-order optimization  |
| Variance subtraction       | Proper uncertainty quantification, Bayesian methods  |
| Covariance in variance sum | Feature correlation analysis, portfolio optimization |
| Mode vs expectation        | Decision theory, cost-sensitive learning             |
| PDF vs probability         | Kernel density estimation, generative models         |

## Section 10: PRACTICAL TIPS

*(Zero Hidden Meanings Edition — Real Code, Real Numbers)*

### TIP 1: STANDARDIZING DATA (Z-SCORES)

#### The Problem (With Real Numbers)

You have a dataset with three features:

| Person | Age | Income (\$) | Pixel Value |
|--------|-----|-------------|-------------|
| A      | 25  | 50,000      | 200         |
| B      | 45  | 120,000     | 50          |
| C      | 30  | 80,000      | 180         |

Raw data:

Age: 25, 45, 30 range: 20  
Income: 50k, 120k, 80k range: 70,000  
Pixel: 200, 50, 180 range: 150

What gradient descent sees:

- Income numbers are 1000× bigger than age
- Gradient for income dominates

- Age becomes irrelevant
- **Training breaks or becomes very slow**

## The Solution: Z-Score Standardization

Formula:

$$Z = \frac{X - \mu}{\sigma}$$

What each symbol means:

| Symbol   | What It Is                    | How to Calculate                                 |
|----------|-------------------------------|--------------------------------------------------|
| X        | Original raw value            | Your data point                                  |
| $\mu$    | Mean of the feature           | $\mu = \frac{1}{n} \sum X_i$                     |
| $\sigma$ | Standard deviation of feature | $\sigma = \sqrt{\frac{1}{n} \sum (X_i - \mu)^2}$ |
| Z        | Standardized value            | Result after transformation                      |

## Complete Numerical Example: Standardizing Age

**Data:** Ages = {25, 45, 30}

**Step 1: Calculate mean**

$$\mu = \frac{25 + 45 + 30}{3} = \frac{100}{3} \approx 33.33$$

**Step 2: Calculate variance**

$$\begin{aligned} \text{Var}(X) &= E[X^2] - (E[X])^2 \\ E[X^2] &= \frac{25^2 + 45^2 + 30^2}{3} = \frac{625 + 2025 + 900}{3} = \frac{3550}{3} \approx 1183.33 \\ (E[X])^2 &= (33.33)^2 \approx 1111.11 \\ \text{Var}(X) &= 1183.33 - 1111.11 = 72.22 \end{aligned}$$

**Step 3: Calculate standard deviation**

$$\sigma = \sqrt{72.22} \approx 8.50$$

**Step 4: Standardize each value**

| Original X | $X - \mu$            | $Z = \frac{X - \mu}{\sigma}$       |
|------------|----------------------|------------------------------------|
| 25         | $25 - 33.33 = -8.33$ | $\frac{-8.33}{8.50} \approx -0.98$ |

|    |                      |                                    |
|----|----------------------|------------------------------------|
| 45 | $45 - 33.33 = 11.67$ | $\frac{11.67}{8.50} \approx 1.37$  |
| 30 | $30 - 33.33 = -3.33$ | $\frac{-3.33}{8.50} \approx -0.39$ |

**Result:** Standardized ages = {-0.98, 1.37, -0.39}

## Complete Numerical Example: Standardizing Income

**Data:** Incomes = {50000, 120000, 80000}

**Step 1: Calculate mean**

$$\mu = \frac{50000 + 120000 + 80000}{3} = \frac{250000}{3} \approx 83333.33$$

**Step 2: Calculate variance**

$$E[X^2] = \frac{50000^2 + 120000^2 + 80000^2}{3} = \frac{2.5 \times 10^9 + 14.4 \times 10^9 + 6.4 \times 10^9}{3}$$

$$= \frac{23.3 \times 10^9}{3} \approx 7.77 \times 10^9$$

$$(E[X])^2 = (83333.33)^2 \approx 6.94 \times 10^9$$

$$\text{Var}(X) = 7.77 \times 10^9 - 6.94 \times 10^9 = 0.83 \times 10^9 = 830,000,000$$

**Step 3: Calculate standard deviation**

$$\sigma = \sqrt{830,000,000} \approx 28,809$$

**Step 4: Standardize each value**

| Original X | $X - \mu$                | $Z = \frac{X - \mu}{\sigma}$         |
|------------|--------------------------|--------------------------------------|
| 50000      | $50000 - 83333 = -33333$ | $\frac{-33333}{28809} \approx -1.16$ |
| 120000     | $120000 - 83333 = 36667$ | $\frac{36667}{28809} \approx 1.27$   |
| 80000      | $80000 - 83333 = -3333$  | $\frac{-3333}{28809} \approx -0.12$  |

**Result:** Standardized incomes = {-1.16, 1.27, -0.12}

## Before vs After Comparison

| Feature | Before Standardization | After Standardization |
|---------|------------------------|-----------------------|
| Age     | 25, 45, 30             | -0.98, 1.37, -0.39    |
| Income  | 50000, 120000, 80000   | -1.16, 1.27, -0.12    |



|       |              |                       |
|-------|--------------|-----------------------|
| Pixel | 200, 50, 180 | (calculate similarly) |
|-------|--------------|-----------------------|

**Key observation:** After standardization, ALL features have:

- Mean  $\approx 0$
- Standard deviation  $\approx 1$
- Same numerical scale

## Visual: Why Standardization Matters

BEFORE (raw data):

Age:

0   25   30   45   100

Income:

0   50k   80k   120k   1M

Pixel:

0   50   180   255

Different scales! Income dominates.

AFTER (standardized):

Age:

-1   0   1

Income:

-1   0   1

Pixel:

-1   0   1

Same scale! All features contribute fairly.

## Why This Works (The Math)

**After standardization:**

☒  $E[Z] = \frac{\sum (X_i - \mu)}{n} = 0$

$$\text{Var}(Z) = \text{Var}\left(\frac{X - \mu}{\sigma}\right) = \frac{1}{\sigma^2} \text{Var}(X) = \frac{\sigma^2}{\sigma^2} = 1$$

**Guaranteed result:**  $E[Z] = 0$  and  $\text{Var}(Z) = 1$

---

## TIP 2: WHAT “MINIMIZING VARIANCE” ACTUALLY MEANS IN ML

### The Real Problem

**High variance model behavior:**

Training input: Image of cat (slightly rotated)  
Prediction 1: 0.99 (cat)  
Prediction 2: 0.45 (not sure)  
Prediction 3: 0.01 (definitely not cat)

**Same image, slightly different    completely different predictions.**  
**This is high variance = unstable = overfitting.**

### Low Variance Model Behavior

Training input: Image of cat (slightly rotated)  
Prediction 1: 0.95 (cat)  
Prediction 2: 0.93 (cat)  
Prediction 3: 0.94 (cat)

**Same image, slightly different    similar predictions.**  
**This is low variance = stable = generalizes.**

### Complete Example: Two Linear Models

**True relationship:**  $y = 2x + 1 + \text{noise}$

**Training data (5 points):**

| x | y    |
|---|------|
| 1 | 3.2  |
| 2 | 4.8  |
| 3 | 7.1  |
| 4 | 8.9  |
| 5 | 11.2 |

**Model A (High Variance — Overfit):**

- Fits a wiggly curve through every point
- Equation:  $y = 0.1x^3 - 0.5x^2 + 3.2x - 0.8$

#### Model B (Low Variance — Simple):

- Fits a straight line
- Equation:  $y = 2.01x + 0.95$

#### Test on new point $x = 6$ :

- True value:  $y \approx 13$
- Model A predicts:  
 $y = 0.1(216) - 0.5(36) + 3.2(6) - 0.8 = 21.6 - 18 + 19.2 - 0.8 = 22$  WAY OFF
- Model B predicts:  $y = 2.01(6) + 0.95 = 12.06 + 0.95 = 13.01$  CLOSE

**Model A has low bias (fits training perfectly) but high variance (unstable on new data).**

**Model B has slightly higher bias but much lower variance (generalizes better).**

## What Researchers Actually Do to Reduce Variance

| Technique                | How It Works                      | Effect on Variance                                          |
|--------------------------|-----------------------------------|-------------------------------------------------------------|
| <b>L2 Regularization</b> | Penalizes large weights           | Pulls weights toward zero   simpler model<br>lower variance |
| <b>Dropout</b>           | Randomly disables neurons         | Prevents co-adaptation   more robust<br>lower variance      |
| <b>Data Augmentation</b> | Creates artificial training data  | More diverse training   less memorization<br>lower variance |
| <b>Early Stopping</b>    | Stops training before overfitting | Less time to memorize noise   lower variance                |
| <b>Ensemble Methods</b>  | Average multiple models           | Errors cancel out   lower variance                          |

## Complete L2 Regularization Example

**Without regularization:**

$$\text{Loss} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

**With L2 regularization:**

$$\text{Loss} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^m w_j^2$$

**The extra term  $\lambda \sum w_j^2$ :**

- Penalizes large weights
- Forces model to use smaller weights
- Smaller weights → simpler function → lower variance

**Numerical example:**

- Without regularization: weights = {10, -8, 15, 3} → complex model
  - With regularization ( $\lambda = 0.1$ ): weights = {2, -1, 3, 0.5} → simpler model
- 

## TIP 3: COMPUTATIONAL SHORTCUT (The Fast Way)

### The Slow Way (Multiple Passes)

**Algorithm:**

1. First pass: compute mean  $\mu$
2. Second pass: compute  $(X_i - \mu)^2$  for each point
3. Third pass: average the squared deviations

**For 1 billion data points:** 3 passes = very slow.

### The Fast Way (Two Passes)

**Formula:**

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

**Algorithm:**

1. First pass: compute  $E[X]$  (sum all values, divide by n)
2. Second pass: compute  $E[X^2]$  (sum all squared values, divide by n)
3. Subtract:  $\text{Var}(X) = E[X^2] - (E[X])^2$

**For 1 billion data points:** 2 passes = 33% faster.

### Complete Numerical Example

**Data:** {2, 4, 6, 8, 10}

**Slow method:**

- $\mu = \frac{2+4+6+8+10}{5} = 6$
- Deviations: {-4, -2, 0, 2, 4}
- Squared deviations: {16, 4, 0, 4, 16}
- Variance:  $\frac{16+4+0+4+16}{5} = \frac{40}{5} = 8$

**Fast method:**

- $\checkmark = 6$
- $E[X^2] = \frac{4+16+36+64+100}{5} = \frac{220}{5} = 44$
- $(E[X])^2 = 36$
- $\text{Var}(X) = 44 - 36 = 8$

Same answer, fewer operations.

**Why NumPy/PyTorch Use This**

```
# NumPy implementation (simplified)
def variance_fast(data):
    n = len(data)
    mean = np.sum(data) / n      # First pass
    mean_sq = np.sum(data**2) / n # Second pass (vectorized!)
    return mean_sq - mean**2

# vs slow method
def variance_slow(data):
    mean = np.sum(data) / n
    deviations = data - mean     # Extra memory allocation
    squared_dev = deviations ** 2 # Extra computation
    return np.sum(squared_dev) / n
```

**The fast method:**

- Uses vectorized operations (data\*\*2 is fast)
- No intermediate arrays (deviations, squared\_dev)
- Cache-friendly (sequential memory access)

**TIP 4: UNIT CHECKING (Critical for Real Engineering)**

**Complete Unit Analysis**

| Quantity       | Formula                | Units               | Example                          |
|----------------|------------------------|---------------------|----------------------------------|
| X              | Raw data               | meters              | Height = 1.7 m                   |
| $E[X]$         | $\frac{1}{n} \sum X_i$ | meters              | Average height = 1.75 m          |
| $X - E[X]$     | Deviation              | meters              | $1.7 - 1.75 = -0.05 \text{ m}$   |
| $(X - E[X])^2$ | Squared deviation      | meters <sup>2</sup> | $(-0.05)^2 = 0.0025 \text{ m}^2$ |

|                                 |                               |                     |                     |
|---------------------------------|-------------------------------|---------------------|---------------------|
| $\text{Var}(X)$                 | Average of squared deviations | meters <sup>2</sup> | 0.01 m <sup>2</sup> |
| $\sigma = \sqrt{\text{Var}(X)}$ | Standard deviation            | meters              | 0.1 m               |

## Why Variance in Squared Units Is Confusing

**Example:** Human heights

- Mean: 1.75 meters
- Variance: 0.01 m<sup>2</sup>
- Standard deviation: 0.1 m

**Interpretation:**

- “Average height is 1.75 meters” Clear
- “Variance is 0.01 square meters” What does that mean?
- “Typical person is 0.1 meters from average” Clear!

**This is why we report standard deviation, not variance, in practice.**

## Real-World Reporting

| Field    | What They Report          | Why                                       |
|----------|---------------------------|-------------------------------------------|
| Medicine | “Mean ± SD”               | Standard deviation in original units      |
| Finance  | “Volatility”              | Usually standard deviation of returns     |
| Weather  | “Temperature: 25°C ± 2°C” | Standard deviation                        |
| ML       | “Accuracy: 85% ± 3%”      | Standard deviation or confidence interval |

**Nobody says:** “The variance of temperature is 4 degrees squared.”

**Everybody says:** “The standard deviation is 2 degrees.”

# HANDWRITTEN NOTES CHEAT SHEET

## Standardization (Z-Scores)

- **Formula:**  $Z = \frac{x - \mu}{\sigma}$
- **Result:**  $E[Z] = 0, \text{Var}(Z) = 1$
- **Why:** Makes all features comparable
- **When:** Before feeding data to neural networks, SVMs, k-means

## Minimizing Variance in ML

- **Meaning:** Make predictions stable, not sensitive to noise
- **High variance:** Overfitting, memorizes training data
- **Low variance:** Generalizes well to new data
- **Techniques:** L2 regularization, dropout, data augmentation, ensembles

## Computational Shortcut

- **Fast formula:**  $\text{Var}(X) = E[X^2] - (E[X])^2$
- **Why faster:** 2 passes instead of 3
- **Used in:** NumPy, PyTorch, TensorFlow
- **Key:** Vectorizable, cache-friendly

## Unit Checking

- **Expectation:** Same units as data (meters, dollars, etc.)
  - **Variance:** Squared units (meters<sup>2</sup>, dollars<sup>2</sup>)
  - **Standard deviation:** Original units (meters, dollars)
  - **Report:** Use standard deviation for communication
- 

## MEMORY HOOKS

### Standardization = Translating Languages

- Before: Features speak different languages (Age speaks “years,” Income speaks “dollars”)
- After: All features speak “standard deviations from mean”
- Result: Gradient descent can understand everyone

### Variance = Fragility

- High variance = Fragile crystal vase (breaks easily)
- Low variance = Durable plastic cup (handles bumps)
- Minimizing variance = Making your model more durable

### Computational Shortcut = Two-Step Recipe

- Instead of: measure   subtract mean   square   average (3 steps)
- Use: measure   square   average   subtract (2 steps)
- Same cake, faster baking

### Units = Dress Code

- Variance shows up in a ball gown (fancy squared units)
  - Standard deviation shows up in casual clothes (original units)
  - You can understand casual clothes better
- 

## CONNECTIONS TO ADVANCED TOPICS

| Practical Tip           | Leads To                                      |
|-------------------------|-----------------------------------------------|
| Z-score standardization | Batch normalization, layer normalization      |
| Variance minimization   | Bias-variance tradeoff, regularization theory |
| Computational shortcut  | Online algorithms, streaming statistics       |
| Unit checking           | Dimensional analysis, physics-informed ML     |

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## Section 11: MINI QUIZ — VERIFIED SOLUTIONS

*(Zero Hidden Meanings Edition — Every Answer Fully Explained)*

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### HOW TO USE THIS QUIZ

**Don't just read the answers.** Cover the solution, try to solve it yourself, then check. Each question tests ONE specific concept from the previous sections.

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### Q1 — What Is a Random Variable?

#### Question

What is a Random Variable mapping?

#### Options:

- (A) Numbers   probability
  - (B) Physical events   numbers
  - © Variance   expectation
- 

**Correct Answer: (B) Physical events   numbers**



## Complete Explanation

A random variable is a **FUNCTION**:

$$X : \Omega \rightarrow \mathbb{R}$$

What this means:

- **Input:** Something from the real world ( Heads, Tails, rain, sun)
- **Output:** A number (1, 0, 1, 0)

It's **NOT**:

- A random number by itself
- A probability
- A variance or expectation

## Complete Example

**Experiment:** Flip a coin

| Physical Event ( $\Omega$ ) | Random Variable X | Output |
|-----------------------------|-------------------|--------|
| Heads                       | X(Heads)          | 1      |
| Tails                       | X(Tails)          | 0      |

The random variable is the **RULE** “assign 1 to Heads, 0 to Tails.”

## Why Other Answers Are Wrong

| Option                         | Why It's Wrong                                                                                                   |
|--------------------------------|------------------------------------------------------------------------------------------------------------------|
| <b>(A) Numbers probability</b> | That's what the PMF/PDF does, not the random variable. The random variable comes BEFORE probability is assigned. |
| <b>© Variance expectation</b>  | These are both properties OF a random variable, not what a random variable IS.                                   |

## Visual

| WRONG (A): |             | CORRECT (B): |         |
|------------|-------------|--------------|---------|
| Numbers    | Probability | Physical     | Numbers |
| 1          | 0.5         | Heads        | 1       |
| 2          | 0.3         | Tails        | 0       |
| 3          | 0.2         | Rain         | 1       |

This is PMF!

This is Random Variable!

## Q2 — What Does Variance = 0 Mean?

### Question

If variance is exactly 0, what does it mean?

**Answer: The system is completely deterministic — no randomness at all.**

### Complete Explanation

From the definition:

$$\text{Var}(X) = E[(X - \mu)^2] = 0$$

For this to equal 0:

- $(X - \mu)^2 = 0$  for ALL possible outcomes
- Therefore  $X = \mu$  for ALL possible outcomes
- Therefore  $X$  is ALWAYS the same value

### Complete Example

Distribution:

| X             | P (X = x) |
|---------------|-----------|
| 5             | 1.0       |
| anything else | 0.0       |

Properties:

- ☒ = 5
- $\text{Var}(X) = (5 - 5)^2(1.0) = 0$
- $\sigma = 0$

Visual:

Probability

1

5

X

Single spike at 5  
No spread whatsoever  
100% certainty

## ML Interpretation

| Aspect          | Meaning                                       |
|-----------------|-----------------------------------------------|
| Good side       | Predictions are perfectly stable              |
| Bad side        | Model has no flexibility                      |
| When it happens | Underfitting (model too simple)               |
| Real example    | Model that always predicts the majority class |

## Q3 — Why Do We Square Distances in Variance?

### Question

Why is variance defined using squared distance?

**Answer: Two reasons: (1) Prevent cancellation, (2) Amplify large deviations**

### Complete Explanation with Numbers

**Reason 1: Prevent Cancellation**

**Without squaring:**

| X | $X - \mu$    | Deviation |
|---|--------------|-----------|
| 1 | $1 - 3 = -2$ | -2        |

|   |             |    |
|---|-------------|----|
| 5 | $5 - 3 = 2$ | +2 |
|---|-------------|----|

Average deviation:  $\frac{-2+2}{2} = 0$

**Wrong conclusion:** "No spread!" (but there IS spread)

**With squaring:**

| X | $(X - \mu)^2$ | Squared Deviation |
|---|---------------|-------------------|
| 1 | $(-2)^2$      | 4                 |
| 5 | $(2)^2$       | 4                 |

Average squared deviation:  $\frac{4+4}{2} = 4$

**Correct conclusion:** "There IS spread!"

**Reason 2: Amplify Large Deviations**

| Deviation | Squared Deviation | Amplification |
|-----------|-------------------|---------------|
| 1         | 1                 | 1×            |
| 2         | 4                 | 4×            |
| 3         | 9                 | 9×            |
| 10        | 100               | 100×          |

**A deviation of 10 is penalized 100× more than a deviation of 1.**

**Why this matters in ML:** Large errors are much worse than small errors. Squaring makes the model care more about big mistakes.

## Visual Comparison

Without squaring (WRONG):

-3 -2 -1 0 1 2 3

-2 +2 cancel 0 (no spread detected)

With squaring (CORRECT):

0 1 4 9 16 25 36

4 4 average 4 (spread detected!)

## Q4 — Find $E[X^2]$ Given $E[X]$ and $\text{Var}(X)$

### Question

Given:

- $\checkmark = 5$
- $\text{Var}(X) = 2$

Find  $E[X^2]$ .

---

**Answer: 27**

### Complete Step-by-Step Solution

**Step 1:** Write the variance shortcut formula

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

**Step 2:** Substitute known values

$$2 = E[X^2] - (5)^2$$

**Step 3:** Calculate  $(E[X])^2$

$$(5)^2 = 25$$

**Step 4:** Solve for  $E[X^2]$

$$2 = E[X^2] - 25$$

$$E[X^2] = 2 + 25$$

$$E[X^2] = 27$$

### Verification

If  $E[X^2] = 27$  and  $\checkmark = 5$ :

$$\text{Var}(X) = 27 - 25 = 2$$

### What $E[X^2]$ Actually Represents

| Term            | Value | Meaning                          |
|-----------------|-------|----------------------------------|
| $E[X]$          | 5     | Center of distribution           |
| $(E[X])^2$      | 25    | "Center energy"                  |
| $E[X^2]$        | 27    | "Total energy" (center + spread) |
| $\text{Var}(X)$ | 2     | "Pure spread" (total - center)   |

#### Visual:

$$E[X^2] = 27$$

Center effect =  $(E[X])^2 = 25$   
(squared mean)

Spread =  $\text{Var}(X) = 2$   
(variance)

0    25    27

## Q5 — Expectation of {10, 20}

### Question

Given:

- Values: {10, 20}
- Probabilities: {0.5, 0.5}

Find  $E[X]$ .

**Answer: 15**

### Complete Step-by-Step Solution

**Step 1:** Write the formula

✓  $= \sum x_i P(X=x_i)$

**Step 2:** Expand for both values

$$\checkmark = 10 \cdot P(X=10) + 20 \cdot P(X=20)$$

**Step 3:** Substitute probabilities

$$\checkmark = 10(0.5) + 20(0.5)$$

**Step 4:** Calculate each term

- $10 \times 0.5 = 5$
- $20 \times 0.5 = 10$

**Step 5:** Add

$$\checkmark = 5 + 10 = 15$$

## Key Insight

| Observation                  | Explanation                                          |
|------------------------------|------------------------------------------------------|
| 15 is not in {10, 20}        | Expectation is NOT required to be a possible outcome |
| 15 is exactly halfway        | Because probabilities are equal (0.5 each)           |
| If $P(10)=0.8$ , $P(20)=0.2$ | Then $\checkmark = 8 + 4 = 12$ (pulled toward 10)    |

**Visual balance:**

|               |                     |     |
|---------------|---------------------|-----|
| 10            | 15                  | 20  |
| 0.5           |                     | 0.5 |
| Equal weights | balance at midpoint |     |

## Q6 — Scaling by 3

### Question

What happens if we multiply a dataset by 3?

**Answer: Expectation  $\times 3$ , Variance  $\times 9$**

### Complete Explanation with Proof

**Given:** Dataset with  $E[X]$  and  $\text{Var}(X)$

**Define:**  $Y = 3X$

**Effect on Expectation:**

**Step 1:** Use linearity

$$E[Y] = E[3X] = 3E[X]$$

**Result:** Expectation is multiplied by 3

**Effect on Variance:**

**Step 1:** Use variance scaling property

$$\text{Var}(Y) = \text{Var}(3X) = 3^2 \cdot \text{Var}(X) = 9 \cdot \text{Var}(X)$$

**Result:** Variance is multiplied by 9

## Complete Numerical Example

**Original data:** {1, 3, 5}

| Property        | Calculation        | Value |
|-----------------|--------------------|-------|
| $E[X]$          | $\frac{1+3+5}{3}$  | 3     |
| $E[X^2]$        | $\frac{1+9+25}{3}$ | 11.67 |
| $\text{Var}(X)$ | $11.67 - 9$        | 2.67  |

**Scaled data:** {3, 9, 15} (each  $\times 3$ )

| Property         | Calculation          | Value                  |
|------------------|----------------------|------------------------|
| $E[3X]$          | $\frac{3+9+15}{3}$   | 9 (= 3 $\times$ 3)     |
| $E[(3X)^2]$      | $\frac{9+81+225}{3}$ | 105                    |
| $\text{Var}(3X)$ | $105 - 81$           | 24 (= 9 $\times$ 2.67) |

**Check:**  $24 = 9 \times 2.67$

## Visual

|           |                        |
|-----------|------------------------|
| Original: | Scaled ( $\times 3$ ): |
| 5         | 15                     |
| 3         | 9                      |



1

3

Spread: 2 units

Spread: 6 units ( $\times 3$ )

Variance: 2.67

Variance: 24 ( $\times 9$ )

## Why Variance Scales by $a^2$ (Not $a$ )

Variance measures squared distances:

- Original distance:  $X - \mu$
- Scaled distance:  $3X - 3\mu = 3(X - \mu)$
- Squared scaled distance:  $[3(X - \mu)]^2 = 9(X - \mu)^2$

The “3” gets squared because variance already involves squaring.

## Q7 — PMF vs PDF

### Question

What is the difference between PMF and PDF?

**Answer: PMF gives exact probabilities for discrete variables. PDF gives density for continuous variables — probability requires integration (area).**

### Complete Side-by-Side Comparison

| Feature                     | PMF (Discrete)            | PDF (Continuous)             |
|-----------------------------|---------------------------|------------------------------|
| Full name                   | Probability Mass Function | Probability Density Function |
| Used for                    | Discrete random variables | Continuous random variables  |
| Notation                    | $P(X = x)$                | $f(x)$                       |
| Gives probability directly? | <b>YES</b>                | <b>NO</b>                    |
| How to get probability      | Read the value            | Calculate area under curve   |
| $P(X = \text{exact value})$ | Can be $> 0$              | Always = 0                   |
| Total                       | $\sum P(X = x) = 1$       | $\int f(x)dx = 1$            |

|        |                  |              |
|--------|------------------|--------------|
| Visual | Bars (histogram) | Smooth curve |
|--------|------------------|--------------|

## Complete PMF Example

Fair die:

| X | P (X = x) |
|---|-----------|
| 1 | 1/6       |
| 2 | 1/6       |
| 3 | 1/6       |
| 4 | 1/6       |
| 5 | 1/6       |
| 6 | 1/6       |

$P(X = 4) = \frac{1}{6}$  Exact probability, read directly

## Complete PDF Example

Uniform [0, 1]:

$$f(x) = \begin{cases} 1 & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

$f(0.5) = 1$  This is DENSITY, not probability!

$P(X = 0.5) = 0$  Probability of exact point is ZERO

$P(0.4 \leq X \leq 0.6) = \int_{0.4}^{0.6} 1 \, dx = 0.2$  Probability of interval

## Visual Comparison

PMF (Discrete):

PDF (Continuous):

0.17

1 2 3 4 5 6

Bars = exact probabilities    Curve = density

$$P(X=3) = 0.17$$

$$f(0.5) = 1 \text{ (not probability!)}$$

$$P(0.4 \leq X \leq 0.6) = \text{area} = 0.2$$

## BONUS PRACTICE PROBLEMS

### Problem 1: Find Variance from Raw Data

**Data:** {2, 4, 6, 8, 10}

**Solution:**

- $\bar{x} = \frac{2+4+6+8+10}{5} = 6$
- $E[X^2] = \frac{4+16+36+64+100}{5} = 44$
- $\text{Var}(X) = 44 - 36 = 8$
- $\sigma = \sqrt{8} \approx 2.83$

### Problem 2: Effect of Adding a Constant

**Given:**  $\bar{x} = 10$ ,  $\text{Var}(X) = 4$

**Find:**  $E[X + 5]$  and  $\text{Var}(X + 5)$

**Solution:**

- $\bar{x} + 5 = 15$  (shifts mean)
- $\text{Var}(X + 5) = \text{Var}(X) = 4$  (spread unchanged!)

### Problem 3: Biased Coin

**Coin:**  $P(\text{Heads}) = 0.7$ ,  $P(\text{Tails}) = 0.3$

**Random Variable:**  $X = 1$  if Heads,  $0$  if Tails

**Find:**  $E[X]$  and  $\text{Var}(X)$

**Solution:**

- $\bar{x} = 1(0.7) + 0(0.3) = 0.7$
- $E[X^2] = 1^2(0.7) + 0^2(0.3) = 0.7$
- $\text{Var}(X) = 0.7 - (0.7)^2 = 0.7 - 0.49 = 0.21$

### Problem 4: Sum of Independent Variables

**Given:**  $X$  and  $Y$  independent,  $\text{Var}(X) = 3$ ,  $\text{Var}(Y) = 5$

**Find:**  $\text{Var}(X + Y)$

**Solution:**

- $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) = 3 + 5 = 8$  (because independent, covariance = 0)

### Problem 5: Sum of Dependent Variables

**Given:**  $Y = 2X$ ,  $\text{Var}(X) = 3$

**Find:**  $\text{Var}(X + Y)$

**Solution:**

- $\text{Var}(X + Y) = \text{Var}(X + 2X) = \text{Var}(3X) = 9\text{Var}(X) = 27$
  - (NOT  $3 + 12 = 15$ !)
- 

## HANDWRITTEN NOTES CHEAT SHEET

### Q1: Random Variable

- **Answer:** (B) Physical events   numbers
- **Key:** It's a FUNCTION, not a number itself

### Q2: Variance = 0

- **Answer:** Completely deterministic, no randomness
- **Key:**  $X = \mu$  always, single point mass

### Q3: Why Square?

- **Answer:** Prevent cancellation + amplify large deviations
- **Key:**  $(-2)^2 + 2^2 = 8$  (not 0)

### Q4: Find $E[X^2]$

- **Formula:**  $\text{Var}(X) = E[X^2] - (E[X])^2$
- **Solution:**  $E[X^2] = \text{Var}(X) + (E[X])^2 = 2 + 25 = 27$

### Q5: Expectation

- **Formula:**  $\sum x \cdot P(x) = 10(0.5) + 20(0.5) = 15$
- **Key:** Not required to be a possible outcome

### Q6: Scaling

- **Expectation:**  $E[aX] = aE[X]$  (linear)
- **Variance:**  $\text{Var}(aX) = a^2\text{Var}(X)$  (quadratic)

## Q7: PMF vs PDF

- **PMF:** Discrete, exact probability,  $P(X = x) > 0$  possible
- **PDF:** Continuous, density only,  $P(X = x) = 0$  always

## MEMORY HOOK (Quiz Review)

| Question                | One-Line Answer                      |
|-------------------------|--------------------------------------|
| What is RV?             | Function: reality → numbers          |
| Var = 0?                | No randomness, ever                  |
| Why square?             | Stop cancellation, punish outliers   |
| Find $E[X^2]$ ?         | Use $\text{Var} = E[X^2] - (E[X])^2$ |
| Expectation of {10,20}? | Balance point = 15                   |
| Scale by 3?             | Mean ×3, variance ×9                 |
| PMF vs PDF?             | Discrete exact vs continuous density |

## CONNECTIONS TO ADVANCED TOPICS

| Quiz Concept                | Leads To                              |
|-----------------------------|---------------------------------------|
| Random variable as function | Measure theory, sigma algebras        |
| Variance = 0                | Dirac delta, degenerate distributions |
| Squaring in variance        | $L_2$ norm, least squares, MSE        |
| Scaling properties          | Feature engineering, normalization    |
| PMF vs PDF                  | Discrete vs continuous optimization   |
| $E[X^2]$ calculation        | Moment generating functions           |

# Section 12: MASTER CHEAT SHEET: RANDOM VARIABLES, EXPECTATION & VARIANCE

(Complete Structural Understanding — No Memorization Required)

## THE ONE SENTENCE THAT EXPLAINS EVERYTHING

Probability theory is the study of how uncertainty behaves when you turn it into numbers, find its center, and measure its spread.

Everything else is just details of this sentence.

## 1. RANDOM VARIABLE

### The Formula

$$X : \Omega \rightarrow \mathbb{R}$$

### What Each Symbol Means

| Symbol        | Name            | What It Actually Is                             |
|---------------|-----------------|-------------------------------------------------|
| $X$           | Random Variable | The <b>rule</b> that converts events to numbers |
| $\Omega$      | Sample Space    | The bag of all possible physical outcomes       |
| $\mathbb{R}$  | Real Numbers    | The number line                                 |
| $\rightarrow$ | "Maps to"       | "Takes input and gives output"                  |

### What It Actually Does

| PHYSICAL WORLD | MATH WORLD               |
|----------------|--------------------------|
| Heads          | Random Variable X      1 |
| Tails          |                          |
|                | 0                        |
| Rain           | Random Variable Y      1 |
|                |                          |

|           |                                                      |
|-----------|------------------------------------------------------|
| Sun       | 0                                                    |
| Cat image | Random Variable Z [0.2, 0.1, 0.7]<br>(probabilities) |
| Dog image |                                                      |

## Why It Exists

**Math only works with numbers.** Reality gives us words, objects, and events. Random variables are the **translator**.

## Complete Example

**Experiment:** Roll two dice. Let  $X$  = sum of the two dice.

| Physical Outcome | $X$ (Sum) |
|------------------|-----------|
| (1,1)            | 2         |
| (1,2)            | 3         |
| ...              | ...       |
| (6,6)            | 12        |

$X$  is **NOT** the dice.  $X$  is the **rule** “add the two numbers.”

## ML Connection

| ML Task                | Random Variable | What It Represents         |
|------------------------|-----------------|----------------------------|
| Image classification   | $Y$             | Class label (0=cat, 1=dog) |
| Regression             | $\hat{Y}$       | Predicted house price      |
| Language model         | $W$             | Next word token            |
| Reinforcement learning | $R$             | Reward signal              |

## 2. EXPECTATION (DISCRETE)

### The Formula

$$E[X] = \sum_i x_i \cdot P(X = x_i)$$

## What Each Symbol Means

| Symbol       | What It Is                       | Example                                                |
|--------------|----------------------------------|--------------------------------------------------------|
| $E[X]$       | The expected value we're finding | <input checked="" type="checkbox"/> = 3.5 for fair die |
| $\sum$       | "Add up all of these"            | Sum over all outcomes                                  |
| $x_i$        | The $i$ -th possible value       | $x_1 = 1, x_2 = 2, \dots$                              |
| $P(X = x_i)$ | Probability of that value        | $P(X = 4) = \frac{1}{6}$                               |

## What It Actually Computes

The "balance point" of probability mass.

If probability were physical weight placed on a number line,  $E[X]$  is where the line would balance.

## Complete Example: Fair Die

**Step 1:** List values and probabilities

| $x_i$ | $P(X = x_i)$  | $x_i \cdot P(X = x_i)$ |
|-------|---------------|------------------------|
| 1     | $\frac{1}{6}$ | $\frac{1}{6}$          |
| 2     | $\frac{1}{6}$ | $\frac{2}{6}$          |
| 3     | $\frac{1}{6}$ | $\frac{3}{6}$          |
| 4     | $\frac{1}{6}$ | $\frac{4}{6}$          |
| 5     | $\frac{1}{6}$ | $\frac{5}{6}$          |
| 6     | $\frac{1}{6}$ | $\frac{6}{6}$          |

**Step 2:** Add the weighted values

$$\text{✓} = \frac{1+2+3+4+5+6}{6} = \frac{21}{6} = 3.5$$

**Critical insight:** You can NEVER roll 3.5. But if you roll a million times, your average will be very close to 3.5.

## Visual (Balance Point)

1   2   3   4   5   6



1/6 1/6 1/6 1/6 1/6 1/6 Equal weights

$E[X]=3.5$   
(Balance point)

## ML Connection

| ML Concept         | Expectation Role                 |
|--------------------|----------------------------------|
| Loss function      | $E[\text{Loss}]$ = average error |
| Risk minimization  | Minimize expected loss           |
| Policy gradient    | Maximize expected reward         |
| Bayesian inference | Expected posterior value         |

## 3. EXPECTATION (CONTINUOUS)

### The Formula

$$E[X] = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

### What Each Symbol Means

| Symbol                | What It Is     | Analogy                 |
|-----------------------|----------------|-------------------------|
| $\int$                | Integration    | "Continuous $\sum$ "    |
| $-\infty$ to $\infty$ | Limits         | "All possible values"   |
| $x$                   | Variable value | Current point           |
| $f(x)$                | PDF            | "Density at point $x$ " |
| $dx$                  | Tiny interval  | "Infinitesimal slice"   |

### Why This Formula Exists

**Continuous variables have infinitely many values.** You can't list them all. Integration is "adding up infinitely many tiny pieces."

### Complete Example: Uniform [0, 2]

**PDF:**  $f(x) = \frac{1}{2}$  for  $0 \leq x \leq 2$

**Step 1:** Set up integral

$$\checkmark = \int_0^2 x \cdot \frac{1}{2} \, dx$$

**Step 2:** Pull out constant

$$= \frac{1}{2} \int_0^2 x \, dx$$

**Step 3:** Integrate ( $\int x \, dx = \frac{x^2}{2}$ )

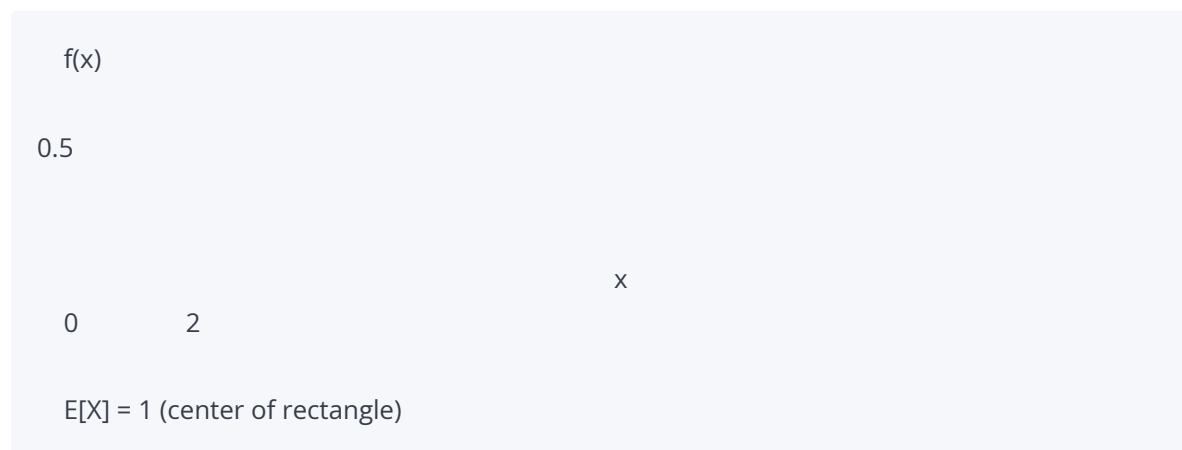
$$= \frac{1}{2} \left[ \frac{x^2}{2} \right]_0^2$$

**Step 4:** Evaluate

$$= \frac{1}{2} \left( \frac{4}{2} - 0 \right) = \frac{1}{2} \times 2 = 1$$

**Answer:**  $\checkmark = 1$  (center of  $[0, 2]$ )

## Visual



## 4. LINEARITY OF EXPECTATION

### The Formulas

$$E[aX + b] = aE[X] + b$$

$$E[X + Y] = E[X] + E[Y]$$

### What This Means

Expectation behaves “nicely” under transformation:

- Scaling: multiply expectation by same factor
- Shifting: add same constant
- Adding: expectations add (even if dependent!)

## Complete Proof: $\checkmark + b$

**Step 1:** Start with definition

$$E[aX + b] = \sum_i (ax_i + b)P(X = x_i)$$

**Step 2:** Split the sum

$$= \sum_i ax_i P(X = x_i) + \sum_i bP(X = x_i)$$

**Step 3:** Pull out constants

$$= a \sum_i x_i P(X = x_i) + b \sum_i P(X = x_i)$$

**Step 4:** Recognize pieces

- $\sum_i x_i P(X = x_i) = E[X]$
- $\sum_i P(X = x_i) = 1$

**Step 5:** Final result

$\checkmark + b$

## Complete Numerical Example

**Given:**  $\checkmark = 10$

**Find:**  $E[3X + 7]$

$$E[3X + 7] = 3E[X] + 7 = 3(10) + 7 = 37$$

**No need to know the full distribution!**

## ML Connection

| Application     | Why Linearity Helps                                    |
|-----------------|--------------------------------------------------------|
| Batch loss      | $E[\sum L_i] = \sum E[L_i]$ — decompose complex losses |
| Feature scaling | Predict how transformed features affect output         |
| Ensemble models | Average of expectations = expectation of average       |

# 5. VARIANCE

## The Definition

$$\text{Var}(X) = E[(X - \mu)^2]$$

## The Shortcut (Use This)

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

## What Each Symbol Means

| Symbol          | What It Is        | Why It's There                         |
|-----------------|-------------------|----------------------------------------|
| $\text{Var}(X)$ | Variance          | Spread measure                         |
| $X$             | Random variable   | The uncertain quantity                 |
| $\mu = E[X]$    | Mean              | Center point                           |
| $X - \mu$       | Deviation         | Distance from center                   |
| $(X - \mu)^2$   | Squared deviation | Makes positive; amplifies large errors |
| $E[\cdot]$      | Average           | "Average squared deviation"            |

## Complete Proof of Shortcut

Step 1: Start with definition

$$\text{Var}(X) = E[(X - \mu)^2]$$

Step 2: Expand square

$$(X - \mu)^2 = X^2 - 2X\mu + \mu^2$$

Step 3: Apply expectation

$$= E[X^2] - E[2X\mu] + E[\mu^2]$$

Step 4: Pull out constants ( $\mu$  is a fixed number)

☒  $+ \mu^2$

Step 5: Substitute ☒  $= \mu$

$$= E[X^2] - 2\mu^2 + \mu^2$$

Step 6: Simplify

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

## Complete Example

**Distribution:**

| X | P(X = x) |
|---|----------|
| 1 | 0.5      |
| 5 | 0.5      |

**Step 1:**  $\checkmark = 1(0.5) + 5(0.5) = 3$

**Step 2:**  $E[X^2] = 1(0.5) + 25(0.5) = 13$

**Step 3:**  $(E[X])^2 = 9$

**Step 4:**  $\text{Var}(X) = 13 - 9 = 4$

**Step 5:**  $\sigma = \sqrt{4} = 2$

## Visual (What Variance Measures)

1      3      5

-2       $\mu=3$       +2

Distance from center:  $\pm 2$

Squared: 4

Average: 4

Variance = 4

## ML Connection

| Concept               | Variance Role                                 |
|-----------------------|-----------------------------------------------|
| <b>Overfitting</b>    | High variance = model memorizes noise         |
| <b>Regularization</b> | Penalizes large weights    reduces variance   |
| <b>Dropout</b>        | Randomly disables neurons    reduces variance |
| <b>Ensembles</b>      | Averaging reduces variance                    |

|                       |                                              |
|-----------------------|----------------------------------------------|
| <b>Xavier/He init</b> | Controls weight variance for stable training |
|-----------------------|----------------------------------------------|

## 6. VARIANCE SCALING

### The Formula

$$\text{Var}(aX + b) = a^2\text{Var}(X)$$

### What This Means

| Operation     | Effect on Variance    | Why?                                           |
|---------------|-----------------------|------------------------------------------------|
| Multiply by a | Variance $\times a^2$ | Spread scales by factor, variance uses squares |
| Add b         | No effect             | Shifting doesn't change spread                 |

### Complete Proof

**Step 1:** Start with definition

$$\text{Var}(aX + b) = E[(aX + b - E[aX + b])]^2$$

**Step 2:** Use linearity of expectation

$$E[aX + b] = a\mu + b$$

**Step 3:** Substitute

$$= E[(aX + b - a\mu - b)^2]$$

**Step 4:** Cancel b

$$= E[(aX - a\mu)^2] = E[a^2(X - \mu)^2]$$

**Step 5:** Factor out  $a^2$

$$= E[a^2(X - \mu)^2] = a^2E[(X - \mu)^2]$$

**Step 6:** Recognize variance

$$\text{Var}(aX + b) = a^2\text{Var}(X)$$

### Complete Example

**Given:**  $\text{Var}(X) = 4$

**Find:**  $\text{Var}(3X + 5)$

$$\text{Var}(3X + 5) = 3^2 \times \text{Var}(X) = 9 \times 4 = 36$$

**The +5 does nothing!**

## Visual

| Original:       | Scaled ( $\times 3$ ):            |
|-----------------|-----------------------------------|
| 5               | 15                                |
| 3               | 9                                 |
| 1               | 3                                 |
| Spread: 2 units | Spread: 6 units ( $\times 3$ )    |
| Variance: 4     | Variance: 36 ( $\times 9 = 3^2$ ) |

## 7. STANDARD DEVIATION

### The Formula

$$\sigma = \sqrt{\text{Var}(X)}$$

### What This Means

**Variance is in “squared units”** (meters<sup>2</sup>, dollars<sup>2</sup>). **Standard deviation restores original units** (meters, dollars).

### Complete Example

| Quantity        | Value | Units               | Interpretation                |
|-----------------|-------|---------------------|-------------------------------|
| Height $X$      | —     | meters              | Raw measurement               |
| $E[X]$          | 1.75  | meters              | Average height                |
| $\text{Var}(X)$ | 0.01  | meters <sup>2</sup> | Spread in squared units       |
| $\sigma$        | 0.1   | meters              | Typical distance from average |

**“People are typically 0.1 meters from average height”** Clear!

**“Variance is 0.01 square meters”** Confusing!

## Visual

Variance:  $0.01 \text{ m}^2$  (abstract)

Square root

Standard deviation:  $0.1 \text{ m}$  (concrete)

"Typical person is 10cm from average"

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## MASTER STRUCTURE: THE UNCERTAINTY HIERARCHY

### LEVEL 1: RANDOM VARIABLE

"Convert reality into numbers"

$X: \Omega$

Example: Heads 1, Tails 0

### LEVEL 2: EXPECTATION

"Find the center/balance point"

$E[X] = \sum x \cdot P(x) \text{ or } \int x \cdot f(x) dx$

Example:  $E[\text{die}] = 3.5$

### LEVEL 3: VARIANCE

"Measure the spread/instability"

$\text{Var}(X) = E[X^2] - (E[X])^2$

Example:  $\text{Var}(\text{die}) = 2.92$

### LEVEL 4: STANDARD DEVIATION

"Make spread interpretable"

$\sigma = \sqrt{\text{Var}(X)}$

Example:  $\sigma(\text{die}) \approx 1.71$



LEVEL 5: TRANSFORMATION LAWS

"How do center and spread change?"

$E[aX+b] = aE[X]+b$  (linear)

$Var(aX+b) = a^2Var(X)$  (quadratic)

## THE PHYSICS ANALOGY (Master Memory Hook)

| Probability Concept | Physics Equivalent      | Why They Match                        |
|---------------------|-------------------------|---------------------------------------|
| Random Variable     | Coordinate system       | Maps physical reality to numbers      |
| Expectation         | Center of mass          | Balance point of probability “mass”   |
| Variance            | Moment of inertia       | Resistance to rotation = spread       |
| Standard Deviation  | Radius of gyration      | Typical distance from axis            |
| Linearity           | Galilean transformation | Physics laws hold under translation   |
| Scaling             | Scaling transformation  | Mass distribution changes predictably |

## ONE-PAGE FORMULA REFERENCE

| Concept                  | Formula                        | When to Use              |
|--------------------------|--------------------------------|--------------------------|
| Expectation (discrete)   | $E[X] = \sum x_i P(x_i)$       | Countable outcomes       |
| Expectation (continuous) | $E[X] = \int x f(x) dx$        | Uncountable outcomes     |
| Linearity                | $E[aX + b] = aE[X] + b$        | Transforming variables   |
| Variance (definition)    | $Var(X) = E[(X - \mu)^2]$      | Understanding spread     |
| Variance (shortcut)      | $Var(X) = E[X^2] - (E[X])^2$   | Fast computation         |
| Variance scaling         | $Var(aX + b) = a^2Var(X)$      | Feature scaling          |
| Std deviation            | $\sigma = \sqrt{Var(X)}$       | Reporting/interpretation |
| Sum of independent       | $Var(X + Y) = Var(X) + Var(Y)$ | Ensemble models          |

## FINAL INSIGHT

You don't need to memorize formulas. You need to understand that:

1. Reality is uncertain — we map it to numbers (RV)
2. Numbers have a center — we find it (Expectation)
3. Numbers spread around center — we measure it (Variance)
4. Transformations change center and spread predictably (Linearity & Scaling)

*Everything in ML — loss functions, optimization, regularization, uncertainty — is just an application of these four ideas.*

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